

Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning

Lukas Brunke*, Melissa Greeff*, Adam W. Hall*, Zhaocong Yuan*, Siqi Zhou*, Jacopo Panerati, and Angela P. Schoellig

*Equal contribution

Institute for Aerospace Studies, University of Toronto, Toronto, Ontario, Canada, M3H 5T6; University of Toronto Robotics Institute, Toronto, Ontario, Canada, M5S 1A4; Vector Institute for Artificial Intelligence, Toronto, Ontario, Canada, M5G 1M1; emails: {firstname.lastname}@robotics.utias.utoronto.ca

Published in the Annual Review of
Control, Robotics, and Autonomous
Systems.

This unofficial version compiled on:
December 8, 2021.

Keywords

safe learning, robotics, robot learning, learning-based control, safe reinforcement learning, adaptive control, robust control, model predictive control, machine learning, benchmarks

Abstract

The last half-decade has seen a steep rise in the number of contributions on safe learning methods for real-world robotic deployments from both the control and reinforcement learning communities. This article provides a concise but holistic review of the recent advances made in using machine learning to achieve safe decision making under uncertainties, with a focus on unifying the language and frameworks used in control theory and reinforcement learning research. Our review includes: learning-based control approaches that safely improve performance by learning the uncertain dynamics, reinforcement learning approaches that encourage safety or robustness, and methods that can formally certify the safety of a learned control policy. As data- and learning-based robot control methods continue to gain traction, researchers must understand when and how to best leverage them in real-world scenarios where safety is imperative, such as when operating in close proximity to humans. We highlight some of the open challenges that will drive the field of robot learning in the coming years, and emphasize the need for realistic physics-based benchmarks to facilitate fair comparisons between control and reinforcement learning approaches.

Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning

Lukas Brunke*, Melissa Greeff*, Adam W. Hall*, Zhaocong Yuan*, Siqi Zhou*, Jacopo Panerati, and Angela P. Schoellig

*加拿大安大略省多伦多市, M3H 5T6, 多伦多大学航空航天研究所, 贡献相等; 加拿大安大略省多伦多市, M5S 1A4, 多伦多大学机器人研究所; 加拿大安大略省多伦多市, M5G 1M1, Vector人工智能研究所; 电子邮件: {firstname.lastname}@robotics.utias.utoronto.ca

发表于《控制、机器人与自主系统年刊》。

此非官方版本编译于: 2021年12月8日。

Keywords

安全学习、机器人学、机器人学习、基于学习的控制、安全强化学习、自适应控制、鲁棒控制、模型预测控制、机器学习、基准测试

Abstract

在过去的五年中, 关于用于现实世界机器人部署的安全学习方法的研究贡献数量迅速增加, 来自控制领域和强化学习领域的学者均有所贡献。本文提供了一个简明但全面的综述, 总结了近年来使用机器学习在不确定性下实现安全决策方面的进展, 重点在于统一控制理论与强化学习研究中使用的语言和框架。我们的综述包括: 通过学习不确定动力学安全地提高性能的基于学习的控制方法、鼓励安全性或鲁棒性的强化学习方法, 以及能够正式验证学习控制策略安全性的方法。随着基于数据和学习的机器人控制方法继续获得关注, 研究人员必须理解在现实世界场景中——例如在接近人类操作时——何时以及如何最佳利用这些方法以确保安全。我们强调

Contents

1. INTRODUCTION	2
2. PRELIMINARIES AND BACKGROUND OF SAFE LEARNING CONTROL.....	3
2.1. Problem Statement	4
2.2. A Control Theory Perspective	6
2.3. A Reinforcement Learning Perspective	8
2.4. Bridging Control Theory and RL for Safe Learning Control	9
3. SAFE LEARNING CONTROL APPROACHES.....	10
3.1. Learning Uncertain Dynamics to Safely Improve Performance.....	11
3.2. Encouraging Safety and Robustness in Reinforcement Learning	14
3.3. Certifying Learning-Based Control Under Dynamics Uncertainty	18
4. BENCHMARKS	22
5. DISCUSSION AND PERSPECTIVES FOR FUTURE DIRECTIONS.....	25
A. SUMMARY OF REVIEWED LITERATURE	34
B. REVISION HISTORY	36

1. INTRODUCTION

Robotics researchers strive to design systems that can operate autonomously in increasingly complex scenarios, often in close proximity to humans. Examples include self-driving vehicles (1), aerial delivery (2), and the use of mobile manipulators for service tasks (3). However, the dynamics of these complex applications are often uncertain or only partially known—for example, the mass distribution of a carried payload might not be given *a priori*. Uncertainties arise from various sources. For example, the robot dynamics may not be perfectly modeled, sensor measurements may be noisy, and/or the operating environment may not be well-characterized or may include other agents whose dynamics and plans are not known.

In these real-world applications, robots must *make decisions despite having only partial knowledge of the world*. In recent years, the research community has multiplied its efforts to leverage data-based approaches to address this problem. This was motivated in part by the success of machine learning in other areas such as computer vision and natural language processing.

A crucial, domain-specific challenge of *learning for robot control* is the need to implement and formally guarantee safety of the robot’s behavior, not only for the optimized policy (or controller, which is essential for the certification of systems that interact with humans) but also during learning, to avoid costly hardware failures and improve convergence. Ultimately, these safety guarantees can only be derived from the assumptions and structure captured by the problem formalization.

Both control theory and machine learning—reinforcement learning (RL) in particular—have proposed approaches to tackle this problem. Control theory has traditionally taken a model-driven approach (see **Figure 1**): it leverages a given dynamics model and provides guarantees with respect to known operating conditions. RL has traditionally taken a data-driven approach, which makes it highly adaptable to new contexts at the expense of providing formal guarantees. Combining model-driven and data-driven approaches, and leveraging the advantages of each, is a promising direction for safe learning in robotics. The methods we review encourage **robustness** (by accounting for the worst-case scenarios and taking conservative actions), enable **adaptation** (by learning from online observations and adapting to unknown situations), and build and leverage **prediction models** (based on a combination of domain knowledge, real-world data, and high-fidelity simulators).

While control is still the bedrock of all current robot applications, the body of safe RL

Contents

1. INTRODUCTION	2
2. PRELIMINARIES AND BACKGROUND OF SAFE LEARNING CONTROL.....	3
2.1. Problem Statement	4
2.2. A Control Theory Perspective	6
2.3. A Reinforcement Learning Perspective	8
2.4. Bridging Control Theory and RL for Safe Learning Control	9
3. SAFE LEARNING CONTROL APPROACHES.....	10
3.1. Learning Uncertain Dynamics to Safely Improve Performance.....	11
3.2. Encouraging Safety and Robustness in Reinforcement Learning	14
3.3. Certifying Learning-Based Control Under Dynamics Uncertainty	18
4. BENCHMARKS	22
5. DISCUSSION AND PERSPECTIVES FOR FUTURE DIRECTIONS.....	25
A. SUMMARY OF REVIEWED LITERATURE	34
B. REVISION HISTORY	36

1. INTRODUCTION

机器人研究人员努力设计能够在日益复杂的场景中自主运行的系统，这些场景通常与人类靠近。例如，自动驾驶车辆、自主空中投递以及用于服务任务的移动操作臂。然而，这些复杂应用的动力学通常不确定或仅部分已知——例如，所携带负载的质量分布可能不明确。各种来源都会产生不确定性。例如，机器人动力学可能无法被完美建模，传感器测量可能存在噪声，操作环境可能未被充分描述，或者可能包含其他动力学和计划未知的主体。

在这些现实世界的应用中，机器人必须 *make decisions despite having only partial knowledge of the world*。近年来，研究界加倍努力利用基于数据的方法来解决这个问题。这部分是受到机器学习在其他领域（如计算机视觉和自然语言处理）成功的推动。

learning for robot control 的一个关键的、特定领域的挑战是需要实现并正式保证机器人的行为安全，这不仅仅针对优化后的策略（或控制器，这对于与人类互动的系统认证至关重要），还包括在学习过程中，以避免昂贵的硬件故障并提高收敛性。最终，这些安全保证只能从问题形式化所捕捉的假设和结构中获得。

控制理论和机器学习——特别是强化学习（RL）——都提出了应对这一问题的方法。控制理论传统上采取模型驱动的方法（见 **Figure 1**）：它利用给定的动力学模型，并针对已知的运行条件提供保证。强化学习传统上采取数据驱动的方法，这使其在新环境中高度适应，但以无法提供形式化保证为代价。结合模型驱动和数据驱动的方法，并利用各自的优势，是机器人安全学习的一个有前景的方向。我们回顾的方法通过考虑最坏情况并采取保守措施来鼓励 **robustness** ()，通过从在线观察中学习并适应未知情境来实现 **adaptation** ()，并构建和利用 **prediction models** (，基于领域知识、现实世界数据和高保真模拟器的结合)。

尽管控制正在 是所有当前机器人应用的基石，s 的主体

安全强化学习

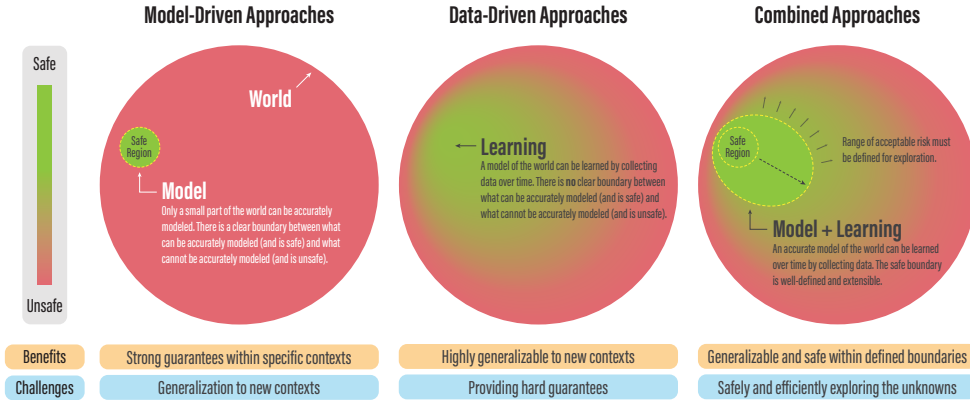


Figure 1: A comparison of model-driven, data-driven, and combined approaches.

literature has ballooned from tens to over a thousand publications in just a few years since its most recent review (4). Physics-based simulation (5), which we leverage in our open-source benchmark implementation¹, has played an important role in the recent progress of RL, however, the transfer to real systems remains a research area in itself (6).

Previous review works focused on specific techniques—for example, learning-based model predictive control (MPC) (7), iterative learning control (ILC) (8, 9), model-based RL (10), data-efficient policy search (11), imitation learning (12), or the use of RL in robotics (13, 14) and in optimal control (15)—without emphasizing the safety aspect. Recent surveys on safe learning control focus on either control-theoretic (16) or RL approaches (17), and do not provide a unifying perspective.

In this article, we provide a bird’s eye view of the most recent work in learning-based control and reinforcement learning that implement safety and provide safety guarantees for robot control. We focus on safe learning control approaches where the data generated by the robot system is used to learn or modify the feedback controller (or policy). We hope to help shrink the gap between the control and RL communities by creating a common vocabulary and introducing benchmarks for algorithm evaluation that can be leveraged by both (18, 19). Our target audience are researchers, with either a control or RL background, who are interested in a concise but holistic perspective on the problem of safe learning control. While we do not cover perception, estimation, planning, or multi-agent systems, we do connect our discussion to these additional challenges and opportunities.

2. PRELIMINARIES AND BACKGROUND OF SAFE LEARNING CONTROL

In this review, we are interested in the problem of *safe decision making under uncertainties using machine learning* (i.e., *safe learning control*). Intuitively, in safe learning control, our goal is to allow a robot to fulfil a task while respecting a set of safety constraints despite the uncertainties present in the problem. In this section, we define the safe learning control problem (Sec. 2.1) and provide an overview of how the problem of *decision making under uncertainties* has traditionally been tackled by the control (Sec. 2.2) and RL communities (Sec. 2.3). We highlight the main limitations of these approaches and articulate how novel data-based, safety-focused methods can address these gaps (Sec. 2.4).

¹Safe control benchmark suite on GitHub: <https://github.com/utiasDSL/safe-control-gym>

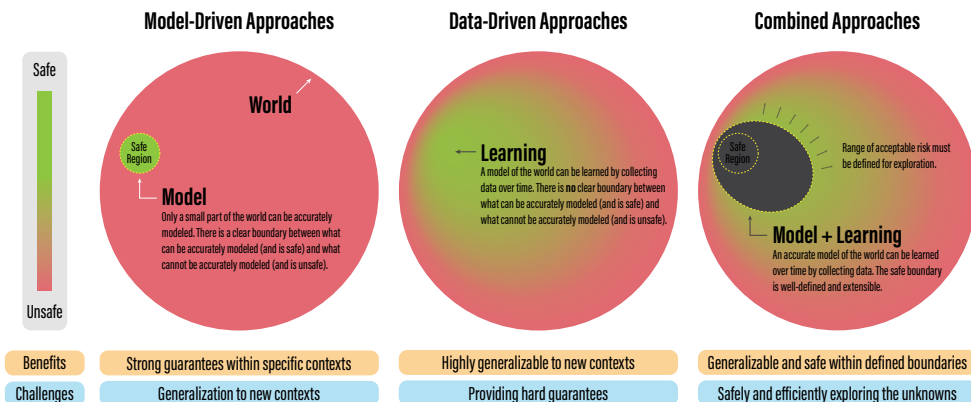


图 1：模型驱动、数据驱动和组合方法的比较。

自最近一次审查以来，文献数量在短短几年内从几十篇激增到一千多篇(4)。基于物理的仿真(5)，我们在开源基准实现¹中利用了它，在强化学习的近期进展中发挥了重要作用，然而，将其转移到真实系统仍然是一个独立的研究领域(6)。

以往的综述工作侧重于具体技术——例如，基于学习的模型预测控制（MPC）（7）、迭代学习控制（ILC）（8, 9）、基于模型的强化学习（10）、高效数据的策略搜索（11）、模仿学习（12），或在机器人（13, 14）和最优控制（15）中使用强化学习——而没有强调安全方面。近期关于安全学习控制的综述则侧重于控制理论（16）或强化学习方法（17），并未提供统一的视角。

在本文中，我们提供了基于学习的控制和强化学习领域中最新工作的概览，这些工作实现了安全性并为机器人控制提供了安全保障。我们重点关注安全学习控制方法，其中由机器人系统生成的数据用于学习或修改反馈控制器（或策略）。我们希望通过创建一个共同的词汇并引入可以被两者利用的算法评估基准（18, 19），帮助缩小控制与强化学习社区之间的差距。我们的目标读者是对安全学习控制问题感兴趣的研究人员，无论其背景是控制还是强化学习，他们希望获得简明但全面的视角。虽然我们不涉及感知、估计、规划或多智能体系统，但我们确实将讨论与这些额外的挑战和机遇联系起来。

2. PRELIMINARIES AND BACKGROUND OF SAFE LEARNING CONTROL

在本综述中，我们关注于 *safe decision making under uncertainties using machine learning* (的问题，即 *safe learning control*)。直观地说，在安全学习控制中，我们的目标是在存在问题不确定性的情况下，使机器人在遵守一系列安全约束的前提下完成任务。在本节中，我们定义了安全学习控制问题（第2.1节），并概述了控制界（第2.2节）和强化学习界（第2.3节）传统上是如何解决 *decision making under uncertainties* 问题的。我们强调了这些方法的主要局限性，并阐述了新型基于数据、以安全为重点的方法如何能够弥补这些不足（第2.4节）。

¹安全控制本

chmark suite on GitHub: <https://github.com/utiasDSL/safe-control-gym>

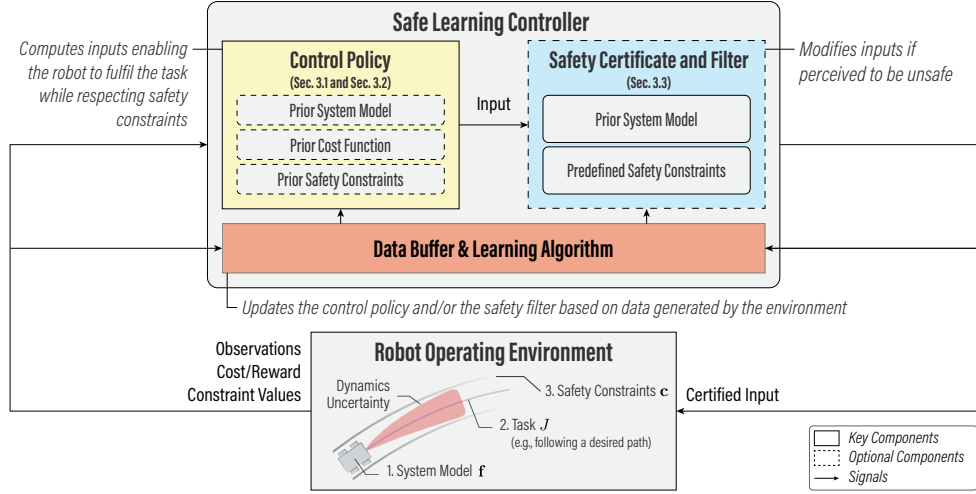


Figure 2: The safe learning control problem is defined by the cost function J , the system model $\mathbf{f}$, and the constraints $\mathbf{c}$, which may all be initially unknown. Data is used to update the control policy (see Sec. 3.1 and Sec. 3.2) or the safety filter (see Sec. 3.3).

M1. Transition

Probability Function:

The dynamics model in Equation 1 can be equivalently represented as a state transition probability function $T_k(\mathbf{x}_{k+1} | \mathbf{x}_k, \mathbf{u}_k)$, commonly seen in RL approaches (20).

2.1. Problem Statement

We formulate the safe learning control problem as an optimization problem to capture the efforts of both the control and RL communities. The optimization problem has three main components (see Figure 2): (i) a system model that describes the dynamic behavior of the robot, (ii) a cost function that defines the control objective or task goal, and (iii) a set of constraints that specify the safety requirements. The goal is to find a *controller* or *policy* that computes commands (also called *inputs*) that enable the system to fulfil the task while respecting given safety constraints. In general, any of the components could be initially unknown or only partially known. Below, we first introduce each of the three components and conclude by stating the overall safe learning control problem.

2.1.1. System Model. We consider a robot whose dynamics can be represented by the following discrete-time model:

$$\mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k), \quad 1.$$

where $k \in \mathbb{Z}_{\geq 0}$ is the discrete-time index, $\mathbf{x}_k \in \mathbb{X}$ is the state with $\mathbb{X}$ denoting the state space, $\mathbf{u}_k \in \mathbb{U}$ is the input with $\mathbb{U}$ denoting the input space, $\mathbf{f}_k$ is the dynamics model of the robot, $\mathbf{w}_k \in \mathbb{W}$ is the process noise distributed according to a distribution $\mathcal{W}$ (see also M1). Throughout this review, we assume direct access to (possibly noisy) measurements of the robot state $\mathbf{x}_k$ and neglect the problem of state estimation. Equation 1 represents many common robot platforms (e.g., quadrotors, manipulators, and ground vehicles). More complex models (e.g., partial differential equations) may be necessary for other robot designs.

2.1.2. Cost Function. The robot's task is defined by a cost function. We consider a finite-horizon optimal control problem with time horizon N . Given an initial state $\mathbf{x}_0$, the cost is computed based on the sequence of states $\mathbf{x}_{0:N} = \{\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N\}$ and the sequence of inputs $\mathbf{u}_{0:N-1} = \{\mathbf{u}_0, \mathbf{u}_1, \dots, \mathbf{u}_{N-1}\}$:

$$J(\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1}) = l_N(\mathbf{x}_N) + \sum_{k=0}^{N-1} l_k(\mathbf{x}_k, \mathbf{u}_k), \quad 2.$$

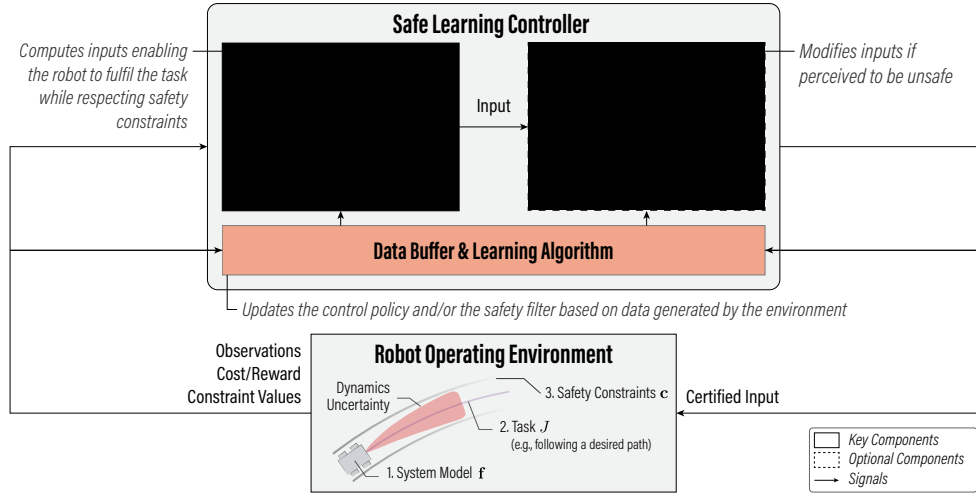


图2: 安全学习控制问题由成本函数 J 、系统模型 f 和约束 c 定义, 这些可能最初都是未知的。数据用于更新控制策略 (参见第3.1节和第3.2节) 或安全滤波器 (参见第3.3节)。

M1. Transition

Probability Function: 方程1中的动力学模型可以等效地表示为一个状态转移概率函数

$T_k(\mathbf{x}_{k+1} | \mathbf{x}_k, \mathbf{u}_k)$, 这在强化学习方法中很常见 (20)。

2.1. Problem Statement

我们将安全学习控制问题表述为一个优化问题, 以涵盖控制和强化学习两个领域的努力。该优化问题有三个主要组成部分 (见 **Figure 2**) : (i) 描述机器人动态行为的系统模型, (ii) 定义控制目标或任务目标的代价函数, 以及 (iii) 指定安全要求的一组约束条件。目标是找到一个 *controller* 或 *policy*, 该方法计算出命令 (也称为 *inputs*) , 使系统在遵守给定安全约束的同时完成任务。通常, 任何一个组成部分在初始阶段可能都是未知的或仅部分已知的。下面, 我们首先介绍这三个组成部分, 最后总结整体的安全学习控制问题。

2.1.1. System Model. 我们考虑一个其动力学可以用以下离散时间模型表示的机器人:

$$\mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k), \quad 1.$$

其中 $k \in \mathbb{Z}_{\geq 0}$ 是离散时间索引, $\mathbf{x}_k \in \mathbb{X}$ 是状态, $\mathbb{X}$ 表示状态空间, $\mathbf{u}_k \in \mathbb{U}$ 是输入, $\mathbb{U}$ 表示输入空间, $\mathbf{f}_k$ 是机器人的动力学模型, $\mathbf{w}_k \in \mathbb{W}$ 是根据分布 $\mathcal{W}$ (分布的过程噪声, 参见 M1) 。在本综述中, 我们假设可以直接获得机器人的状态 $\mathbf{x}_k$ (可能含噪声) 的测量值, 并忽略状态估计的问题。方程 1 代表了许多常见的机器人平台 (例如四旋翼、机械臂和地面车辆)。对于其他机器人设计, 可能需要更复杂的模型 (例如偏微分方程)。

2.1.2. Cost Function. 机器人的任务由成本函数定义。我们考虑一个有限时间范围的最优控制问题, 时间范围为 N 。给定初始状态 $\mathbf{x}_0$, 成本是基于状态序列 $\mathbf{x}_{0:N} = \{\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N\}$ 和输入序列 $\mathbf{u}_{0:N-1} = \{\mathbf{u}_0, \mathbf{u}_1, \dots, \mathbf{u}_{N-1}\}$ 计算的:

$$J(\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1}) = l_N(\mathbf{x}_N) + \sum_{k=0}^{N-1} l_k(\mathbf{x}_k, \mathbf{u}_k), \quad 2.$$

where $l_k : \mathbb{X} \times \mathbb{U} \mapsto \mathbb{R}$ is the stage cost incurred at each time step k , and $l_N : \mathbb{X} \mapsto \mathbb{R}$ is the terminal cost incurred at the end of the N -step horizon (see also M2). The stage and terminal cost functions map the state and input sequences, which may be random variables, to a real number, and may, for example, include the expected value or variance operators.

2.1.3. Safety Constraints. Safety constraints ensure, or encourage, the safe operation of the robot and include: (i) *state constraints* $\mathbb{X}_c \subseteq \mathbb{X}$, which define the set of safe operating states (e.g., the lane in self-driving), (ii) *input constraints* $\mathbb{U}_c \subseteq \mathbb{U}$ (e.g., actuation limits), and (iii) *stability guarantees* (e.g., the robot's motion converging to a desired path, see M3).

To encode the safety constraints, we define n_c constraint functions: $\mathbf{c}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \in \mathbb{R}^{n_c}$ with each constraint c_k^j being a real-valued, time-varying function. Starting with the strongest guarantee, we introduce three levels of safety: hard, probabilistic, and soft constraints (illustrated in Figure 3). In practice, safety levels are often mixed. For example, input constraints are typically hard constraints but state constraints may be soft constraints.

Safety Level III: Constraint Satisfaction Guaranteed. The system satisfies hard constraints:

$$c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \leq 0, \quad 3.$$

for all times $k \in \{0, \dots, N\}$ and constraint indexes $j \in \{1, \dots, n_c\}$.

Safety Level II: Constraint Satisfaction with Probability p . The system satisfies probabilistic constraints:

$$\Pr(c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \leq 0) \geq p^j, \quad 4.$$

where $\Pr(\cdot)$ denotes the probability and $p^j \in (0, 1)$ defines the likelihood of the j -th constraint being satisfied, with $j \in \{1, \dots, n_c\}$ and for all times $k \in \{0, \dots, N\}$. The chance constraint in Equation 4 is identical to the hard constraint in Equation 3 for $p^j = 1$.

Safety Level I: Constraint Satisfaction Encouraged. The system encourages constraint satisfaction. This can be achieved in different ways. One way is to add a penalty term to the objective function that discourages the violation of constraints with a high cost. A non-negative ϵ_j is added to the right-hand side of the inequality Equation 3, for all times $k \in \{0, \dots, N\}$ and $j \in \{1, \dots, n_c\}$,

$$c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \leq \epsilon_j, \quad 5.$$

and an appropriate penalty term $l_\epsilon(\epsilon) \geq 0$ with $l_\epsilon(\epsilon) = 0 \iff \epsilon = \mathbf{0}$ is added to the objective function. The vector ϵ includes all elements ϵ_j and is an additional variable of the optimization problem. Alternatively, although $c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k)$ is a step-wise quantity, some approaches only aim to provide guarantees on its expected value $E[\cdot]$ on a trajectory level:

$$J_{c^j} = E \left[\sum_{k=0}^{N-1} c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \right] \leq d_j, \quad 6.$$

where J_{c^j} represents the expected total constraint cost, and d_j defines the corresponding constraint threshold. The constraint function can optionally be discounted as $\gamma^k c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k)$, similar to the stage cost (see M2).

2.1.4. Formulation of the Safe Learning Control Problem. The functions introduced above, i.e., the system model $\mathbf{f}$, the constraints $\mathbf{c}$, and the cost function J , represent the *true* functions of the robot control problem. In practice, $\mathbf{f}$, $\mathbf{c}$, and J may be unknown or partially

M2. Discounted Rewards:

The stage cost can be related to the discounted rewards in RL:

$$l_k = -\gamma^k r_k(\mathbf{x}_k, \mathbf{u}_k),$$

where $r_k : \mathbb{X} \times \mathbb{U} \mapsto \mathbb{R}$ is the reward function, and $\gamma \in [0, 1]$ is the discount factor (20). We formulate a cost minimization problem for safe learning control, while RL typically solves a reward maximization problem.

M3. Stability:

There are different notions of stability in the control literature (21). Stability generally implies boundedness of the system state. One common notion of stability is Lyapunov stability, which requires the system to either stay close to (*stable*) or converge to (*asymptotically stable*) a desired state.

其中 $l_k: \mathbb{X} \times \mathbb{U} \mapsto \mathbb{R}$ 是在每个时间步 k 产生的阶段成本, $l_N: \mathbb{X} \mapsto \mathbb{R}$ 是在 N 步期末产生的终端成本 (参见 M2)。阶段和终端成本函数将状态和输入序列 (可能是随机变量) 映射为实数, 例如, 可能包括期望值或方差算子。

2.1.3. Safety Constraints. 安全约束确保或鼓励机器人的安全操作, 包括:

(i) *state constraints* $\mathbb{X}_c \subseteq \mathbb{X}$, 定义安全操作状态的集合 (例如自动驾驶中的车道),
(ii) *input constraints* $\mathbb{U}_c \subseteq \mathbb{U}$ (例如, 驱动限制), 以及 (iii) *stability guarantees* (例如, 机器人的运动收敛到所需路径, 参见 M3)。

为了编码安全约束, 我们定义了 n_c 个约束函数: $c_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \in \mathbb{R}^{n_c}$, 每个约束 c_k^j 都是实值、随时间变化的函数。从最强的保证开始, 我们引入了三种安全级别: 硬约束、概率约束和软约束 (如 **Figure 3** 所示)。在实际应用中, 安全级别通常是混合的。例如, 输入约束通常是硬约束, 而状态约束可能是软约束。

Safety Level III: Constraint Satisfaction Guaranteed. 该系统满足硬性约束:

$$c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \leq 0, \quad 3.$$

对于所有时间 $k \in \{0, \dots, N\}$ 和约束索引 $j \in \{1, \dots, n_c\}$ 。

Safety Level II: Constraint Satisfaction with Probability p . 该系统满足概率约束:

$$\Pr(c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \leq 0) \geq p^j, \quad 4.$$

其中 $\Pr(\cdot)$ 表示概率, $p^j \in (0, 1)$ 定义第 j 个约束被满足的可能性, 其中 $j \in \{1, \dots, n_c\}$ 并且对于所有时间 $k \in \{0, \dots, N\}$ 。公式 4 中的机会约束与公式 3 中的硬约束在 $p^j = 1$ 时相同。

Safety Level I: Constraint Satisfaction Encouraged. 该系统鼓励约束满足。这可以通过不同的方式实现。一种方式是在目标函数中添加一个惩罚项, 以降低高成本约束违反的可能性。在不等式方程 3 的右侧添加一个非负的 ϵ_j , 对于所有时间 $k \in \{0, \dots, N\}$ 和 $j \in \{1, \dots, n_c\}$,

$$c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \leq \epsilon_j, \quad 5.$$

并且将一个适当的惩罚项 $l_\epsilon(\epsilon) \geq 0$ 与 $l_\epsilon(\epsilon) = 0 \iff \epsilon = \mathbf{0}$ 添加到目标函数中。向量 ϵ 包含所有元素 ϵ_j , 并且是优化问题的一个额外变量。或者, 尽管 $c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k)$ 是一个分步量, 一些方法仅旨在对其在轨迹层面的期望值 $E[\cdot]$ 提供保证:

$$J_{c^j} = E \left[\sum_{k=0}^{N-1} c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \right] \leq d_j, \quad 6.$$

其中 J_{c^j} 表示预期的总约束成本, d_j 定义相应的约束阈值。约束函数可以选择性地折现为 $\gamma^k c_k^j(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k)$, 类似于阶段成本 (见 M2)。

2.1.4. Formulation of the Safe Learning Control Problem. 上面介绍的函数, 即系统模型 $\mathbf{f}$ 、约束 $\mathbf{c}$ 和代价函数 J , 表示机器人控制问题的 *true* 函数。在实践中, $\mathbf{f}$ 、 $\mathbf{c}$ 和 J 可能是未知的或部分已知的

M2. Discounted Rewards: 阶段成本可以与强化学习中的折扣奖励相关:

$$l_k = -\gamma^k r_k(\mathbf{x}_k, \mathbf{u}_k),$$

其中 r_k :

$\mathbb{X} \times \mathbb{U} \mapsto \mathbb{R}$ 是奖励函数, 且 $\gamma \in [0, 1]$ 是折扣因子 (20)。我们将安全学习控制问题表述为成本最小化问题, 而强化学习通常求解奖励最大化问题。

M3. Stability: 在控制文献中有不同的稳定性概念 (21)。稳定性通常意味着系统状态的有界性。一种常见的稳定性概念是李雅普诺夫稳定性, 它要求系统要么保持接近 (*stable*), 要么收敛到 (*asymptotically*) 期望状态。

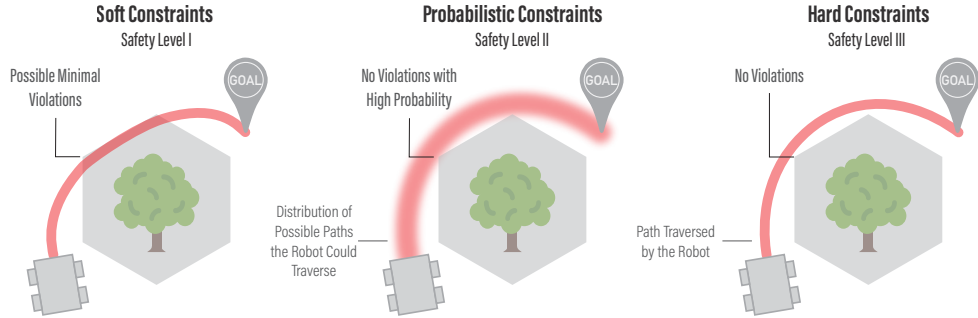


Figure 3: Illustration of the different safety levels.

known. Without loss of generality, we assume that each of the true functions $\mathbf{f}$, $\mathbf{c}$, and J can be decomposed into a nominal component $(\bar{\cdot})$, reflecting our prior knowledge, and an unknown component $(\hat{\cdot})$, to be learned from data. For instance, the dynamics model $\mathbf{f}$ can be decomposed as

$$\mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) = \bar{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k) + \hat{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k), \quad 7.$$

where $\bar{\mathbf{f}}$ is the prior dynamics model and $\hat{\mathbf{f}}$ are the uncertain dynamics.

Safe learning control (SLC) leverages our prior knowledge $\mathcal{P} = \{\bar{\mathbf{f}}, \bar{\mathbf{c}}, \bar{J}\}$ and the data collected from the system $\mathcal{D} = \{\mathbf{x}^{(i)}, \mathbf{u}^{(i)}, \mathbf{c}^{(i)}, l^{(i)}\}_{i=0}^D$ to find a policy (or controller) $\pi_k(\mathbf{x}_k)$ that achieves the given task while respecting all safety constraints,

$$\text{SLC} : (\mathcal{P}, \mathcal{D}) \mapsto \pi_k, \quad 8.$$

where $(\cdot)^{(i)}$ denotes a sample of a quantity $(\cdot)_k$ and D is the data set size. More specifically, we aim to find a policy π_k that best approximates the true optimal policy π_k^* , which is the solution to the following optimization problem:

$$J^{\pi^*}(\bar{\mathbf{x}}_0) = \min_{\pi_{0:N-1}, \epsilon} J(\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1}) + l_\epsilon(\epsilon) \quad 9a.$$

$$\text{s.t. } \mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k), \quad \mathbf{w}_k \sim \mathcal{W}, \quad \forall k \in \{0, \dots, N-1\}, \quad 9b.$$

$$\mathbf{u}_k = \pi_k(\mathbf{x}_k), \quad 9c.$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0, \quad 9d.$$

$$\text{Safety constraints according to either} \quad 9e.$$

Equation 3, Equation 4, Equation 5 or Equation 6, and $\epsilon \geq 0$,

where $\bar{\mathbf{x}}_0 \sim \mathcal{X}_0$ is the initial state with $\mathcal{X}_0$ being the initial state distribution, and ϵ and l_ϵ are introduced to account for the soft safety constraint case (Safety Level I, Equation 5) and are set to zero, for example, if only hard and probabilistic safety constraints are considered (Safety Levels II and III).

2.2. A Control Theory Perspective

Safe decision making under uncertainty has a long history in the field of control theory. Typical assumptions are that a model of the system is available and it is either parameterized by an unknown parameter or it has bounded unknown dynamics and noise. While the control approaches are commonly formulated using continuous-time dynamic models, they are usually implemented in discrete time with sampled inputs and measurements (see M4).

Adaptive control considers systems with parametric uncertainties (see M5) and adapts the controller parameters online to optimize performance. Adaptive control requires knowledge of the parametric form of the uncertainty (23) and typically considers a dynamics

M4. Continuous Time:

Many classical control approaches rely on a continuous-time formulation. Consider the uncertain time-varying, continuous-time model $\dot{\mathbf{x}} = \mathbf{f}_t(\mathbf{x}, \mathbf{u}, \mathbf{w})$, where $\mathbf{x}$, $\mathbf{u}$ and $\mathbf{w}$ are functions of time t . We can recover our discrete state through sampling $\mathbf{x}_k = \mathbf{x}(k\Delta t)$, where Δt is the sampling time. Similarly, the discrete control input is $\mathbf{u}_k = \mathbf{u}(k\Delta t)$ and often kept constant (22) over the time interval $[k\Delta t, (k+1)\Delta t)$.

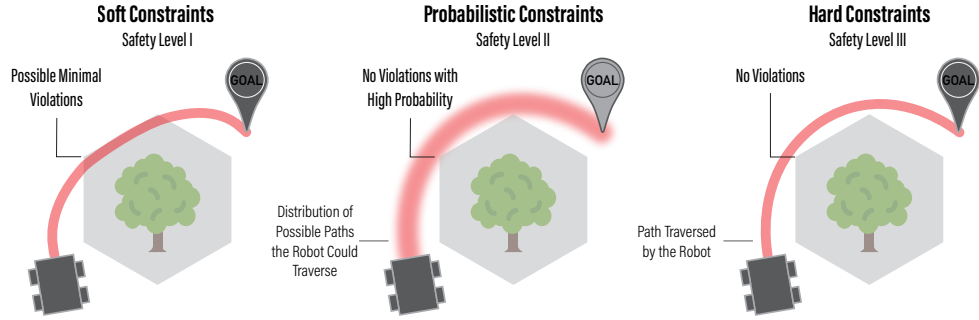


图3: 不同安全级别的示意图。

已知。在不失一般性的情况下，我们假设每个真实函数 $\mathbf{f}$ 、 $\mathbf{c}$ 和 J 都可以分解为一个名义成分 $\bar{(\cdot)}$ ，反映我们的先验知识，以及一个未知成分 $\hat{(\cdot)}$ ，将从数据中学习。例如，动力学模型 $\mathbf{f}$ 可以分解为

$$\mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) = \bar{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k) + \hat{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k), \quad 7.$$

其中 $\bar{\mathbf{f}}$ 是先前的动态模型， $\hat{\mathbf{f}}$ 是不确定的动态。

Safe learning control (SLC) 利用我们已有的知识 $\mathcal{P} = \{\bar{\mathbf{f}}, \bar{\mathbf{c}}, \bar{J}\}$ 以及从系统收集的数据 $\mathcal{D} = \{\mathbf{x}^{(i)}, \mathbf{u}^{(i)}, \mathbf{c}^{(i)}, l^{(i)}\}_{i=0}^D$ 来寻找一个策略（或控制器） $\pi_k(\mathbf{x}_k)$ ，该策略在实现给定任务的同时遵守所有安全约束，

$$\text{SLC} : (\mathcal{P}, \mathcal{D}) \mapsto \pi_k, \quad 8.$$

其中 $(\cdot)^{(i)}$ 表示一个量 $(\cdot)_k$ 的样本， D 是数据集的大小。更具体地说，我们旨在找到一个策略 π_k ，它最接近真实最优策略 π_k^* ，该策略是以下优化问题的解：

$$J^{\pi^*}(\bar{\mathbf{x}}_0) = \min_{\pi_{0:N-1}, \epsilon} J(\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1}) + l_\epsilon(\epsilon) \quad 9a.$$

$$\text{s.t. } \mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k), \quad \mathbf{w}_k \sim \mathcal{W}, \quad \forall k \in \{0, \dots, N-1\}, \quad 9b.$$

$$\mathbf{u}_k = \pi_k(\mathbf{x}_k), \quad 9c.$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0, \quad 9d.$$

$$\text{根据任一方的安全约束} \quad 9e.$$

$$\text{方程3、方程4、方程5或方程6, 以及 } \epsilon \geq 0,$$

其中 $\bar{\mathbf{x}}_0 \sim \mathcal{X}_0$ 是初始状态， $\mathcal{X}_0$ 是初始状态分布， ϵ 和 l_ϵ 的引入是为了考虑软安全约束情况（安全级别 I，公式 5），并且被设为零，例如，如果只考虑硬性和概率性安全约束（安全级别 II 和 III）。

2.2. A Control Theory Perspective

在不确定性下进行安全决策在控制理论领域有着悠久的历史。典型的假设是系统模型是可用的，并且它要么由未知参数参数化，要么具有有界的未知动态和噪声。虽然控制方法通常使用连续时间动态模型来制定，但它们通常以离散时间实现，并使用采样的输入和测量（参见 M4）。

Adaptive control 考虑具有参数不确定性的系统（见 M5），并在线调整控制器参数以优化性能。自适应控制需要了解不确定性的参数形式（23），并且通常考虑一个动力学

M4. Continuous

Time: 许多经典控制方法依赖于连续时间的表述。考虑不确定的时变连续时间模型 $\dot{\mathbf{x}} = \mathbf{f}_t(\mathbf{x}, \mathbf{u}, \mathbf{w})$ ，其中 $\mathbf{x}$ 、 $\mathbf{u}$ 和 $\mathbf{w}$ 是时间 t 的函数。我们可以通过采样 $\mathbf{x}_k = \mathbf{x}(k\Delta t)$ 恢复离散状态，其中 Δt 是采样时间。类似地，离散控制输入为 $\mathbf{u}_k = \mathbf{u}(k\Delta t)$ ，并且通常在时间区间 $[k\Delta t, (k+1)\Delta t)$ 内保持不变 (22)。

model that is affine in $\mathbf{u}$ and the uncertain parameters $\boldsymbol{\theta} \in \Theta$:

$$\mathbf{x}_{k+1} = \bar{\mathbf{f}}_{\mathbf{x}}(\mathbf{x}_k) + \bar{\mathbf{f}}_{\mathbf{u}}(\mathbf{x}_k)\mathbf{u}_k + \bar{\mathbf{f}}_{\boldsymbol{\theta}}(\mathbf{x}_k)\boldsymbol{\theta}, \quad (10)$$

where $\bar{\mathbf{f}}_{\mathbf{x}}$, $\bar{\mathbf{f}}_{\mathbf{u}}$, and $\bar{\mathbf{f}}_{\boldsymbol{\theta}}$ are known functions often derived from first principles and Θ is a possibly bounded parameter set. The control input is $\mathbf{u}_k = \boldsymbol{\pi}(\mathbf{x}_k, \hat{\boldsymbol{\theta}}_k)$, which is parameterized by $\hat{\boldsymbol{\theta}}_k$. The parameter $\hat{\boldsymbol{\theta}}_k$ is adapted by using either a *Lyapunov function* to guarantee that the closed-loop system is stable (see M6) or *Model Reference Adaptive Control (MRAC)* to make the system behave as a predefined stable reference model (23). Adaptive control is typically limited to parametric uncertainties and relies on a specific model structure. Moreover, adaptive control approaches tend to “overfit” to the latest observations and convergence to the true parameters is generally not guaranteed (23, 24). These limitations motivate the learning-based adaptive control approaches in Sec. 3.1.1.

Robust control is a control design technique that guarantees stability for pre-specified bounded disturbances, which can include unknown dynamics and noise. In contrast to adaptive control, which adapts to the parameters currently present, robust control finds a suitable controller for all possible disturbances and keeps the controller unchanged after the initial design. Robust control is largely limited to linear, time-invariant (LTI) systems with linear nominal model $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{A}}\mathbf{x}_k + \bar{\mathbf{B}}\mathbf{u}_k$ and unknown dynamics $\hat{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) = \hat{\mathbf{A}}\mathbf{x}_k + \hat{\mathbf{B}}\mathbf{u}_k + \mathbf{w}_k \in \mathbb{D}$, with $\mathbb{D}$ being known and bounded, and $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, $\hat{\mathbf{A}}$, $\hat{\mathbf{B}}$ being static matrices of appropriate size. That is,

$$\mathbf{x}_{k+1} = (\bar{\mathbf{A}} + \hat{\mathbf{A}})\mathbf{x}_k + (\bar{\mathbf{B}} + \hat{\mathbf{B}})\mathbf{u}_k + \mathbf{w}_k. \quad (11)$$

Robust control design techniques, such as robust H_{∞} - and H_2 -control design (25), yield controllers that are robustly stable for all $\hat{\mathbf{f}}_k \in \mathbb{D}$. Robust control can be extended to nonlinear systems whose dynamics can be decomposed into a linear nominal model $\bar{\mathbf{f}}$ and a nonlinear function $\hat{\mathbf{f}}$ with known bound $\hat{\mathbf{f}} \in \mathbb{D}$ (26).

Robust Model Predictive Control (MPC) extends classical adaptive and robust control by additionally guaranteeing state and input constraints, $\mathbf{x}_k \in \mathbb{X}_c$ and $\mathbf{u}_k \in \mathbb{U}_c$, for all possible bounded disturbances $\hat{\mathbf{f}} \in \mathbb{D}$. At every time step k , MPC solves a constrained optimization problem over a control input sequence $\mathbf{u}_{k:k+H-1}$ for a finite horizon H , and applies the first optimal control input to the system and then re-solves the optimization problem in the next time step based on the current state (27). A common approach in robust MPC (RMPC) is *tube-based MPC* (28), which uses a nominal prediction model $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k)$ in the MPC optimization and tightens the constraints to account for unmodeled dynamics. A stabilizing controller keeps the true state inside a bounded set of states around the nominal state, called *tube*, for all possible disturbances. Since the nominal states satisfy the tightened constraints and the true states stay inside the tube around the nominal states, constraint satisfaction for the true states is guaranteed. Tube-based MPC typically considers a linear nominal model $\bar{\mathbf{f}}(\bar{\mathbf{x}}_k, \bar{\mathbf{u}}_k) = \bar{\mathbf{A}}\bar{\mathbf{x}}_k + \bar{\mathbf{B}}\bar{\mathbf{u}}_k$ with nominal state $\bar{\mathbf{x}}_k$ and input $\bar{\mathbf{u}}_k$. In its simplest implementation, prior knowledge of set $\mathbb{D}$ is combined with a known stabilizing linear controller, $\mathbf{u}_{k,\text{stab}} = \mathbf{K}(\mathbf{x}_k - \bar{\mathbf{x}}_k)$ with gain $\mathbf{K}$, to determine the bounded tube Ω_{tube} from the matrices $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, $\mathbf{K}$, and the set $\mathbb{D}$. The stabilizing controller $\mathbf{u}_{k,\text{stab}}$ keeps all potential errors within the tube, $\mathbf{x}_k - \bar{\mathbf{x}}_k \in \Omega_{\text{tube}}$, for all k . For the nominal model, tube-based MPC solves the following constrained optimization problem at every time step k to obtain the optimal sequence $\bar{\mathbf{u}}_{0:H-1}^*$:

$$J_{\text{RMPC}}^*(\bar{\mathbf{x}}_k) = \min_{\bar{\mathbf{u}}_{0:H-1}} l_H(\mathbf{z}_H) + \sum_{i=0}^{H-1} l_i(\mathbf{z}_i, \bar{\mathbf{u}}_i) \quad (12a)$$

$$\text{s.t. } \mathbf{z}_{i+1} = \bar{\mathbf{A}}\mathbf{z}_i + \bar{\mathbf{B}}\bar{\mathbf{u}}_i, \quad \forall i \in \{0, \dots, H-1\} \quad (12b)$$

$$\mathbf{z}_i \in \mathbb{X}_c \ominus \Omega_{\text{tube}}, \quad \bar{\mathbf{u}}_i \in \mathbb{U}_c \ominus \mathbf{K}\Omega_{\text{tube}}, \quad (12c)$$

$$\mathbf{z}_0 = \bar{\mathbf{x}}_k, \quad \mathbf{z}_H \in \mathbb{X}_{\text{term}}, \quad (12d)$$

M5. Parametric Uncertainty: A

parametric model depends on a finite number of parameters that may have a physical interpretation or reflect our prior knowledge about the system structure in other ways. *Parametric uncertainty* is the uncertainty in the model parameters.

M6. Lyapunov Function:

Lyapunov functions are used to analyze the stability of a dynamical system (21). Consider a closed-loop system under some policy $\boldsymbol{\pi}(\mathbf{x})$: $\mathbf{x}_{k+1} = \mathbf{f}_{\boldsymbol{\pi}}(\mathbf{x}_k) = \mathbf{f}(\mathbf{x}_k, \boldsymbol{\pi}(\mathbf{x}_k))$ with $\mathbf{f}_{\boldsymbol{\pi}}(\mathbf{x}_k)$ being Lipschitz continuous. A Lipschitz continuous positive definite function $L: \mathbb{X} \rightarrow \mathbb{R}_{\geq 0}$ with $L(\mathbf{0}) = 0$ and $L(\mathbf{x}) > 0, \forall \mathbf{x} \neq \mathbf{0}$, is a Lyapunov function, if L maps states under closed-loop state feedback to strictly smaller values (i.e., $\Delta L(\mathbf{x}) = L(\mathbf{f}_{\boldsymbol{\pi}}(\mathbf{x})) - L(\mathbf{x}) < 0$). This implies that the state converges to the equilibrium at the origin. There exists an analogous formulation for continuous-time systems based on the derivatives of L (21).

在 $\mathbf{u}$ 和不确定参数 $\boldsymbol{\theta} \in \Theta$ 上是仿射的模型:

$$\mathbf{x}_{k+1} = \bar{\mathbf{f}}_{\mathbf{x}}(\mathbf{x}_k) + \bar{\mathbf{f}}_{\mathbf{u}}(\mathbf{x}_k)\mathbf{u}_k + \bar{\mathbf{f}}_{\boldsymbol{\theta}}(\mathbf{x}_k)\boldsymbol{\theta}, \quad 10.$$

其中 $\bar{\mathbf{f}}_{\mathbf{x}}$, $\bar{\mathbf{f}}_{\mathbf{u}}$ 和 $\bar{\mathbf{f}}_{\boldsymbol{\theta}}$ 是已知函数, 通常由第一性原理推导而来, Θ 是可能有界的参数集。控制输入是 $\mathbf{u}_k = \pi(\mathbf{x}_k, \hat{\boldsymbol{\theta}}_k)$, 它由 $\hat{\boldsymbol{\theta}}_k$ 参数化。参数 $\hat{\boldsymbol{\theta}}_k$ 的自适应通过使用 *Lyapunov function* 来保证闭环系统的稳定性 (见 M6), 或者使用 *Model Reference Adaptive Control (MRAC)* 来使系统表现为预定义的稳定参考模型 (23)。自适应控制通常仅限于参数不确定性, 并依赖于特定的模型结构。此外, 自适应控制方法往往会“过度拟合”最新观测值, 并且通常不能保证收敛到真实参数 (23, 24)。这些局限性促使了第 3.1.1 节中的基于学习的自适应控制方法的提出。

Robust control 是一种控制设计技术, 它可以保证在预先指定的有界扰动下的稳定性, 这些扰动可以包括未知的动态和噪声。与自适应控制不同, 自适应控制会根据当前存在的参数进行调节, 而鲁棒控制则为所有可能的扰动寻找一个合适的控制器, 并在初始设计后保持控制器不变。鲁棒控制主要限于具有线性标称模型 $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{A}}\mathbf{x}_k + \bar{\mathbf{B}}\mathbf{u}_k$ 和未知动态 $\hat{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) = \hat{\mathbf{A}}\mathbf{x}_k + \hat{\mathbf{B}}\mathbf{u}_k + \mathbf{w}_k \in \mathbb{D}$ 的线性时不变 (LTI) 系统, 其中 $\mathbb{D}$ 是已知且有界的, 而 $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, $\hat{\mathbf{A}}$, $\hat{\mathbf{B}}$ 是适当尺寸的静态矩阵。也就是说,

$$\mathbf{x}_{k+1} = (\bar{\mathbf{A}} + \hat{\mathbf{A}})\mathbf{x}_k + (\bar{\mathbf{B}} + \hat{\mathbf{B}})\mathbf{u}_k + \mathbf{w}_k. \quad 11.$$

稳健控制设计技术, 例如稳健的 H_{∞} 和 H_2 控制设计 (25), 能够产生在所有 $\hat{\mathbf{f}}_k \in \mathbb{D}$ 下都稳健稳定的控制器。稳健控制可以扩展到非线性系统, 其动力学可以分解为线性标称模型 $\bar{\mathbf{f}}$ 和具有已知界 $\hat{\mathbf{f}} \in \mathbb{D}$ (的非线性函数 $\hat{\mathbf{f}}$ (26))。

Robust Model Predictive Control (MPC) 在经典自适应和鲁棒控制的基础上进一步扩展, 通过额外保证状态和输入约束 $\mathbf{x}_k \in \mathbb{X}_c$ 和 $\mathbf{u}_k \in \mathbb{U}_c$, 适用于所有可能的有界扰动 $\hat{\mathbf{f}} \in \mathbb{D}$ 。在每一个时间步 k , MPC 会对一个有限时间范围 H 的控制输入序列 $\mathbf{u}_{k:k+H-1}$ 求解约束优化问题, 并将第一个最优控制输入应用到系统, 然后在下一时间步根据当前状态重新求解优化问题 (27)。一种常见的方法是

鲁棒模型预测控制 (RMPC) 是 *tube-based MPC* (28), 它在 MPC 优化中使用名义预测模型 $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k)$, 并收紧约束以考虑未建模的动力学。一个稳定控制器将真实状态保持在围绕名义状态的有界状态集合内, 称为 *tube*, 以应对所有可能的扰动。由于名义状态满足收紧后的约束, 并且真实状态保持在名义状态周围的管道内, 因此保证了真实状态的约束满足。基于管道的 MPC 通常考虑具有名义状态 $\bar{\mathbf{x}}_k$ 和输入 $\bar{\mathbf{u}}_k$ 的线性名义模型 $\bar{\mathbf{f}}(\bar{\mathbf{x}}_k, \bar{\mathbf{u}}_k) = \bar{\mathbf{A}}\bar{\mathbf{x}}_k + \bar{\mathbf{B}}\bar{\mathbf{u}}_k$ 。在其最简单的实现中, 将集合 $\mathbb{D}$ 的先验知识与已知的稳定线性控制器 $\mathbf{u}_{k,\text{stab}} = \mathbf{K}(\mathbf{x}_k - \bar{\mathbf{x}}_k)$ (增益为 $\mathbf{K}$) 结合起来, 以从矩阵 $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, $\mathbf{K}$ 和集合 $\mathbb{D}$ 确定有界管道 Ω_{tube} 。稳定控制器 $\mathbf{u}_{k,\text{stab}}$ 将所有潜在误差保持在管道内, 即 $\mathbf{x}_k - \bar{\mathbf{x}}_k \in \Omega_{\text{tube}}$, 适用于所有 k 。对于名义模型, 基于管道的 MPC 求解如下约束

$$J_{\text{RMPC}}^*(\bar{\mathbf{x}}_k) = \min_{\bar{\mathbf{u}}_{0:H-1}} l_H(\mathbf{z}_H) + \sum_{i=0}^{H-1} l_i(\mathbf{z}_i, \bar{\mathbf{u}}_i) \quad 12a.$$

$$\text{s.t. } \mathbf{z}_{i+1} = \bar{\mathbf{A}}\mathbf{z}_i + \bar{\mathbf{B}}\bar{\mathbf{u}}_i, \quad \forall i \in \{0, \dots, H-1\} \quad 12b.$$

$$\mathbf{z}_i \in \mathbb{X}_c \ominus \Omega_{\text{tube}}, \quad \bar{\mathbf{u}}_i \in \mathbb{U}_c \ominus \mathbf{K}\Omega_{\text{tube}}, \quad 12c.$$

$$\mathbf{z}_0 = \bar{\mathbf{x}}_k, \quad \mathbf{z}_H \in \mathbb{X}_{\text{term}}, \quad 12d.$$

M5. Parametric Uncertainty: A

parametric model 取决于有限数量的参数, 这些参数可能具有物理意义, 或以其他方式反映我们对系统结构的先验知识。*Parametric uncertainty* 是模型参数的不确定性。

M6. Lyapunov

Function: 李雅普诺夫函数用于分析动力系统的稳定性 (21)。考虑一个在某个策略 $\pi(\mathbf{x})$ 下的闭环系统: $\mathbf{x}_{k+1} = \mathbf{f}_{\pi}(\mathbf{x}_k) =, \mathbf{f}_{\pi}(\mathbf{x}_k)$ 是利普希茨连续的。一个利普希茨连续的正定函数 $L: \mathbb{X} \rightarrow \mathbb{R}_{\geq 0}$, 并且满足 $L(\mathbf{0}) = 0$ 和 $L(\mathbf{x}) > 0, \forall \mathbf{x} \neq \mathbf{0}$, 如果 L 在闭环状态反馈下将状态映射到严格更小的值 (即 $\Delta L(\mathbf{x}) = L(\mathbf{f}_{\pi}(\mathbf{x})) - L(\mathbf{x}) < 0$) 则它是一个李雅普诺夫函数。这意味着状态会收敛到位于原点的平衡点。对于基于 L (21) 导数的连续时间系统, 也存在类似的形式化表达。

M7. Constraint

Tightening: The constraint sets $\mathbb{X}_c$ and $\mathbb{U}_c$ are tightened by an error tube Ω_{tube} using the Pontryagin difference, $\mathbb{X}_c \ominus \Omega_{\text{tube}} = \{\mathbf{x} \in \mathbb{X} : \mathbf{x} + \boldsymbol{\omega} \in \mathbb{X}_c, \forall \boldsymbol{\omega} \in \Omega_{\text{tube}}\}$ and $\mathbb{U}_c \ominus \mathbf{K}\Omega_{\text{tube}}$, where $\mathbf{K}$ maps all elements in Ω_{tube} to the control input space $\mathbb{U}$ with $\mathbf{K}\Omega_{\text{tube}}$.

M8. Value and Action-Value

Functions: The value function of state $\mathbf{x}_k$ under policy π_k , $V^{\pi_k}(\mathbf{x}_k)$, is the expected return J of applying π_k from $\mathbf{x}_k$. Similarly, the action-value function of action $\mathbf{u}_k$ in state $\mathbf{x}_k$ under π_k , $Q^{\pi_k}(\mathbf{x}_k, \mathbf{u}_k)$, is the expected return J when taking the action $\mathbf{u}_k$ at $\mathbf{x}_k$, and then following π_k .

M9. Neural

Network: A *neural network* (NN) is a computational model with interconnected layers of neurons (parameterized by weights) that can be used to approximate highly nonlinear functions.

where $\mathbf{z}_i$ is the open-loop nominal state at time step $k+i$ and the state and input constraints are tightened using the bounded tube Ω_{tube} (see M7). Combined with the stabilizing control input $\mathbf{u}_{k,\text{stab}}$, the control input $\mathbf{u}_k = \mathbf{u}_{k,\text{stab}} + \bar{\mathbf{u}}_0^*$ is applied to the system at every time step. Stability and satisfaction of the tightened constraints is guaranteed by selecting the terminal cost l_H in Equation 12a, such that after the prediction horizon H the nominal state $\mathbf{z}_H$ is within a terminal constraint set $\mathbb{X}_{\text{term}}$ (see Equation 12d) at which point a known linear controller can be safely applied (27).

Both robust control and robust MPC are conservative, as they guarantee stability and—in the case of MPC—state and input constraints for the worst-case scenario. This usually yields poor performance, as described in (7). For example, a conservative uncertainty set $\mathbb{D}$ generates a large tube Ω_{tube} resulting in tight hard constraints that are prioritized over cost optimization. Learning-based robust control and RMPC improve performance by using data to (i) learn a less conservative state- and input-dependent uncertainty set $\mathbb{D}$ and/or (ii) learn the unknown dynamics $\hat{\mathbf{f}}$ and, as a result, reduce the remaining model uncertainty, see Sec. 3.1.2 and Sec. 3.1.3 respectively.

2.3. A Reinforcement Learning Perspective

Reinforcement learning (RL) is the standard machine learning framework to address the problem of sequential decision making under uncertainty. Unlike traditional control, RL generally does not rely on an *a priori* dynamics model $\hat{\mathbf{f}}$ and can be directly applied to uncertain dynamics $\mathbf{f}$. However, the lack of explicit assumptions and constraints in many of the works limit their applicability to safe control. RL algorithms attempt to find π^* while gathering data and knowledge of $\mathbf{f}$ from interaction with the system—taking random actions, initially, and improving afterwards. A long-standing challenge of RL, which hampers safety during the learning stages, is the exploration-exploitation dilemma—that is, whether to (i) act greedily with the available data or to (ii) explore, which means taking sub-optimal—possibly unsafe—actions $\mathbf{u}$ to learn a more accurate $\hat{\mathbf{f}}$.

RL typically assumes that the underlying control problem is a Markov decision process (MDP). An MDP comprises a state space $\mathbb{X}$, an input (*action*) space $\mathbb{U}$, stochastic dynamics (also called *transition model*, see M1), and a per-step reward function (see M2). When all the components of an MDP are known (in particular, $\mathbf{f}$ and J in Sec. 2.1), then it solves the problem in Equations 9a, 9b, 9c without the constraints. Dynamic programming (DP) algorithms such as value and policy iteration can be used to find an optimal policy π^* . Many RL approaches, however, make no assumptions on any part of $\mathbf{f}$ being known *a priori*.

We can distinguish model-based RL approaches, which learn an explicit model $\hat{\mathbf{f}}$ of the system dynamics $\mathbf{f}$ and use it to optimize a policy, from model-free RL algorithms. The latter algorithms (29) can be broadly categorized as: (i) value function-based methods, learning a value function (see M8); (ii) policy-search and policy-gradient methods, directly trying to find an optimal policy π^* ; and (iii) actor-critic methods, learning both a value function (*critic*) and the policy (*actor*). We also note that the convergence of these methods has been shown for simple scenarios but is still a challenge for more complex scenarios or when function approximators are used (30).

There are multiple practical hurdles to the deployment of RL algorithms in real-world robotics problems (6), for example, (i) the continuous, possibly high-dimensional $\mathbb{X}$ and $\mathbb{U}$ in robotics (often assumed to be finite, discrete sets in RL), (ii) the stability and convergence of the learning algorithm (31) (necessary, albeit not sufficient, to produce a stable policy, see M3), (iii) learning robust policies from limited samples, (iv) the interpretability of the learned policy—especially in deep RL when leveraging neural networks (NNs) (see M9) for function approximation (29)—and, importantly, (v) providing provable safety guarantees.

The exploration-exploitation dilemma can be mitigated using Bayesian inference. This

M7. Constraint

Tightening: 约束集合 $\mathbb{X}_c$ 和 $\mathbb{U}_c$ 通过使用 Pontryagin 差的误差管 Ω_{tube} 被收紧, $\mathbb{X}_c \ominus \Omega_{\text{tube}} = \{\mathbf{x} \in :$

$\mathbb{X} \quad \mathbf{x} + \boldsymbol{\omega} \in \mathbb{X}_c, \forall \boldsymbol{\omega} \in \Omega_{\text{tube}}\}$ 和 $\mathbb{U}_c \ominus \mathbf{K} \Omega_{\text{tube}}$, 其中 $\mathbf{K}$ 将 Ω_{tube} 中的所有元素映射到控制输入空间 $\mathbb{U}$, 并通过 $\mathbf{K} \Omega_{\text{tube}}$ 。

M8. Value and Action-Value

Functions: 状态 $\mathbf{x}_k$ 在策略 $\pi_k, V^{\pi_k}(\mathbf{x}_k)$ 下的价值函数是从 $\mathbf{x}_k$ 开始应用 π_k 后的期望回报 J 。类似地, 动作 $\mathbf{u}_k$ 在状态 $\mathbf{x}_k$ 下的动作价值函数在 $\pi_k, Q^{\pi_k}(\mathbf{x}_k, \mathbf{u}_k)$ 下是当在 $\mathbf{x}_k$ 采取动作 $\mathbf{u}_k$ 后, 再遵循 π_k 时的期望回报 J 。

M9. Neural

Network: A neural network 是具有相互连接的神经元层 (通过权重参数化) 的计算模型, 可用于近似高度非线性函数。

其中 $\mathbf{z}_i$ 是时间步 $k+i$ 的开环名义状态, 状态和输入约束通过有界管 Ω_{tube} (收紧, 见 M7)。结合稳定控制输入 $\mathbf{u}_{k,\text{stab}}$, 控制输入 $\mathbf{u}_k = \mathbf{u}_{k,\text{stab}} + \bar{\mathbf{u}}_0^*$ 在每个时间步施加到系统上。通过在方程 12a 中选择终端代价 l_H 来保证稳定性和收紧约束的满足, 从而在预测视界 H 之后, 名义状态 $\mathbf{z}_H$ 位于终端约束集 $\mathbb{X}_{\text{term}}$ (内, 见方程 12d), 此时可以安全地应用已知的线性控制器 (27)。

鲁棒控制和鲁棒 MPC 都是保守的, 因为它们保证了稳定性, 并且——在 MPC 的情况下——在最坏情况下保证状态和输入约束。这通常会导致性能较差, 如 (7) 所述。例如, 一个保守的不确定性集 $\mathbb{D}$ 会生成一个大的管 Ω_{tube} , 从而导致严格的硬约束, 这些约束被优先于成本优化。基于学习的鲁棒控制和 RMPC 通过使用数据来 (i) 学习一个较不保守的依赖于状态和输入的不确定性集 $\mathbb{D}$ 和/或 (ii) 学习未知动力学 $\hat{\mathbf{f}}$, 从而减少剩余的模型不确定性, 从而提升性能, 见 3.1.2 节和 3.1.3 节。

2.3. A Reinforcement Learning Perspective

强化学习 (RL) 是解决不确定性下序列决策问题的标准机器学习框架。与传统控制不同, 强化学习通常不依赖于动力学模型, 并且可以直接应用于不确定的动力学。然而, 许多工作的显性假设和约束的缺乏限制了它们在安全控制中的适用性。强化学习算法试图在收集数据和系统交互中获取知识的同时找到最优策略——最初采取随机动作, 然后逐步改进。强化学习的一个长期挑战, 在学习阶段会影响安全性, 是探索-利用困境——也就是, 是利用现有数据贪婪地行动, 还是进行探索, 即采取次优的——可能不安全的——动作以获得更准确的模型。

强化学习通常假设底层控制问题是一个马尔可夫决策过程 (MDP)。MDP 由状态空间 $\mathbb{X}$ 、输入 (action) 空间 $\mathbb{U}$ 、随机动力学 (也称为 *transition model*, 参见 M1) 以及每步奖励函数 (参见 M2) 组成。当已知 MDP 的所有组件 (特别是第 2.1 节中的 $\mathbf{f}$ 和 J) 时, 它将解决没有约束的方程 9a、9b、9c 中的问题。动态规划 (DP) 算法, 如价值迭代和策略迭代, 可以用来找到最优策略 π^* 。然而, 许多强化学习方法对 $\mathbf{f}$ 的任何部分是否已知没有假设 *a priori*。

我们可以区分基于模型的强化学习方法, 这些方法学习系统动力学的显式模型, 并利用它来优化策略, 与无模型的强化学习算法不同。后者算法大致可以分为: 基于价值函数的方法, 学习一个价值函数; 策略搜索和策略梯度方法, 直接尝试找到一个最优策略; 以及演员-评论家方法, 同时学习价值函数和策略。我们还注意到, 这些方法的收敛性已在简单场景中得到证明, 但在更复杂的场景或使用函数近似器时仍然是一个挑战。

在现实世界的机器人问题中部署强化学习算法存在多个实际障碍 (6), 例如, (i) 机器人中的连续、可能是高维的 $\mathbb{X}$ 和 $\mathbb{U}$ (在强化学习中通常假设为有限的、离散的集合), (ii) 学习算法的稳定性和收敛性 (31) (虽然必要, 但不足以产生稳定的策略, 见 M3), (iii) 从有限样本中学习稳健的策略, (iv) 学到的策略的可解释性——特别是在利用神经网络 (NNs) (见 M9) 进行函数逼近 (29) 的深度强化学习中——以及, 重要的是, (v) 提供可证明的安全性保障。

探索

可以使用贝叶斯推断来减轻 n-开发困境

rence。这

is achieved by computing posterior distributions over the states $\mathbb{X}$, possible dynamics $\mathbf{f}$, or total cost J from past observed data (32). These posteriors provide explicit knowledge of the problem’s uncertainty and can be used to guide exploration. In practice, however, full posterior inference is almost always prohibitively expensive and concrete implementations must rely on approximations.

To achieve constraint satisfaction (over states or sequences of states) and robustness to different, possibly noisy dynamics $\mathbf{f}$, constrained MDPs and robust MDPs are extensions of traditional MDPs that more closely resemble the problem statement in Sec. 2.1.

Constrained MDPs (CMDPs) (33) extend simple MDPs with constraints and optimize the problem in Equation 9 when Equation 9e takes the discounted form of *Safety Level I*’s Equation 6. We refer to the discounted constraint cost J_{c_j} under policy π as $J_{c_j}^\pi$. Traditional approaches to solve CMDPs, such as Linear Programming and Lagrangian methods, often assume discrete state-action spaces and cannot properly scale to complex tasks such as robot control. Deep RL promises to mitigate this, yet, applying it to the constrained problem still suffers from the computational complexity of the off-policy evaluation (M10) of trajectory-level constraints J_{c_j} (34). In Sec. 3.2.3, we present some recent advances in CMDP-based work that feature: (i) integration of deep learning techniques in CMDPs for more complex control tasks; (ii) provable constraint satisfaction throughout the exploration or learning process; and (iii) constraint transformation for the efficient evaluation of J_{c_j} from data collected off-policy.

Robust MDPs (35), inspired by robust control (see Sec. 2.2), extend MDPs such that the dynamics can include parametric uncertainties or disturbances, and the cost of the worst-case scenario is optimized. This is captured by the min-max optimization problem:

$$J^\pi(\bar{\mathbf{x}}_0) = \min_{\pi_{0:N-1}} \max_{\hat{\mathbf{f}} \in \mathbb{D}} J(\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1}) \quad 13a.$$

$$\text{s.t.} \quad \text{Equation 9b, Equation 9c, Equation 9d,} \quad 13b.$$

where $\mathbb{D}$ is a given uncertainty set of $\hat{\mathbf{f}}$. To keep solutions tractable, practical implementations typically restrict $\mathbb{D}$ to certain classes of models. This can limit the applicability of robust MDPs beyond toy problems. Recent work (36, 37, 38) applied deep RL to robust decision making, targeting key theoretical and practical hurdles such as (i) how to effectively model uncertainty with deep neural networks, and (ii) how to efficiently solve the min-max optimization (e.g., via sampling or two-player, game-theoretic formulations). These ideas, including adversarial RL and domain randomization, are presented in Sec. 3.2.4.

2.4. Bridging Control Theory and RL for Safe Learning Control

When designing a learning-based controller, we typically have two sources of information: (i) our prior knowledge, and (ii) data generated by the robot system. Control approaches rely on prior knowledge and on assumptions such as parametric dynamics models to provide safety guarantees. RL approaches typically make use of expressive learning models to extract patterns from data that facilitate learning complex tasks, but these can impede the provision of any formal guarantees. In recent literature, we see an effort from both the control and RL communities to develop safe learning control algorithms, with the goal of systematically leveraging expressive models for closed-loop control (see Figure 4). Questions that arise from these efforts include: how can control-theoretic tools be applied to expressive machine learning models? Or, how can expressive models be incorporated into control frameworks?

Expressive learning models can be categorized as deterministic (e.g., standard deep neural networks (DNNs)) or probabilistic (e.g., Gaussian processes (GPs), see M11, and Bayesian linear regression (BLR)). Deep learning techniques such as feed-forward NNs, convolutional NNs, and long short-term memory networks have the advantage of being able

M10. Off-policy, On-policy, and Offline RL:

Off-policy RL improves the value functions estimates with data collected operating under different policies. On-policy methods only use data generated by the current policy. Offline RL uses data previously collected by an unknown policy.

M11. Gaussian Process:

A Gaussian process (GP) is a set of random variables in which any finite number of the random variables can be modeled as a joint Gaussian distribution. Practically, a GP is a probabilistic model specifying a distribution over functions.

是通过计算关于状态 $\mathbf{x}$ 、可能的动态 $\mathbf{f}$ 或总成本 J 的后验分布来实现的，这些分布来自过去观察到的数据 (32)。这些后验分布提供了问题不确定性的明确知识，并可用于指导探索。然而，在实践中，完整的后验推断几乎总是代价高昂的，具体实现必须依赖于近似方法。

为了实现约束满足（在状态或状态序列上）以及对不同、可能有噪声的动态的鲁棒性，受约束的MDP和鲁棒MDP是传统MDP的扩展，更加贴近第2.1节中的问题陈述。

Constrained MDPs (CMDPs) (33) 在简单的 MDP 基础上增加了约束，当方程 9e 采用 *Safety Level I* 的方程 6 的折扣形式时，优化方程 9 中的问题。我们将策略 π 下的折扣约束代价 $J_{c,j}$ 称为 $J_{c,j}^\pi$ 。传统的解决 CMDPs 的方法，如线性规划和拉格朗日方法，通常假设状态-动作空间是离散的，无法适当扩展到机器人控制等复杂任务。深度强化学习有望缓解这一问题，然而，将其应用于有约束的问题仍然受制于轨迹级约束 $J_{c,j}$ (34) 的离策略评估 (M10) 的计算复杂性。在第 3.2.3 节中，我们介绍了一些基于 CMDP 的研究的最新进展，这些进展的特点包括：(i) 在 CMDPs 中整合深度学习技术以处理更复杂的控制任务；(ii) 在探索或学习过程中可证明的约束满足；以及 (iii) 为从离策略收集的数据高效评估 $J_{c,j}$ 的约束转换。

Robust MDPs (35)，受鲁棒控制的启发（参见第2.2节），扩展了MDP，使得动态可以包含参数不确定性或扰动，并优化最坏情况的成本。这可以通过最小-最大优化问题来表达：

$$J^\pi(\bar{\mathbf{x}}_0) = \min_{\pi_{0:N-1}} \max_{\mathbf{f} \in \mathbb{D}} J(\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1}) \quad 13a.$$

$$\text{s.t. Equation 9b, Equation 9c, Equation 9d,} \quad 13b.$$

$\mathbb{D}$ 是给定的不确定性集合 $\hat{\mathbf{f}}$ 。为了保持解的可处理性，实际实现通常将 $\mathbb{D}$ 限制在某些模型类别中。这可能限制了鲁棒 MDP 在玩具问题之外的适用性。近期的工作 (36, 37, 38) 将深度强化学习应用于鲁棒决策，针对关键的理论和实际障碍，如 (i) 如何使用深度神经网络有效地建模不确定性，以及 (ii) 如何高效地求解最小-最大优化（例如，通过采样或双人博弈理论形式）。这些思想，包括对抗性强化学习和领域随机化，在第 3.2.4 节中进行了介绍。

2.4. Bridging Control Theory and RL for Safe Learning Control

在设计基于学习的控制器时，我们通常有两种信息来源：(i) 我们的先验知识，以及 (ii) 由机器人系统生成的数据。控制方法依赖于先验知识以及诸如参数化动力学模型等假设，以提供安全保证。强化学习方法通常利用表达能力强的学习模型，从数据中提取有助于学习复杂任务的模式，但这些可能阻碍提供任何形式化的保证。在近期文献中，我们可以看到控制和强化学习社区都在努力开发安全学习控制算法，目标是系统地利用表达能力强的模型进行闭环控制（参见 **Figure 4**）。这些努力引发的问题包括：控制理论工具如何应用于表达能力强的机器学习模型？或者，如何将表达能力强的模型纳入控制框架？

表达性学习模型可以分为确定性模型（例如，标准深度神经网络（DNNs））或概率性模型（例如，高斯过程（GPs），参见 M11，以及贝叶斯线性回归（BLR））。深度学习技术如前馈神经网络、卷积神经网络和长短期记忆网络的优势在于能够

M10. Off-policy, On-policy, and Offline RL:

离策略强化学习使用在不同策略下收集的数据来改进价值函数的估计。在线策略方法只使用由当前策略生成的数据。离线强化学习使用之前由未知策略收集的数据。

M11. Gaussian Process:

Gaussian process(GP) 是一组随机变量，其中任意有限数量的随机变量都可以被建模为联合高斯分布。在实际中，GP 是一个概率模型，用于指定函数的分布。

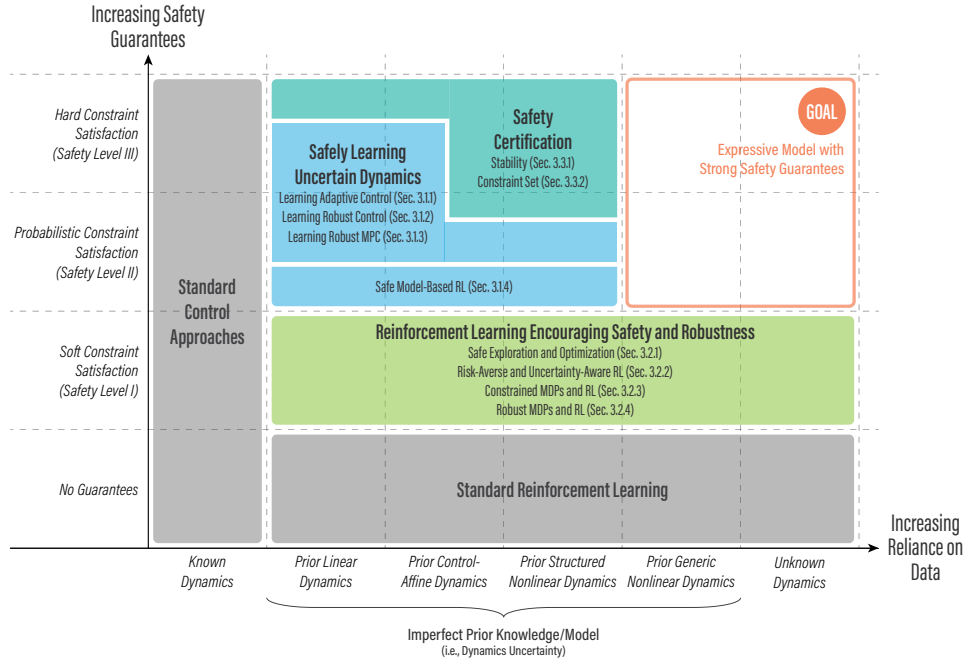


Figure 4: A summary of the safe learning control approaches reviewed in Sec. 3.

to abstract large volumes of data, enabling real-time execution in a control loop. On the other hand, their probabilistic counterparts, such as GPs and BLRs, provide model output uncertainty estimates that can be naturally blended into traditional adaptive and robust control frameworks. We note that there are approaches aiming to combine the advantages of the two types of learning (e.g., Bayesian NNs), and quantifying uncertainty in deep learning is still an active research direction.

In this review, we focus on approaches that address the problem of safe learning control at two stages: (i) *online adaptation or learning*, where online data is used to adjust the parameters of the controller, the robot dynamics model, the cost function, or the constraint functions during closed-loop operation; and (ii) *offline learning*, where data collected from each trial is recorded and used to update a model in a batch manner in between trials of closed-loop operation. In safe learning control, data is generally used to address the issue of uncertainties in the problem formulation and reduce the conservatism in the system design, while the safety aspect boils down to “knowing what is not known” and cautiously accounting for the incomplete knowledge via algorithm design.

3. SAFE LEARNING CONTROL APPROACHES

The ability to guarantee safe robot control is inevitably dependent on the amount of prior knowledge available and the type of uncertainties present in the problem of interest. In this section, we discuss approaches for *safe learning control in robotics* based on the following three categories (see Figure 2 and Figure 4):

- **Learning uncertain dynamics to safely improve performance:** The first set of works we review relies on an *a priori* model of the robot dynamics. The robot’s performance is improved by learning the uncertain dynamics from data. Safety is typically guaranteed based on standard control-theoretic frameworks, achieving Safety

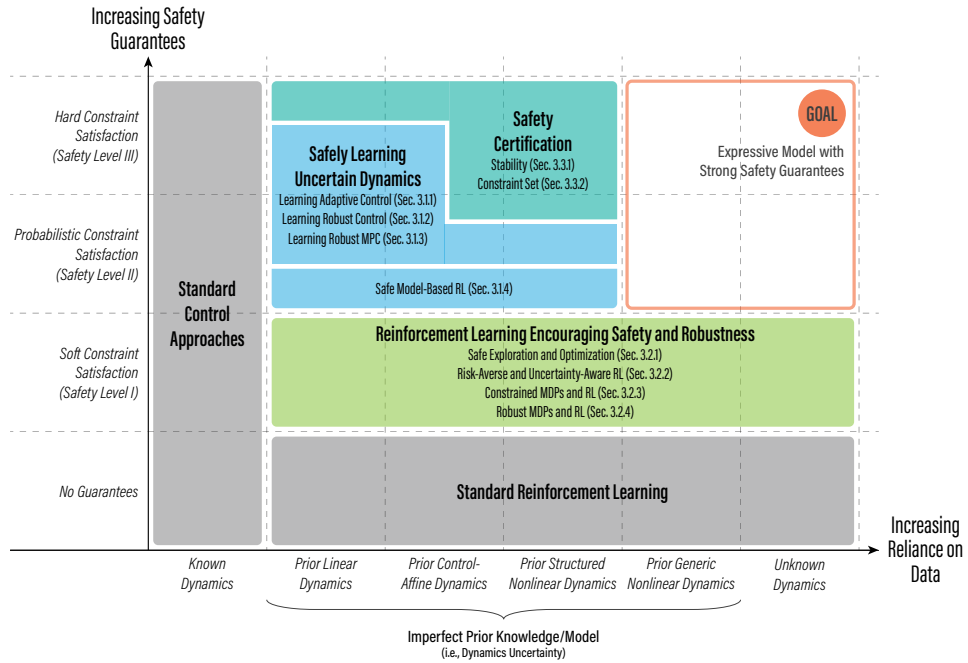


图 4：第 3 节中回顾的安全学习控制方法的总结。

抽取大量数据，实现控制回路中的实时执行。另一方面，它们的概率对应方法，例如高斯过程（GPs）和贝叶斯线性回归（BLRs），提供模型输出的不确定性估计，这些估计可以自然地融合到传统的自适应和稳健控制框架中。我们注意到有一些方法旨在结合这两种学习的优势（例如贝叶斯神经网络），而在深度学习中量化不确定性仍然是一个活跃的研究方向。

在本综述中，我们关注于在两个阶段解决安全学习控制问题的方法：

(i) *online adaptation or learning*，在该阶段中，在线数据用于在闭环操作过程中调整控制器参数、机器人动力学模型、代价函数或约束函数；以及 (ii) *offline learning*，在该阶段中，每次试验收集的数据被记录下来，并用于在闭环操作的试验间隔以批处理方式更新模型。在安全学习控制中，数据通常用于解决问题定义中的不确定性问题，并减少系统设计中的保守性，而安全性方面则归结为“了解未知”，并通过算法设计谨慎地考虑不完全知识。

3. SAFE LEARNING CONTROL APPROACHES

保证机器人安全控制的能力不可避免地依赖于可用的先验知识数量以及所关注问题中存在的的不确定性类型。在本节中，我们将基于以下三类讨论 *safe learning control in robotics* 的方法（参见 Figure 2 和 Figure 4）：

- **Learning uncertain dynamics to safely improve performance:** 我们回顾的第一组工作依赖于机器人动力学的 *a priori* 模型。通过从数据中学习不确定的动力学，可以提高机器人的性能。安全性通常是基于标准控制理论框架来保证的，从而实现安全。

Level II or III.

- **Encouraging safety and robustness in RL:** The second set of works encompasses approaches that usually do not have knowledge of an *a priori* robot model or the safety constraints. Rather than providing hard safety guarantees, these approaches encourage safe robot operation (Safety Level I), for example, by penalizing dangerous actions.
- **Certifying learning-based control under dynamics uncertainty:** The last set of works aims to provide safety certificates for learning-based controllers that do not inherently consider safety constraints. These approaches modify the learning controller output by either constraining the control policy, leveraging a known safe backup controller, or modifying the controller output directly to achieve stability and/or constraint satisfaction. They typically achieve Safety Level II or III.

Figure 4 categorizes the approaches reviewed in this section based on their safety level and reliance on data. A more detailed summary of the approaches can be found in **Table 1** of **Appendix A**.

3.1. Learning Uncertain Dynamics to Safely Improve Performance

In this section, we consider approaches that improve the robot’s performance by learning the uncertain dynamics and provide safety guarantees (Safety Level II or III) via control frameworks such as adaptive control, robust control, and robust MPC (outlined in **Sec. 2.2**). These approaches make assumptions about the unknown parts of the problem (e.g., Lipschitz continuity) and often rely on a known model structure (e.g., control-affine or linear system with bounded uncertainty) to prove stability and/or constraint satisfaction (see **Figure 4**).

3.1.1. Integrating Machine Learning and Adaptive Control. There are three main ideas to incorporating online machine learning into traditional adaptive control (see **Sec. 2.2**), each with its own distinct benefits. These include: (i) using black-box machine learning models to accommodate nonparametric unknown dynamics, (ii) using probabilistic learning and explicitly accounting for the learned model uncertainties, in order to achieve cautious adaptation, and (iii) augmenting adaptive control with deep learning approaches for experience memorization in order to minimize the need for re-adaptation.

Learning Nonparametric Unknown Dynamics with Machine Learning Models. One goal of integrating adaptive control and machine learning is to improve the performance of a robot subject to nonparametric dynamics uncertainties. As opposed to **Equation 10**, we consider a system with nominal dynamics $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{A}}\mathbf{x}_k + \bar{\mathbf{B}}\mathbf{u}_k$ and unknown dynamics $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{B}}\psi_k(\mathbf{x}_k)$, where $\psi_k(\mathbf{x}_k)$ is an unknown nonlinear function without an obvious parametric structure. Learning-based MRAC approaches (39, 40) make the uncertain system behave as the linear nominal model $\bar{\mathbf{f}}$ by using a combination of $\mathcal{L}_1$ adaptation (41) and online learning to approximate $\psi_k(\mathbf{x}_k)$ by $\hat{\psi}_k(\mathbf{x}_k) = \hat{\psi}_{\mathcal{L}_1,k} + \hat{\psi}_{\text{learn},k}(\mathbf{x}_k)$, where $\hat{\psi}_{\mathcal{L}_1,k}$ is the input computed by the $\mathcal{L}_1$ adaptive controller and $\hat{\psi}_{\text{learn},k}$ is the input computed by the learning module, which can be an NN (39) or a GP (40). The estimated $\hat{\psi}_k(\mathbf{x}_k)$ is then used in the controller $\pi_k(\mathbf{x}_k)$ to account for the unknown nonlinear dynamics $\psi_k(\mathbf{x}_k)$, improving a linear nominal control policy designed based on $\bar{\mathbf{f}}$. In these approaches, fast adaptation and stability guarantees are provided by the $\mathcal{L}_1$ adaptation framework (Safety Level III), while the learning module provides additional flexibility to capture the unknown dynamics. As shown in (39), the addition of learning improves the performance of the standard $\mathcal{L}_1$ adaptive controller and allows fast adaptation to be achieved at a lower sampling rate.

Cautious Adaptation with Probabilistic Model Learning. Another stream of adap-

二级或三级。

- **Encouraging safety and robustness in RL:** 第二组工作包括通常不知道 *a priori* 机器人模型或安全约束的方法。这些方法并不提供严格的安全保证，而是鼓励安全的机器人操作（安全级别 I），例如，通过惩罚危险行为。
- **Certifying learning-based control under dynamics uncertainty:** 最后一组工作旨在为不固有考虑安全约束的基于学习的控制器提供安全证书。这些方法通过限制控制策略、利用已知的安全备份控制器或直接修改控制器输出以实现稳定性和/或约束满足，从而修改学习控制器的输出。它们通常达到安全等级 II 或 III。

Figure 4 根据安全性和对数据的依赖程度，对本节中回顾的方法进行了分类。关于这些方法的更详细总结可以在附录A的Table 1中找到。

3.1. Learning Uncertain Dynamics to Safely Improve Performance

在本节中，我们考虑通过学习不确定动力学来提高机器人性能的方法，并通过控制框架（如自适应控制、鲁棒控制和鲁棒MPC（在第2.2节中概述））提供安全保障（安全级别 II 或 III）。这些方法对问题中未知的部分做出假设（例如 Lipschitz 连续性），并且通常依赖已知的模型结构（例如带有有界不确定性的控制仿射或线性系统）来证明稳定性和/或约束满足（见 Figure 4）。

3.1.1. Integrating Machine Learning and Adaptive Control. 将在线机器学习融入传统自适应控制有三个主要思路（见第2.2节），每个思路都有其独特的优势。这些包括：(i) 使用黑箱机器学习模型来适应非参数未知动态，(ii) 使用概率学习并明确考虑所学模型的不确定性，以实现谨慎的自适应，以及 (iii) 使用深度学习方法增强自适应控制，用于经验记忆，从而尽量减少重新自适应的需求。

Learning Nonparametric Unknown Dynamics with Machine Learning Models. 整合自适应控制和机器学习的一个目标是提高机器人在非参数动力学不确定性下的性能。与方程10不同，我们考虑一个具有标称动力学 $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{A}}\mathbf{x}_k + \bar{\mathbf{B}}\mathbf{u}_k$ 和未知动力学 $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{B}}\psi_k(\mathbf{x}_k)$ 的系统，其中 $\psi_k(\mathbf{x}_k)$ 是一个未知的非线性函数，没有明显的参数结构。基于学习的MRAC方法 (39, 40) 通过结合 $\mathcal{L}_1$ 适应 (41) 和在线学习，用 $\hat{\psi}_k(\mathbf{x}_k) = \hat{\psi}_{\mathcal{L}_1,k} + \hat{\psi}_{\text{learn},k}(\mathbf{x}_k)$ 近似 $\psi_k(\mathbf{x}_k)$ ，使不确定系统表现得像线性标称模型 $\bar{\mathbf{f}}$ ，其中 $\hat{\psi}_{\mathcal{L}_1,k}$ 是由 $\mathcal{L}_1$ 自适应控制器计算的输入， $\hat{\psi}_{\text{learn},k}$ 是由学习模块计算的输入，该模块可以是神经网络 (39) 或高斯过程 (40)。然后将估计的 $\hat{\psi}_k(\mathbf{x}_k)$ 用在控制器 $\pi_k(\mathbf{x}_k)$ 中，以考虑未知的非线性动力学 $\psi_k(\mathbf{x}_k)$ ，从而改进基于 $\bar{\mathbf{f}}$ 设计的线性标称控制策略。在这些方法中，快速适应和稳定性保证由 $\mathcal{L}_1$ 适应框架提供。

Cautious Adaptation with Probabilistic Model Learning. 另一个小溪 适应的米

tive control approaches leverage probabilistic models to achieve cautious adaptation by weighting the contribution of the learned model based on the model output uncertainty. We consider a system with nominal dynamics $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{f}}_{\mathbf{x}}(\mathbf{x}_k) + \bar{\mathbf{f}}_{\mathbf{u}}(\mathbf{x}_k)\mathbf{u}_k$ and unknown dynamics of the same form $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \hat{\mathbf{f}}_{\mathbf{x}}(\mathbf{x}_k) + \hat{\mathbf{f}}_{\mathbf{u}}(\mathbf{x}_k)\mathbf{u}_k$, where $\hat{\mathbf{f}}_{\mathbf{x}}(\mathbf{x}_k)$ and $\hat{\mathbf{f}}_{\mathbf{u}}(\mathbf{x}_k)$ are unknown nonparametric nonlinear functions. In a model inversion-based MRAC framework, an approximate feedback linearization is achieved via the nominal model to facilitate the design of MRAC, and a GP-based adaptation approach is used to compensate for feedback linearization errors due to the unknown dynamics (42). To account for the uncertainty in the GP model learning, the controller relies on the GP model only if the confidence in the latter is high: $\boldsymbol{\pi}_k(\mathbf{x}_k) = \boldsymbol{\pi}_{\text{nom}}(\mathbf{x}_k) - \gamma(\mathbf{x}_k, \mathbf{u}_k) \boldsymbol{\pi}_{\text{learn},k}(\mathbf{x}_k)$, where $\boldsymbol{\pi}_{\text{nom}}(\mathbf{x}_k)$ is the control policy designed based on the nominal model, $\boldsymbol{\pi}_{\text{learn},k}(\mathbf{x}_k)$ is the adaptive component designed based on the GP model, and $\gamma(\mathbf{x}_k, \mathbf{u}_k) \in [0, 1]$ is a scaling factor with 0 indicating low confidence in the GP. The stability of the overall system (Safety Level III) is guaranteed via a stochastic stability analysis, and the efficacy of the approach has been demonstrated in quadrotor experiments (43, 42).

Memorizing Experience with Deep Architectures. Apart from compensating for nonlinear and nonparametric dynamics uncertainties, deep learning approaches have also been applied to adaptive control for “memorizing” generalizable feature functions as the system adapts. In particular, an asynchronous DNN adaptation approach is proposed in (44, 45). Similar to (39, 40), we consider a linear nominal model $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{A}}\mathbf{x}_k + \bar{\mathbf{B}}\mathbf{u}_k$ and unknown nonlinear dynamics $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{B}}\psi_k(\mathbf{x}_k)$ with $\psi_k(\mathbf{x}_k)$ being an unknown nonlinear function. In the proposed approach, the last layer of the DNN is updated at a higher frequency for fast adaptation, while the inner layers are updated at a lower frequency to “memorize” pertinent features for the particular operation regimes. To provide safety guarantees, the authors derive an upper bound on the sampling complexity of the DNN to achieve a prescribed level of modelling error and leverage this result to show that Lyapunov stability of the adapted system can be guaranteed (Safety Level III) by ensuring the modelling error of the DNN is lower than a given bound. In contrast to other MRAC approaches, which do not usually retain a memory of the past experience, the inner layers of the asynchronous DNN in (44, 45) store relevant features that facilitate adaptation when similar scenarios arise in the future.

3.1.2. Learning-based Robust Control. Learning-based robust control improves the performance of classical robust control in Sec. 2.2 by using data to improve the linear dynamics model and reduce the uncertainty in Equation 11.

Using a Gaussian Process Dynamics Model for Linear Robust Control. The conservative performance of robust control in Sec. 2.2 is improved by updating the linear dynamics model and uncertainty in Equation 11 with a Gaussian Process (46) (GP). The unknown nonlinear dynamics $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k)$ are learnt as a GP, which is then linearized about an operating point. By linearizing the GP (as opposed to directly fitting a linear model) data close to the operating point are prioritized. The uncertain linear dynamics in Equation 11 are assumed to be modelled as $\tilde{\mathbf{A}} = \mathbf{A}_0 + \tilde{\mathbf{A}}\Delta$, $\tilde{\mathbf{B}} = \mathbf{B}_0 + \tilde{\mathbf{B}}\Delta$, where $\mathbf{A}_0$ and $\mathbf{B}_0$ are obtained from the linearized GP mean, $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$ are obtained from the linearized GP variance (often two standard deviations), and Δ represents a matrix with elements taking any value in the range of $[-1, +1]$. Further performance improvement is achieved by modelling $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$ as state dependent (47). Additionally, (48) was able to achieve better performance than (46) by leveraging the GP’s distribution, while maintaining the same level of safety. The main advantage of these approaches is that they can robustly guarantee Safety Level II stability while improving performance. This is achieved by shrinking the GP uncertainty as more data is added, which improves the linear model $\mathbf{A}_0$ and $\mathbf{B}_0$ and reduces the uncertain component $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$. This has been shown on a quadrotor (46). However, these approaches

主动控制方法利用概率模型通过根据模型输出的不确定性加权学习模型的贡献来实现谨慎的适应。我们考虑一个名义动力学为 $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{f}}_x(\mathbf{x}_k) + \bar{\mathbf{f}}_u(\mathbf{x}_k)\mathbf{u}_k$ 的系统，以及同形式的未知动力学 $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \hat{\mathbf{f}}_x(\mathbf{x}_k) + \hat{\mathbf{f}}_u(\mathbf{x}_k)\mathbf{u}_k$ ，其中 $\hat{\mathbf{f}}_x(\mathbf{x}_k)$ 和 $\hat{\mathbf{f}}_u(\mathbf{x}_k)$ 是未知的非参数非线性函数。在基于模型反演的 MRAC 框架中，通过名义模型实现近似反馈线性化以便于 MRAC 的设计，并使用基于 GP 的适应方法来补偿由未知动力学引起的反馈线性化误差 (42)。为了考虑 GP 模型学习中的不确定性，控制器仅在对 GP 模型的置信度较高时才依赖它：

$\pi_k(\mathbf{x}_k) = \pi_{\text{nom}}(\mathbf{x}_k) - \gamma(\mathbf{x}_k, \mathbf{u}_k) \pi_{\text{learn},k}(\mathbf{x}_k)$ ，其中 $\pi_{\text{nom}}(\mathbf{x}_k)$ 是基于名义模型设计的控制策略， $\pi_{\text{learn},k}(\mathbf{x}_k)$ 是基于 GP 模型设计的自适应分量，而 $\gamma(\mathbf{x}_k, \mathbf{u}_k) \in [0, 1]$ 是一个缩放因子，0 表示对 GP 的低置信度。整体系统的稳定性

Memorizing Experience with Deep Architectures. 除了补偿非线性和非参数动力学的不确定性之外，深度学习方法还被应用于自适应控制，以在系统适应过程中“记忆”可推广的特征函数。特别地，在 (44, 45) 中提出了一种异步 DNN 适应方法。类似于 (39, 40)，我们考虑一个线性名义模型 $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{A}}\mathbf{x}_k + \bar{\mathbf{B}}\mathbf{u}_k$ 和未知非线性动力学 $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{B}}\psi_k(\mathbf{x}_k)$ ，其中 $\psi_k(\mathbf{x}_k)$ 是一个未知的非线性函数。在所提出的方法中，DNN 的最后一层以更高的频率更新以实现快速适应，而内部层则以较低的频率更新，以“记住”特定操作状态下的相关特征。为了提供安全保证，作者推导了 DNN 采样复杂度的上限，以达到规定水平的建模误差，并利用这一结果表明，通过确保 DNN 的建模误差处于可控范围内，可以保证适应系统的 Lyapunov 稳定性（安全等级 III）。

3.1.2. Learning-based Robust Control. 基于学习的鲁棒控制通过使用数据改进线性动力学模型并减少方程 11 中的不确定性，从而提高了第 2.2 节中经典鲁棒控制的性能。

Using a Gaussian Process Dynamics Model for Linear Robust Control. 通过使用高斯过程 (46) (GP) 更新方程 11 中的线性动力学模型和不确定性，第 2.2 节中稳健控制的保守性能得到了改进。未知的非线性动力学 $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k)$ 被学习为 GP，然后在一个操作点附近进行线性化。通过线性化 GP（而不是直接拟合线性模型），接近操作点的数据被优先考虑。方程 11 中的不确定线性动力学被假定建模为 $\hat{\mathbf{A}} = \mathbf{A}_0 + \tilde{\mathbf{A}}\Delta, \hat{\mathbf{B}} = \mathbf{B}_0 + \tilde{\mathbf{B}}\Delta$ ，其中 $\mathbf{A}_0$ 和 $\mathbf{B}_0$ 由线性化 GP 均值获得， $\tilde{\mathbf{A}}$ 和 $\tilde{\mathbf{B}}$ 由线性化 GP 方差（通常为两倍标准差）获得， Δ 表示矩阵，其元素取值在 $[-1, +1]$ 范围内。通过将 $\tilde{\mathbf{A}}$ 和 $\tilde{\mathbf{B}}$ 建模为状态依赖 (47)，进一步提高了性能。此外，(48) 通过利用 GP 的分布，在保持相同安全性水平的同时，实现了比 (46) 更好的性能。该 ma

are limited to stabilization tasks and do not account for state and input constraints.

Exploiting Feedback Linearization for Robust Learning-Based Tracking. Trajectory tracking convergence, as opposed to the simpler stabilization task performed in (46), is guaranteed by exploiting the special structure of exactly feedback linearizable systems (49). This structure assumes that the nonlinear system dynamics in Equation 1 can be described by a linear nominal model, where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ have an integrator chain structure. It also assumes that the unknown dynamics are $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{B}}\psi(\mathbf{x}_k, \mathbf{u}_k)$, where $\psi(\mathbf{x}_k, \mathbf{u}_k)$ is an unknown invertible function. A probabilistic upper bound is obtained for $\psi(\mathbf{x}_k, \mathbf{u}_k)$ by learning this function as a GP. A robust linear controller is designed for the uncertain system based on this learnt probabilistic bound. Further performance improvement is achieved by also updating the feedback linearization (50) through improving the estimate of the inverse $\psi^{-1}(\cdot)$. These approaches have been applied to trajectory tracking, with Safety Level II, on Lagrangian mobile manipulators in (49) and quadrotor models in (50). However, they hinge on this special structure and cannot account for state and input constraints.

3.1.3. Reducing Conservatism in Robust MPC with Learning and Adaptation. The conservative nature of robust MPC (Sec. 2.2) is improved—while still satisfying input and state constraints—through (i) robust adaptive MPC, which adapts to parametric uncertainty, and (ii) learning-based robust MPC, which learns the unknown dynamics $\hat{\mathbf{f}}$ or, in one case, cost $\hat{J}$.

Robust Adaptive MPC. Robust adaptive MPC assumes parametric uncertainties, and (i) uses data to reduce the set of possible parameters over time or (ii) uses an inner-loop adaptive controller and an outer-loop robust MPC. This leads to improved performance compared to standard robust MPC (see Sec. 2.2), while satisfying hard constraints (Safety Level III).

The first set of approaches considers stabilization tasks, where the full system dynamics, Equation 10, are assumed to be linear and stable (51) or linear and unstable (52) with uncertain parameter $\theta \in \Theta_0$:

$$\mathbf{x}_{k+1} = (\bar{\mathbf{A}} + \hat{\mathbf{A}}(\theta))\mathbf{x}_k + (\bar{\mathbf{B}} + \hat{\mathbf{B}}(\theta))\mathbf{u}_k + \mathbf{w}_k. \quad 14.$$

The process noise and the parameters are assumed to be bounded by known sets $\mathbb{W}$ and the initially conservative Θ_0 , respectively. Given $\mathbb{W}$ and Θ_0 , we can derive a conservative upper bound on the uncertain dynamics $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k, \theta) = \hat{\mathbf{A}}(\theta)\mathbf{x}_k + \hat{\mathbf{B}}(\theta)\mathbf{u}_k + \mathbf{w}_k \in \mathbb{D}_0$, where $\mathbb{D}_0$ is a compact set determined from $\mathbb{W}$ and Θ_0 . To guarantee constraint satisfaction, tube-based MPC (see Sec. 2.2) is applied, where the initial tube $\Omega_{\text{tube},0}$ is based on $\mathbb{D}_0$. To reduce the conservatism of the approach, an adaptive control method is introduced to improve the estimate of the parameter set Θ_k and reduce the size of the tube $\Omega_{\text{tube},k}$ at each time step k . This idea has been extended to stochastic process noise for probabilistic constraint satisfaction (Safety Level II) (53), time-varying parameters (54), and linearly parameterized uncertain nonlinear systems (55, 56). Further performance improvement is achieved by combining robust adaptive MPC with iterative learning (57), which updates the terminal constraint set $\mathbb{X}_{\text{term}}$ in Equation 12d and the terminal cost l_H in Equation 12a after every full iteration using the closed-loop state trajectory and cost (58). In the second approach, an underlying MRAC (see Sec. 2.2) is used to make the closed-loop system dynamics resemble a linear reference model with bounded disturbance set $\mathbb{D}$ (59). This linear model and its bounds are then used in an outer-loop robust MPC to achieve fast stabilization in the presence of model errors.

Learning-Based Robust MPC. Learning-based robust MPC uses data to (i) improve the unknown dynamics estimate, (ii) reduce the uncertainty set, or (iii) update the cost to avoid states with high uncertainty. Unlike robust adaptive control, learning-based robust MPC considers nonparameterized systems.

仅限于稳态任务，且不考虑状态和输入约束。

Exploiting Feedback Linearization for Robust Learning-Based Tracking. 轨迹跟踪收敛性，与 (46) 中执行的更简单的稳定化任务不同，通过利用精确反馈线性化系统 (49) 的特殊结构得以保证。该结构假设方程1中的非线性系统动力学可以由线性名义模型描述，其中 $\bar{\mathbf{A}}$ 和 $\bar{\mathbf{B}}$ 具有积分器链结构。它还假设未知动力学为 $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) = \bar{\mathbf{B}}\psi(\mathbf{x}_k, \mathbf{u}_k)$ ，其中 $\psi(\mathbf{x}_k, \mathbf{u}_k)$ 是一个未知的可逆函数。通过将该函数学习为高斯过程 (GP)，可以获得 $\psi(\mathbf{x}_k, \mathbf{u}_k)$ 的概率上界。基于这一学习到的概率界，为不确定系统设计了鲁棒线性控制器。通过改进逆 $\psi^{-1}(\cdot)$ 的估计并更新反馈线性化 (50)，进一步提升了性能。这些方法已应用于带有安全等级II的拉格朗日移动操纵器 (49) 和四旋翼模型 (50) 的轨迹跟踪。然而，它们依赖于这一特殊结构

3.1.3. Reducing Conservatism in Robust MPC with Learning and Adaptation. 通过 (i) 鲁棒自适应MPC (适应参数不确定性) 和 (ii) 基于学习的鲁棒MPC (学习未知动态 $\hat{\mathbf{f}}$ 或在一种情况下，成本 $\hat{J}$)，改善了鲁棒MPC (第2.2节) 保守性的特性，同时仍满足输入和状态约束。

Robust Adaptive MPC. 稳健的自适应MPC假设存在参数不确定性，并且 (i) 利用数据随时间减少可能参数的集合，或者 (ii) 使用内环自适应控制器和外环稳健MPC。与标准稳健MPC相比，这可以提高性能 (参见第2.2节)，同时满足严格约束 (安全等级III)。

第一组方法考虑稳定化任务，其中假设完整系统动力学，方程10，是线性且稳定的 (51) 或线性且不稳定的 (52)，其参数 $\theta \in \Theta_0$ 存在不确定性：

$$\mathbf{x}_{k+1} = (\bar{\mathbf{A}} + \hat{\mathbf{A}}(\theta))\mathbf{x}_k + (\bar{\mathbf{B}} + \hat{\mathbf{B}}(\theta))\mathbf{u}_k + \mathbf{w}_k. \quad 14.$$

假设过程噪声和参数分别被已知集合 $\mathbb{W}$ 和最初的保守集合 Θ_0 有界。给定 $\mathbb{W}$ 和 Θ_0 ，我们可以推导出不确定动力学 $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k, \theta) = \hat{\mathbf{A}}(\theta)\mathbf{x}_k + \hat{\mathbf{B}}(\theta)\mathbf{u}_k + \mathbf{w}_k \in \mathbb{D}_0$ 的保守上界，其中 $\mathbb{D}_0$ 是由 $\mathbb{W}$ 和 Θ_0 决定的紧集。为了保证约束满足，应用基于管的 MPC (参见第 2.2 节)，其中初始管 $\Omega_{\text{tube},0}$ 基于 $\mathbb{D}_0$ 。为了降低方法的保守性，引入自适应控制方法来改进参数集 Θ_k 的估计，并在每个时间步 k 减小管 $\Omega_{\text{tube},k}$ 的大小。这一思想已扩展到具有概率约束满足 (安全水平 II) (53) 的随机过程噪声、时变参数 (54) 以及线性参数化的不确定非线性系统 (55, 56)。通过将鲁棒自适应 MPC 与迭代学习相结合 (57)，进一步提高了性能，这会更新方程中的终端约束集 $\mathbb{X}_{\text{termo}}$

Learning-Based Robust MPC. 基于学习的鲁棒MPC使用数据来 (i) 改善对未知动力学的估计，(ii) 缩小不确定性集合，或 (iii) 更新代价以避免高不确定性的状态。与鲁棒自适应控制不同，基于学习的鲁棒MPC考虑非参数化系统。

M12. Lipschitz Continuity:

A function $\mathbf{h} : \mathbb{A} \mapsto \mathbb{B}$ is said to be *Lipschitz continuous* on $\mathbb{A}$ if $(\exists \rho > 0)$ such that for any $\mathbf{a}_1, \mathbf{a}_2 \in \mathbb{A}$ the condition $\|\mathbf{h}(\mathbf{a}_1) - \mathbf{h}(\mathbf{a}_2)\| \leq \rho \|\mathbf{a}_1 - \mathbf{a}_2\|$ holds, where $\mathbb{A}$ and $\mathbb{B}$ are the domain and codomain of $\mathbf{h}$, and $\|\cdot\|$ denotes any l_p -norm of a vector. The constant ρ is called a *Lipschitz constant* of $\mathbf{h}$.

M13. Region of

Attraction: Let $\mathbf{x}^*$ be an equilibrium of a closed-loop system under some policy $\pi(\mathbf{x})$: $\mathbf{x}_{k+1} = \mathbf{f}_\pi(\mathbf{x}_k) = \mathbf{f}(\mathbf{x}_k, \pi(\mathbf{x}_k))$ with $\mathbf{f}_\pi(\mathbf{x}_k)$ being Lipschitz continuous. The *region of attraction* (ROA) of $\mathbf{x}^*$ is the set of states from which the system will converge to $\mathbf{x}^*$.

Under the assumption of a linear nominal model $\bar{\mathbf{f}}(\mathbf{x}, \mathbf{u}) = \bar{\mathbf{A}}\mathbf{x} + \bar{\mathbf{B}}\mathbf{u}$ and bounded unknown dynamics $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) \in \mathbb{D}$, the unknown dynamics can be safely learned from data, e.g., using a NN (60). Robust constraint satisfaction, Equation 12c, is guaranteed by using tube-based MPC for the linear nominal model and performance improvement is achieved by optimizing over the control inputs for the combined nominal and learned dynamics. If the bounded unknown dynamics are assumed to be state-dependent $\hat{\mathbf{f}}(\mathbf{x}_k) \in \mathbb{D}(\mathbf{x}_k)$, instead of using a constant tube Ω_{tube} , the state constraints in Equation 12c can be tightened based on the state-dependent uncertainty set $\mathbb{D}(\mathbf{x})$ (61). In a numerical stabilization task, a GP is used to model $\hat{\mathbf{f}}(\mathbf{x}_k)$ and its covariance determines $\mathbb{D}(\mathbf{x})$. This yields less conservative, probabilistic state constraints (Safety Level II).

Learning-based robust MPC can be extended to nonlinear nominal models. Typically, a GP is used to learn the unknown dynamics $\hat{\mathbf{f}}(\mathbf{x}, \mathbf{u})$: the mean updates the dynamics model in Equation 12b and the state- and input-dependent uncertainty set $\mathbb{D}(\mathbf{x}, \mathbf{u})$ is derived from the GP's covariance and contains the true uncertainty with high probability. Similarly, the state- and input-dependent tube in Equation 12c is determined from the uncertainty set $\mathbb{D}(\mathbf{x}, \mathbf{u})$. The main challenge, compared to using a linear nominal model, is the uncertainty propagation over the prediction horizon in the MPC because Gaussian uncertainty (obtained from the GP) is no longer Gaussian when propagated through the nominal nonlinear dynamics. Approximation schemes are required, such as using a sigma-point transform (62), linearization (63), exact moment matching (64), or ellipsoidal uncertainty set propagation (65). Additionally, further approximations (e.g., fixing the GP's covariances over the prediction horizon (63)) are usually required to achieve real-time implementation. These approximations can lead to violations of the probabilistic constraints (Safety Level II) in Equation 9e. An alternative approach to address the challenges of GPs uses a NN regression model that predicts the quantile bounds of the tail of a state trajectory distribution for a tube-based MPC (66). However, this approach is hindered by typically nonexhaustive training data sets, which can lead to the underestimation of the tube size.

For repetitive tasks, another approach is to adjust the cost function based on data instead of the system model. The predicted cost error is learned from data using the difference between the predicted cost at each time step and the actual closed-loop cost at execution. By adding this additional term to the cost function, the MPC penalizes states that had previously resulted in higher closed-loop cost than expected Equation 12a (67), resulting in reliable performance despite model errors.

3.1.4. Safe Model-Based RL with A Priori Dynamics. Safe model-based RL augments model-based RL (see Sec. 2.3) with safety guarantees (see Figure 4). Stability can be probabilistically guaranteed (Safety Level II) under the assumption that the known nominal model $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k)$ and the unknown part $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k)$ are Lipschitz continuous with known Lipschitz constants (see M12) (68). Given a Lyapunov function (see M6), an initial safe policy π_0 , and a GP to learn the unknown dynamics $\hat{\mathbf{f}}$, a control policy π_k is chosen so that it maximizes a conservative estimate of the *region of attraction* (ROA) (see M13). The most uncertain states (based on the GP's covariance) inside the ROA, are explored. This reduces the uncertainty over time and allows the ROA to be extended. The practical implementation resorts to discrete states for tractability and retains the stability guarantees while being sub-optimal in (exploration) performance.

3.2. Encouraging Safety and Robustness in Reinforcement Learning

The approaches in this section are safety-augmented variations of the traditional MDP and RL frameworks. In general, these methods do not assume knowledge of an *a priori* nominal model $\bar{\mathbf{f}}$ and some also learn the reward or step cost l (69), or the safety constraints $\mathbf{c}$ (70). Rather than providing strict safety guarantees, these approaches encourage constraint satis-

在假设一个线性名义模型 $\bar{\mathbf{f}}(\mathbf{x}, \mathbf{u}) = \bar{\mathbf{A}}\mathbf{x} + \bar{\mathbf{B}}\mathbf{u}$ 和有界未知动力学 $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k) \in \mathbb{D}$ 的前提下，可以从数据中安全地学习未知动力学，例如，使用神经网络 (60)。通过对线性名义模型使用基于管的 MPC，可以保证鲁棒约束的满足，即方程 12c，并且通过对名义动力学与学习到的动力学的组合控制输入进行优化，可以实现性能提升。如果假定有界未知动力学是依赖状态的 $\hat{\mathbf{f}}(\mathbf{x}_k) \in \mathbb{D}(\mathbf{x}_k)$ ，而不是使用常数管 Ω_{tube} ，则方程 12c 中的状态约束可以基于状态依赖的不确定性集 $\mathbb{D}(\mathbf{x})$ (进行收紧 61)。在一个数值稳定化任务中，使用高斯过程 (GP) 来建模 $\hat{\mathbf{f}}(\mathbf{x}_k)$ ，其协方差决定了 $\mathbb{D}(\mathbf{x})$ 。这产生了不那么保守的概率性状态约束（安全级别 II）。

M12. Lipschitz

Continuity: 如果存在 $(\exists \rho > 0)$ ，使得对于任意 $\mathbf{a}_1, \mathbf{a}_2 \in \mathbb{A}$ 条件

$$\|\mathbf{h}(\mathbf{a}_1) - \mathbf{h}(\mathbf{a}_2)\| \leq \rho \|\mathbf{a}_1 - \mathbf{a}_2\|$$

成立，则映射函数 $\mathbf{h} : \mathbb{A} \mapsto \mathbb{B}$ 在 $\mathbb{A}$ 上是

Lipschitz continuous 的，其中 $\mathbb{A}$ 和 $\mathbb{B}$ 分别是 $\mathbf{h}$ 的定义域和值域， $\|\cdot\|$ 表示向量的任意 l_p 范数。常数 ρ 称为 $\mathbf{h}$ 的 *Lipschitz constant*

基于学习的鲁棒MPC可以扩展到非线性名义模型。通常，使用高斯过程 (GP) 来学习未知的动力学：均值更新方程12b中的动力学模型，而与状态和输入相关的不确定性集由GP的协方差导出，并且以高概率包含真实的不确定性。类似地，方程12c中的与状态和输入相关的管子由不确定性集决定。与使用线性名义模型相比，主要挑战在于在MPC预测地平线上的不确定性传播，因为通过名义非线性动力学传播后，高斯不确定性（由GP获得）不再是高斯分布。需要近似方案，例如使用sigma点变换、线性化、精确矩匹配或椭圆不确定性集传播。此外，通常还需要进一步的近似（例如，在预测地平线上固定GP的协方差）。

M13. Region of

Attraction: 令 $\mathbf{x}^*$ 为在某个策略 $\pi(\mathbf{x})$ 下闭环系统的一个平衡点：

$\mathbf{x}_{k+1} = \mathbf{f}_\pi(\mathbf{x}_k) = \mathbf{f}(\mathbf{x}_k, \pi(\mathbf{x}_k))$ 是 Lipschitz 连续的。 $\mathbf{x}^*$ 的 *region of attraction* (ROA) 是系统将会收敛到 $\mathbf{x}^*$ 的状态集合。

对于重复性任务，另一种方法是根据数据而不是系统模型来调整代价函数。利用每个时间步的预测代价与实际闭环执行代价之间的差异，从数据中学习预测代价误差。通过将这个附加项加入代价函数中，MPC 会对先前导致闭环代价高于预期的状态进行惩罚（公式 12a (67)），从而在模型存在误差的情况下仍能保证性能可靠。

3.1.4. Safe Model-Based RL with A Priori Dynamics. 安全的基于模型的强化学习在基于模型的强化学习（参见第2.3节）的基础上增加了安全性保证（参见Figure 4）。在已知名义模型 $\bar{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k)$ 和未知部分 $\hat{\mathbf{f}}(\mathbf{x}_k, \mathbf{u}_k)$ 具有已知Lipschitz常数且为Lipschitz连续的假设下，可以以概率方式保证稳定性（安全级别 II）（参见M12）（68）。给定Lyapunov函数（参见M6）、初始安全策略 π_0 以及用于学习未知动态的 $\text{GP}\hat{\mathbf{f}}$ ，选择控制策略 π_k 以最大化 *region of attraction* (ROA) (的保守估计值（参见M13）)。对ROA内最不确定的状态（基于GP的协方差）进行探索。这会随着时间的推移减少不确定性，并允许ROA得到扩展。实际实现中，为了可处理性采用离散状态，同时保持稳定性保证，但在（探索）性能上是次优的。

3.2. Encouraging Safety and Robustness in Reinforcement Learning

本节中的方法是传统MDP和IRL框架的安全增强变体。一般来说，这些方法不假设已知一个 *a priori* 名义模型 $\bar{\mathbf{f}}$ ，有些方法还会学习奖励或步骤成本 l (69)，或安全约束 c (70)。这些方法不是提供严格的安全保证，而是鼓励约束的满足-

faction during and after learning, or robustness of the learned control policy π to uncertain dynamics (Safety Level I, see [Figure 4](#)). In plain MDP formulations, (i) states and inputs (or *actions*) are assumed to have known, often discrete and finite, domains but they are not further constrained while searching for an optimal policy π^* and (ii) only loose assumptions are made on the dynamics $\mathbf{f}$, for example, the system being Markovian ([20](#)) (see [M14](#)).

A previous taxonomy of safe RL ([4](#))—covering research published up until 2015—distinguished methods that either modify the exploration process to include external knowledge or modify the optimality criterion J with a safety factor. However, as pointed out in ([71](#)), the number and breadth of publications in RL, including safe RL, has since greatly increased. Because the approaches recently proposed in this area are numerous and diverse, we provide a high-level review of some of the most significant trends with a focus on their applicability to robotics. These include: (i) the safe exploration of MDPs; (ii) risk-aware and cautious RL adaptation; (iii) RL based on constrained MDPs; and (iv) robust RL.

3.2.1. Safe Exploration and Optimization. An RL algorithm’s need to explore poses a challenge to its safety as it must select inputs with unpredictable consequences in order to learn about them.

Safe Exploration. The problem of safely exploring an MDP is tackled in ([73](#)) through the notion of ergodicity (see [M15](#)). Policy updates that preserve the ergodicity of the MDP enable the system to return to an arbitrary state from any state. Thus, the core idea is to restrict the space of eligible policies to those that make the MDP ergodic (with at least a given probability). Exactly solving this problem, however, is NP-hard. A simplified problem is solved in ([73](#)), using a heuristic exploration algorithm ([74](#)), which leads to sub-optimal but safe exploration that only considers a subset of the ergodic policies. In practice, ([73](#)) is demonstrated in two simulated scenarios with discrete, finite $\mathbb{X}$ and $\mathbb{U}$. Rather than designing for a recoverable exploration through ergodicity, safe exploration strategies that provide constraint satisfaction are developed in ([75](#), [70](#)). A *safety layer* is used to convert an optimal but potentially unsafe action $\mathbf{u}_{\text{learn},k}$ produced by an NN policy, to the closest safe action $\mathbf{u}_{\text{safe},k}$ with respect to some safety state constraint. Both ([75](#)) and ([70](#)) involve solving a constrained least squares problem, $\mathbf{u}_{\text{safe},k} = \arg \min_{\mathbf{u}_k} \|\mathbf{u}_k - \mathbf{u}_{\text{learn},k}\|_2^2$, which is akin to the safety filter approaches further discussed in [Sec. 3.3.2](#). Concretely, ([75](#)) assumes full knowledge of the (linear) constraint functions and solves $\mathbf{u}_{\text{safe},k}$ using a differentiable Quadratic Programming (QP) solver. In contrast, ([70](#)) assumes constraints are unknown *a priori* but can be evaluated. Thus, the approach learns the parameters of linear approximations to these constraints and then uses them in the solver. Notably, ([75](#), [70](#)) consider single time-step state constraints. In [Sec. 3.2.3](#), we will also discuss methods that deal with more general trajectory-level constraints ([Equation 6](#)).

Safe Optimization. Several works address the problem of safely optimizing an unknown function (typically the cost function), often exploiting GP models ([76](#)). Safety refers to sampling inputs that do not violate a given safety threshold ([Equation 3](#), [Equation 4](#)). These approaches fall under the category of Bayesian optimization ([77](#)) and include *SafeOpt* ([78](#)); *SafeOpt-MC* ([79](#)), an extension towards multiple constraints; *StageOpt* ([80](#)), a the more efficient two-stage implementation; and *GoSafe* ([81](#)), to explore beyond the initial safe region. In particular, *SafeOpt* infers two subsets of the safe set from the GP model: (i) one with candidate inputs to extend the safe set, and (ii) one with candidate input to optimize the unknown function—from which it greedily picks the most uncertain. The ideas pioneered by *SafeOpt* were applied to MDPs in *SafeMDP* ([69](#)) resulting in safe exploration of MDPs with an unknown cost function $l(\mathbf{x}, \mathbf{u})$, which ([69](#)) models as a GP. In *SafeMDP*, the single-step reward represents the safety feature that should not violate a given threshold. Another extension of ([78](#)), *SafeExpOpt-MDP* ([82](#)), treats the safety feature c^j and the MDP’s cost l as two separate, unknown functions, allowing for the constraint of the first and the optimization

M14. Markov

Property: The probability for a system to be in state $\mathbf{x}_k$ at time k only depends on the state at time $k-1$, $\mathbf{x}_{k-1}$ and, in MDPs, $\mathbf{u}_{k-1}$.

M15. Ergodicity:

An MDP is *ergodic* if, by following a given policy, any state is reachable from any other state. Ergodic properties of dynamical systems are key to establish conditions for their controllability ([72](#)).

学习期间及之后的派系，或所学控制策略 π 对不确定动力学的稳健性（安全级别 I，见 Figure 4）。在简单的 MDP 表述中，(i) 状态和输入（或 *actions*）假设具有已知的、通常是离散和有限的域，但在寻找最优策略 π^* 时并没有进一步约束，同时 (ii) 只对动力学 f 做了宽松的假设，例如系统是马尔可夫的 (20)（见 M14）。

以往对安全强化学习 (RL) 的分类 (4) ——涵盖截至2015年的研究——区分了两类方法：一类是通过在探索过程中引入外部知识进行修改，另一类是通过加入安全因子来修改最优性准则 J 。然而，正如 (71) 中指出的那样，自那以后，RL 领域（包括安全 RL）的出版物数量和广度都大幅增加。由于近期在这一领域提出的方法众多且多样，我们提供了一些最重要趋势的高层次回顾，重点关注它们在机器人学中的适用性。这些包括：(i) MDPs 的安全探索；(ii) 风险感知与谨慎 RL 适应；(iii) 基于约束 MDPs 的 RL；以及 (iv) 鲁棒 RL。

3.2.1. Safe Exploration and Optimization. 强化学习算法需要探索，这对其安全性构成了挑战，因为它必须选择具有不可预测后果的输入以便了解它们。

Safe Exploration. 安全探索马尔可夫决策过程 (MDP) 的问题在 (73) 中通过遍历性（参见 M15）的概念进行了解决。保持 MDP 遍历性的策略更新使系统能够从任意状态返回到任意状态。因此，核心思想是将可选策略空间限制为能够使 MDP 具有遍历性的策略（至少以给定概率）。然而，完全解决这个问题是 NP 困难的。(73) 中解决了一个简化的问题，使用启发式探索算法 (74)，这导致次优但安全的探索，只考虑遍历性策略的子集。在实践中，(73) 在两个离散有限的模拟场景 $\mathbb{X}$ 和 $\mathbb{U}$ 中得到了验证。与通过遍历性设计可恢复探索不同，(75, 70) 中开发了提供约束满足的安全探索策略。使用 *safety layer* 将由神经网络策略产生的最优但可能不安全的动作 $\mathbf{u}_{\text{learn},k}$ 转换为最近的安全动作 $\mathbf{u}_{\text{safe},k\circ}$

Safe Optimization. 有若干工作关注如何安全地优化一个未知函数（通常是成本函数），通常利用高斯过程 (GP) 模型 (76)。安全性指的是采样输入时不违反给定的安全阈值（公式 3，公式 4）。这些方法属于贝叶斯优化 (77) 的范畴，包括 *SafeOpt* (78)；*SafeOpt-MC* (79)，一个面向多约束的扩展；*StageOpt* (80)，一个更高效的两阶段实现；以及 *GoSafe* (81)，用于探索初始安全区域之外的区域。特别地，*SafeOpt* 从 GP 模型中推断安全集的两个子集：(i) 一个用于扩展安全集的候选输入集，(ii) 一个用于优化未知函数的候选输入集——从中贪婪地选择最不确定的输入。*SafeOpt* 开创的思想被应用于 MDPs 中的 *SafeMDP* (69)，从而实现了具有未知成本函数的 MDPs 的安全探索 $l(\mathbf{x}, \mathbf{u})$ ，(69) 将其建模为 GP。在 *SafeMDP* 中，单步奖励表示安全特征，这

M14. Markov

Property: 系统在时间 k 处于状态 $\mathbf{x}_k$ 的概率仅取决于时间 $k-1$ 、 $\mathbf{x}_{k-1}$ 的状态，以及在马尔可夫决策过程 (MDPs) 中 $\mathbf{u}_{k-1}$ 。

M15. Ergodicity:

如果通过遵循给定的策略，任何状态都可以从任何其他状态到达，则一个 MDP 是 *ergodic* 的。动力系统的遍历性质是建立其可控性条件的关键 (72)。

M16. Conservative Q-learning (CQL):

A recent offline RL approach (83) to mitigate the overestimation problem in value functions. Q-learning is known to overestimate action-state values because of its maximization operator, and offline RL can lead to overestimated values of the actions that are more likely under the policy used to collect the data. CQL tackles this by using a novel update rule with a simple regularizer to learn value lower bounds instead.

M17. Model Ensembles:

Collections of learned models can be used to mitigate noise or better capture the stochasticity of a system. *Probabilistic Ensembles with Trajectory Sampling* (PETS) (84) is a model-based RL method that uses model ensembles and probabilistic networks to capture uncertainties in the system dynamics.

M18. Conditional Value at Risk

(CVaR): CVaR $_{\alpha}$ is a risk measure that is equal to the average of the worst α -percentile of the total cost J (88).

of the latter. A recent survey of these techniques was provided in (76), highlighting the distinction between safe learning in regression—i.e., minimizing the selection and evaluation of non-safe training inputs—and safe exploration in MDPs and stochastic dynamical systems like Equation 9b—i.e., selecting action inputs that also preserve ergodicity).

Learning a Safety Critic. A safety critic is a learnable action-value function Q_{safe}^{π} that can be used to detect if a proposed action can lead to unsafe conditions. The works (85, 86, 87) make use of such a critic and then resort to various fallback schemes to determine a safer alternative input. These works differ from those in Sec. 3.3.2 in that the filtering criterion depends on a model-free, learned value function, which can only grant the satisfaction of Safety Level I.

In (85), *safety Q-functions for reinforcement learning* (SQRL) are used to (i) learn a safety critic from only abstract, sparse safety labels (e.g., a binary indicator), and (ii) transfer knowledge of safe action inputs to new but similar tasks. SQRL trains Q_{safe}^{π} to predict the future probability of failure in a trajectory, and uses it to filter out unsafe actions from the policy π . The knowledge transfer is achieved by pre-training Q_{safe}^{π} and π in simulations, and then fine-tuning π on the new task (with similar dynamics $\mathbf{f}$ and safety constraints), still in simulation, while reusing Q_{safe}^{π} to discriminate unsafe inputs. However, the success of the final safe policy still depends on the task- and environment-specific hyperparameters (85) (which must be found *via* parameter search, in simulation, prior to the actual experiment). Building on (85), *Recovery RL* (86) additionally learns a recovery policy π_{rec} to produce fallback actions for Q_{safe}^{π} as an alternative to filtering out unsafe inputs and resorting to potentially sub-optimal ones in π . In (87), the authors extend Conservative Q-Learning (CQL, see M16) and propose Conservative Safety Critic (CSC). Similarly to (85), CSC assumes sparse safety labels and uses Q_{safe}^{π} for action filtering, but it trains Q_{safe}^{π} to upper bound the probability of failure and ensures provably safe policy improvement at each iteration.

3.2.2. Risk-Averse RL and Uncertainty-Aware RL. Safety in RL can also be encouraged by deriving and using risk (or uncertainty) estimates during learning. These estimates are typically computed for the system dynamics or the overall cost function, and leveraged to produce more conservative (and safer) policies.

Risk can be defined as the probability of collision for a robot performing a navigation task (89, 90). A collision model, captured by a neural network ensemble trained with Monte Carlo dropout, predicts the probability distribution of a collision, given the current state and a sequence of future actions. The collision-averse behavior is then achieved by incorporating the collision model in a MPC planner.

In (91), the authors build upon PETS (84) (see M17) to propose *cautious adaptation for safe RL* (CARL). CARL is composed of two training steps: (i) a pre-training phase that is not risk-aware, where a PETS agent is trained on multiple different system dynamics, and (ii) an adaptation phase, where the agent is fine-tuned on the target system by taking risk-averse actions. CARL also proposed two notions of risk, one to avoid low-reward trajectories and another to avoid *catastrophes* (e.g., irrecoverable states or constraint violations). In SAVED (92), a PETS (84) agent is used to predict the probability of a robot's collision and to evaluate a chance constraint for safe exploration. Similarly to the methods in the last paragraph of Sec. 3.2.1, SAVED also learns a value function from sparse costs, which it uses as a terminal cost estimate.

To learn risk-averse policies using only offline data, the approach in (93) optimizes for a risk measure of the cost such as Conditional Value-at-Risk (CVaR) (see M18). Instead of using model ensembles as in (89, 90, 91), the work in (93) uses distributional RL (see M19) to explicitly model the distribution of the total cost of the task (control of a simulated 1D car), and offline learning to improve scalability.

M16. Conservative Q-learning (CQL):

最近的一种离线强化学习方法 (83) 用于缓解价值函数中的过高估计问题。众所周知，Q学习由于其最大化算子会高估动作-状态值，而离线强化学习可能导致在用于收集数据的策略下更可能采取的动作的价值被高估。CQL 通过使用带有简单正则项的新型更新规则来学习价值下界，从而解决了这一问题。

M17. Model Ensembles:

已训练模型的集合可以用来减少噪声或更好地捕捉系统的随机性。ProbabilisticPET (84) 是一种基于模型集成和概率网络的强化学习方法，它使用模型集成和概率网络来捕捉系统动力学中的不确定性。

M18. Conditional Value at Risk (CVaR):

CVaR $_{\alpha}$ is a risk measure that is equal to the average of the worst α -percentile of the total cost J (88).

后者的相关情况。在文献 (76) 中提供了对这些技术的最新综述，强调了回归中安全学习的区别——即最小化对非安全训练输入的选择和评估——以及MDP和随机动力系统（如方程9b）中的安全探索——即选择也能保持遍历性的动作输入。

Learning a Safety Critic. 安全评论者是一个可学习的动作价值函数 Q_{safe}^{π} ，可用于检测所提议的动作是否可能导致不安全的情况。文献 (85, 86, 87) 使用了这样的评论者，然后采用各种备用方案来确定更安全的替代输入。这些工作与第 3.3.2 节中的工作不同，因为过滤标准依赖于无模型的、学习到的价值函数，而这种函数只能满足一级安全要求。

在 (85) 中，*safety Q-functions for reinforcement learning* (SQRL) 被用来从仅有的抽象、稀疏的安全标签（例如二元指示器） (i) 学习安全评估器，并将安全动作输入的知识转移到新的但类似的任务中。SQRL 训练 Q_{safe}^{π} 预测轨迹中未来的失败概率，并使用它从策略中 π 过滤掉不安全的动作。知识转移是通过在仿真中预训练 Q_{safe}^{π} 和 π 实现的，然后新的任务上 π 进行微调（具有类似的动力学 f 和安全约束），仍然在仿真中，同时重复使用 Q_{safe}^{π} 来区分不安全的输入。然而，最终安全策略的成功仍然取决于任务和环境特定的超参数 (85)（必须在实际实验之前通过仿真进行参数搜索 *via*）。在 (85) 的基础上，*Recovery RL* (86) 还学习了一个恢复策略 π_{rec} 来生成备用动作 Q_{safe}^{π} ，作为过滤不安全输入的替代方法，并依赖于潜在的...

3.2.2. Risk-Averse RL and Uncertainty-Aware RL. 在强化学习中，也可以通过在学习过程中获取并使用风险（或不确定性）估计来促进安全性。这些估计通常是针对系统动力学或整体成本函数计算的，并用于产生更保守（更安全）的策略。

风险可以定义为机器人执行导航任务时发生碰撞的概率 (89, 90)。碰撞模型由使用蒙特卡罗丢弃法训练的神经网络集成捕捉，它根据当前状态和一系列未来动作预测碰撞的概率分布。然后，通过在MPC规划器中引入碰撞模型，实现了对碰撞的规避行为。

在(91)中，作者在PETS (84)的基础上（见M17）提出了 *cautious adaptation for safe RL* (CARL)。CARL由两个训练步骤组成：(i) 一个不考虑风险的预训练阶段，在该阶段中PETS代理会在多个不同的系统动力学上进行训练，以及(ii) 一个适应阶段，在该阶段中代理通过采取规避风险的操作来对目标系统进行微调。CARL还提出了两种风险概念，一种用于避免低回报轨迹，另一种用于避免 *catastrophes*（例如，不可恢复状态或约束违规）。在SAVED (92)中，使用PETS (84)代理来预测机器人碰撞的概率，并评估用于安全探索的机会约束。与第3.2.1节最后一段中的方法类似，SAVED也从稀疏成本中学习价值函数，并将其用作终端成本估计。

为了仅使用离线数据学习风险规避策略，(93) 中的方法对成本的风险度量（如条件风险价值 CVaR，参见 M18）进行优化。(93) 的工作没有像 (89, 90, 91) 中那样使用模型集成，而是使用分布式强化学习（参见 M19）来显式建模任务总成本的分布（模拟一维汽车的控制），并通过离线学习提高可扩展性。

3.2.3. Constrained MDPs and RL. The CMDP framework is frequently used in safe RL, as it introduces constraints that can be used to express arbitrary safety notions in the form of Equation 6. RL for CMDPs, however, faces two important challenges (see Sec. 2.3): (i) how to incorporate and enforce constraints in the RL algorithm; and (ii) how to efficiently solve the constrained RL problem—especially when using deep learning models, which are the *de facto* standard in non-safe RL. Works in this section that address these questions include (i) Lagrangian methods for RL, (ii) generalized Lyapunov functions for constraints, and (iii) Backward Value Functions. Nonetheless, much of the work in this section remains confined to rather naive simulated tasks, motivating further investigation into the applicability of constrained RL in real-world control.

Lagrangian Methods in RL Optimization. In (88, 95), the CMDP optimization problem (see Sec. 2.3) is first transformed into an unconstrained optimization one (see M20) over the primal variable π and dual variable λ using the Lagrangian $\mathcal{L}(\pi, \lambda)$, and RL is used as a subroutine in the primal-dual updates for Equation 15b:

$$\mathcal{L}(\pi, \lambda) = J^\pi + \sum_j \lambda_j (J_{c_j}^\pi - d^j) \quad 15a.$$

$$(\pi^*, \lambda^*) = \arg \max_{\lambda \geq 0} \min_{\pi} \mathcal{L}(\pi, \lambda) \quad \text{s.t.} \quad \text{Equation 9b, 9c, 9d.} \quad 15b.$$

In particular, (88) defines a constraint on the CVaR of cost J^π , and uses policy gradient or actor-critic methods to update the policy in Equation 15b. In subsequent work (95), the authors improve upon (88) by incorporating off-policy updates of the dual variable (with the on-policy primal-dual updates). This is empirically shown to achieve better sample efficiency and faster convergence. In (34), the authors extend a standard trust-region (see M21) RL algorithm (96) to CMDPs using a novel bound that relates the expected cost of two policies to their state-averaged divergence (see M21). The key idea is performing primal-dual updates with surrogates/approximations of the cost J^π and constraint cost $J_{c_j}^\pi$ derived from the bound. The benefits are two-fold: (i) the surrogates can be estimated with only state-action data, bypassing the challenge of trajectory evaluation from off-policy data; and (ii) the updates guarantee monotonic policy improvement and near constraint satisfaction at each iteration (Safety Level I). However, unlike (88, 95), each update involves solving the dual variables from scratch, which can be computationally expensive.

A Lyapunov Approach to Safe RL. Lyapunov functions (see M6) are used extensively in control to analyze system stability and they are a powerful tool to translate a system’s global properties into local conditions. In (97, 98), Lyapunov functions are used to transform the trajectory-level constraints $J_{c_j}^\pi$ in Equation 6 into step-wise, state-based constraints. This allows a more efficient computation of $J_{c_j}^\pi$ and mitigates the cost of off-policy evaluation. These approaches, however, require the system to start from a baseline policy π_0 that already satisfies the constraints. In (97), four different algorithms are proposed to solve CMDPs by combining traditional RL methods and the Lyapunov constraints, but they are only applicable to discrete input spaces (with continuous state spaces). In subsequent work (98), the authors of (97) extend the approach to continuous input spaces and standard policy gradient methods, addressing its computational tractability.

Learning Backward Value Functions. The work in (99) proposes Backward Value Functions (BVF) as a way to overcome the excessive computational cost of the approaches in the previous paragraphs (34, 98). Similarly to a (forward) value function V^π that estimates the total future cost from each state, a BVF $V^{b,\pi}$ estimates the accumulated cost so far (up to the current state). We can decompose a trajectory-level constraint at any time step k as sum of $V_{c_j}^\pi(\mathbf{x}_k)$ and $V_{c_j}^{b,\pi}(\mathbf{x}_k)$ for the constraint cost $J_{c_j}^\pi$. This decomposition also alleviates the problem of off-policy evaluation as these value functions can be learned concurrently and efficiently via Temporal Difference (TD) methods (see M22). In practice, $V_{c_j}^\pi, V_{c_j}^{b,\pi}$,

M19. Distributional RL: Distributional RL (94) focuses on the modeling and learning of the *distribution* of the cost J , rather than its expected value. The greater descriptive power can enable richer predictions and risk-aware behaviours.

M20. Lagrangian Methods: To solve a constrained optimization problem $\min_x f(x)$ s.t. $g(x) \leq 0$, *Lagrangian methods* define a Lagrangian function $\mathcal{L}(x, \lambda) = f(x) + \lambda g(x)$ on primal variable x , dual variable λ , and solve the equivalent unconstrained problem $\max_{\lambda \geq 0} \min_x \mathcal{L}(x, \lambda)$. In practice, numerical approaches, such as gradient methods, are often used to perform the primal-dual updates (88).

3.2.3. Constrained MDPs and RL. CMDP 框架在安全强化学习中经常使用，因为它引入了可以用于以方程 6 形式表达任意安全概念的约束。然而，针对 CMDP 的强化学习面临两个重要挑战（参见第 2.3 节）：(i) 如何在强化学习算法中引入并执行约束；以及 (ii) 如何高效地解决受约束的强化学习问题——尤其是在使用深度学习模型时，这些模型是非安全强化学习中的 *de facto* 标准。本节中解决这些问题的方法包括 (i) 强化学习的拉格朗日方法、(ii) 用于约束的广义李雅普诺夫函数以及 (iii) 逆向值函数。尽管如此，本节的大部分工作仍局限于相当简单的模拟任务，这促使人们进一步研究受约束强化学习在现实控制中的适用性。

Lagrangian Methods in RL Optimization. 在 (88, 95) 中，CMDP 优化问题（见第 2.3 节）首先使用拉格朗日函数 $\mathcal{L}(\pi, \lambda)$ 将问题转化为对原始变量 π 和对偶变量 λ 的无约束优化问题（见 M20），并且在方程 15b 的原-对偶更新中使用 RL 作为子程序：

$$\mathcal{L}(\pi, \lambda) = J^\pi + \sum_j \lambda_j (J_{c_j}^\pi - d^j) \quad 15a.$$

$$(\pi^*, \lambda^*) = \arg \max_{\lambda \geq 0} \min_{\pi} \mathcal{L}(\pi, \lambda) \quad \text{s.t.} \quad \text{Equation 9b, 9c, 9d.} \quad 15b.$$

特别地，(88) 定义了成本 J^π 的 CVaR 限制，并使用策略梯度或 actor-critic 方法来更新方程 15b 中的策略。在后续工作 (95) 中，作者通过结合对偶变量的离策略更新（以及原策略的原始-对偶更新）改进了 (88)。实验证明，这样可以实现更高的样本效率和更快的收敛。在 (34) 中，作者将一个标准的信赖域（参见 M21）强化学习算法 (96) 扩展到带约束的马尔可夫决策过程（CMDPs），使用一个新颖的界限将两种策略的期望成本与它们的状态平均散度关联起来（参见 M21）。关键思想是对从界中导出的成本 J^π 和约束成本 $J_{c_j}^\pi$ 的替代物/近似进行原始-对偶更新。其好处有两方面：(i) 替代物可以仅通过状态-动作数据进行估计，从而绕过对离策略数据轨迹评估的挑战；以及 (ii) 更新保证了策略的单调改进和在 ϵ 处近似满足约束。

A Lyapunov Approach to Safe RL. 李雅普诺夫函数（参见 M6）在控制中被广泛用于分析系统稳定性，它们是将系统的全局特性转化为局部条件的强大工具。在 (97, 98) 中，李雅普诺夫函数被用于将方程 6 中的轨迹级约束 $J_{c_j}^\pi$ 转化为逐步、基于状态的约束。这允许更高效地计算 $J_{c_j}^\pi$ 并降低离策略评估的成本。然而，这些方法要求系统从一个已经满足约束的基线策略 π_0 开始。在 (97) 中，提出了四种不同的算法，通过结合传统的强化学习方法和李雅普诺夫约束来求解 CMDPs，但它们仅适用于离散输入空间（具有连续状态空间）。在后续工作 (98) 中，(97) 的作者将该方法扩展到连续输入空间和标准策略梯度方法，从而解决了其计算可行性的问题。

Learning Backward Value Functions. 在 (99) 的工作中，提出了后向价值函数（Backward Value Functions, BVF）作为一种克服前面几段 (34, 98) 方法计算成本过高的问题的方式。类似于估计每个状态未来总成本的（前向）价值函数 V^π ，BVF $V^{b,\pi}$ 估计到目前为止的累计成本（直到当前状态）。我们可以将任意时间步的轨迹级约束 k 分解为约束成本 $J_{c_j}^\pi$ 的 $V_{c_j}^\pi(\mathbf{x}_k)$ 和 $V_{c_j}^{b,\pi}(\mathbf{x}_k)$ 的和。这种分解也缓解了离策略评估的问题，因为这些价值函数可以通过时序差分（Temporal Difference, TD）方法（参见 M22）同时高效地学习。在实践中， $V_{c_j}^\pi$ ， $V_{c_j}^{b,\pi}$ ，

M19. Distributional RL: 分布式强化学习 (94) 关注于成本 J 的 *distribution* 的建模和学习，而不是其期望值。更强的描述能力可以实现更丰富的预测和风险感知的行为。

M20. Lagrangian Methods: 为了解决约束优化问题 $\min_x f(x)$ ，其约束条件为 $g(x) \leq 0$ ，Lagrangian 方法定义原变量 x 和对偶变量 λ 上的拉格朗日函数 $f(x) + \lambda g(x)$ ，并求解等价的无约束问题 $\max_{\lambda \geq 0} \min_x \mathcal{L}(x, \lambda)$ 。在实际中，经常使用数值方法，比如梯度方法，来执行原-对偶更新 (88)。

M21. Divergence:

Divergence quantifies the similarity between probability distributions. It may not satisfy the symmetry and the triangle inequality of a distance metric. In RL, commonly used ones include Kullback-Leibler divergence (KL), total variation distance (TV), and the Wasserstein distance, which is also a valid distance metric. Divergence measures can be used to constrain policy updates in trust-region RL algorithms (96).

M22. Temporal Difference Learning:

Temporal Difference (TD) learning (20) is an important class of model-free RL methods that updates the value function by bootstrapping from the current estimate using the Bellman equation to formulate sampling-based, iterative updates. For example, given a control step data tuple $(\mathbf{x}_k, l_k, \mathbf{x}_{k+1})$ we can perform a learning step as $V^\pi(\mathbf{x}_k) \leftarrow V^\pi(\mathbf{x}_k) + \alpha(l_k + \gamma V^\pi(\mathbf{x}_{k+1}) - V^\pi(\mathbf{x}_k))$ where α, γ are the step size and discount factor.

V^π are jointly learned (99) and used for policy improvement at each time step, allowing to implement Safety Level I. The approach is intended for discrete action spaces but can be adapted to continuous ones as in (70).

3.2.4. Robust MDPs and RL. Works in this section aim to implement robustness in RL, specifically, learning policies that can operate under disturbances and generalize across similar tasks or robotic systems. This is typically done by framing the learning problem as a robust MDP (Equation 13). In (100), Robust RL (RRL) implements an Actor-Disturber-Critic architecture. The authors observe that the learned policy and value function coincide with those derived analytically from H_∞ control theory for linear systems (see Sec. 2.2). However, more recent robust RL literature often abstains from assumptions on dynamics or disturbances, and applies model-free RL (101) to seek empirically robust performance at the expense of theoretical guarantees. In the following paragraphs, we introduce two lines of work: (i) robust adversarial RL, which explicitly models the min-max problem in Equation 13 in a game theoretic fashion; and (ii) domain randomization, which approximates the same problem in Equation 13 by learning on a set of randomized perturbed dynamics.

Robustness Through Adversarial Training. Combining RL with adversarial learning (102) results in robust adversarial RL (RARL) (36), where the robust optimization problem (Equation 13) is set up as a two-player, discounted zero-sum Markov game in which an agent (protagonist) learns policy π to control the system and another agent (adversary) learns a separate policy to destabilize the system. The two agents learn in an alternated fashion (each is updated while fixing the other), attempting to progressively improve both the robustness of the protagonist's policy and the strength of its adversary. The work of (37) extends (36) with risk-aware agents (see Sec. 3.2.2), with the protagonist being risk-averse and the adversary being risk-seeking. This method learns an ensemble of Deep Q-Networks (DQN) (103) and defines the risk of an action based on the variance of its value predictions. In another extension of (36), a population of adversaries (rather than a single one) is trained (38), leading to the resulting protagonist being less exploitable by new adversaries. Finally, the work in (104) proposes certified lower bounds for the value predictions from a DQN (103), given bounded observation perturbations. The action selection is based on these value lower bounds, assuming adversarial perturbation.

Robustness Through Domain Randomization. Domain randomization methods attempt to learn policies that empirically generalize to a wider range of tasks or robot systems. Instead of considering worst-case disturbances or scenarios, learning happens on systems with randomly perturbed parameters (e.g., inertial properties, friction coefficients). These parameters often have pre-specified ranges, effectively inducing a robust set that (i) approximates the uncertainty set $\mathbb{D}$ in Equation 13, and (ii) allows the system to reuse any standard RL algorithm. As an example, in (105), a quadrotor learns vision-based flight purely in simulation from scenes with randomized properties. Using this model-free policy in the real world resulted in improved collision avoidance performance. Instead of learning a policy directly, the work in (106) uses learned visual predictions with a MPC controller to enable efficient and scalable real-world performance. Besides the uniform randomization in (105, 106), adaptive randomization strategies such as Bayesian search (107) are also a promising direction. In (108), a discriminator is adversarially trained and used to guide the randomization process that generates systems that are less explored or exploited by the current policy.

3.3. Certifying Learning-Based Control Under Dynamics Uncertainty

In this section, we review methods providing certification to learning-based control approaches that do not inherently account for safety constraints, see Figure 2. We divide the

M21. Divergence:**Divergence**

量化概率分布之间的相似性。它可能不满足距离度量的对称性和三角不等式。在强化学习中，常用的有Kullback-Leibler散度（KL）、全变差距离（TV）和Wasserstein距离，后者也是一个有效的距离度量。散度测度可用于约束信任域强化学习算法中的策略更新（96）。

M22. Temporal Difference Learning:**Temporal Difference (TD) learning** (20)

是一类重要的无模型强化学习方法，它通过使用贝尔曼方程从当前估计进行自举来更新价值函数，从而形成基于采样的迭代更新。例如，给定一个控制步骤数据元组 $(\mathbf{x}_k, l_k, \mathbf{x}_{k+1})$ ，我们可以执行一个学习步骤为

$$V^\pi(\mathbf{x}_k) \leftarrow V^\pi(\mathbf{x}_k) + \alpha(l_k + \gamma V^\pi(\mathbf{x}_{k+1}) - V^\pi(\mathbf{x}_k))$$

（其中 α, γ 是步长和折扣因子）。

V^π 它们是联合学习的 (99) 并在每个时间步用于策略改进，从而实现安全等级 I。该方法适用于离散动作空间，但可以像在 (70) 中那样适配连续动作空间。

3.2.4. Robust MDPs and RL. 本节的工作旨在在强化学习（RL）中实现鲁棒性，特别是学习能够在扰动下运行并在类似任务或机器人系统中实现泛化的策略。这通常通过将学习问题框定为鲁棒马尔可夫决策过程（Robust MDP）（方程13）来实现。在 (100) 中，鲁棒强化学习（Robust RL, RRL）实现了演员-扰动器-评论家（Actor-Disturber-Critic）架构。作者观察到，学习到的策略和价值函数与从线性系统的 H_∞ 控制理论分析得出的结果一致（见第2.2节）。然而，更近期的鲁棒强化学习文献通常避免对动力学或扰动做假设，并采用无模型强化学习（model-free RL）（101）以寻求经验上的鲁棒性能，但以牺牲理论保证为代价。在以下段落中，我们介绍两类工作：*(i)* 鲁棒对抗强化学习（robust adversarial RL），通过博弈论的方式显式地建模方程13中的极小-极大问题；以及 *(ii)* 域随机化（domain randomization），通过学习来近似方程13中的相同问题。

Robustness Through Adversarial Training. 将强化学习（RL）与对抗学习（102）结合会产生稳健对抗强化学习（RARL）（36），其中稳健优化问题（方程13）被设定为一个双人折扣零和马尔可夫博弈，其中一个智能体（主角）学习策略 π 来控制系统，而另一个智能体（对手）学习一个独立的策略来破坏系统。这两个智能体交替学习（在固定另一个智能体的情况下更新各自策略），尝试逐步提高主角策略的稳健性和其对手的能力。(37) 的工作在 (36) 的基础上扩展了风险感知智能体（见第3.2.2节），主角厌恶风险，而对手寻求风险。该方法学习了一组深度Q网络（DQN）（103），并基于动作价值预测的方差来定义动作的风险。在 (36) 的另一扩展中，训练了一群对手（而不是单个对手）（38），从而使得到的主角表现为

Robustness Through Domain Randomization. 域随机化方法试图学习能够在经验上推广到更广泛任务或机器人系统的策略。学习并非考虑最坏情况的干扰或场景，而是在参数随机扰动的系统上进行（例如惯性属性、摩擦系数）。这些参数通常具有预先设定的范围，有效地引入了一个稳健集合，该集合近似于公式13中的不确定性集合，并且允许系统重用任何标准的强化学习算法。例如，在(105)中，一架四旋翼无人机在纯仿真的环境中从具有随机化属性的场景中学习基于视觉的飞行。在真实世界中使用这种无模型策略可以改善碰撞避免性能。与直接学习策略不同，(106)中的工作使用学习到的视觉预测结合MPC控制器来实现高效且可扩展的真实世界性能。除了(105, 106)中的均匀随机化，还存在自适应随机化策略，比如B

3.3. Certifying Learning-Based Control Under Dynamics Uncertainty

在本节中，我们回顾了为基于学习的控制方法提供认证的方法，这些方法本身并未考虑安全约束，参见 **Figure 2**。我们将其划分为

discussion into two parts: (i) stability certification, and (ii) constraint set certification. The works in this section leverage an *a priori* dynamics model of the system and provide hard or probabilistic safety guarantees (Safety Level II or III) under dynamics uncertainties (see [Figure 4](#)).

3.3.1. Stability Certification. This section introduces certification approaches that guarantee closed-loop stability under a learning-based control policy.

Lipschitz-Based Safety Certification for DNN-Based Learning Controllers. These approaches exploit the expressive power of DNNs for policy parametrization and guarantee closed-loop stability through a Lipschitz constraint on the DNN. Let ρ be a Lipschitz constant of a DNN policy (see [M12](#)), an upper bound on the Lipschitz constant ρ that guarantees closed-loop stability (Safety Level III) can be established by using a small-gain stability analysis ([109](#)), solving a semi-definite program (SDP) ([110](#)), or applying a sliding mode control framework ([111](#)). This bound on ρ can be used in either (i) a passive, iterative enforcement approach where the Lipschitz constant ρ is first estimated (e.g., via an SDP-based estimation ([112](#))) and then used to guide re-training until the Lipschitz constraint is satisfied, or (ii) an active enforcement approach where the Lipschitz constraint is directly enforced by the training algorithm of the DNN (e.g., via spectral normalization ([111](#))). While guaranteeing stability, the Lipschitz-based certification approaches often rely on the particular structure of the system dynamics (e.g., a control-affine structure or linear structure with additive nonlinear uncertainty) to find the certifying Lipschitz constant. It remains to be explored how this idea can be extended to more generic robot systems.

Learning Regions of Attraction for Safety Certification. The region of attraction (ROA) of a closed-loop system (see [M13](#)) is used in the learning-based control literature as a means to guarantee safety. For a nonlinear system with a given state-feedback controller, the ROA is the set of states that is guaranteed to converge to the equilibrium, which is treated as a safe state. This notion of safety provides a mean to certify a learning-based controller. It guarantees that there is a region in state space from which the controller can drive the system back to the safe state (Safety Level III) ([113](#)). ROAs can, for example, be used to guide data acquisition for model or controller learning ([68, 114](#)).

We consider deterministic closed-loop systems $\mathbf{x}_{k+1} = \mathbf{f}_\pi(\mathbf{x}_k) = \mathbf{f}(\mathbf{x}_k, \pi(\mathbf{x}_k))$ with $\mathbf{f}_\pi(\mathbf{x}_k)$ being Lipschitz continuous. A Lyapunov neural network (LNN) can be used to iteratively learn the ROA of a controlled nonlinear system from the system’s input-output data ([113](#)). As compared to the typical Lyapunov functions in control (e.g., quadratic Lyapunov functions), the proposed method uses the LNN as a more flexible Lyapunov function representation to provide a less conservative estimate of the system’s ROA. The necessary properties of a Lyapunov function are preserved via the network’s architectural design. In ([114](#)), an ROA estimation approach for high-dimensional systems is presented, combining a sum of squares (SOS) programming method for the ROA computation ([115](#)) and a dynamics model order reduction technique to curtail computational complexity ([116](#)).

3.3.2. Constraint Set Certification. This section summarizes approaches that provide constraint set certification to a learning-based controller based on the notion of *robust positive control invariant* (RPCI) safe sets $\Omega_{\text{safe}} \subseteq \mathbb{X}_c$ (see [M23](#)). Certified learning, which can be achieved through a safety filter ([7](#)) or shielding ([117](#)), finds the minimal modification of a learning-based control input $\mathbf{u}_{\text{learn}}$ (see [Figure 2](#)) such that the system’s state stays inside

M23. Robust Positive Control Invariant Safe Set:

A *robust positive control invariant* (RPCI) safe set Ω_{safe} is a set contained in $\mathbb{X}_c$ such that, if we start within the safe set, there exists a feedback policy $\pi(\mathbf{x})$ to keep the system in the safe set despite all possible model errors and process noise, captured by $\mathbb{D}$.

讨论分为两个部分：(i) 稳定性认证，以及 (ii) 约束集合认证。本节的工作利用系统的 *a priori* 动力学模型，并在动力学不确定性下提供严格或概率安全保障（安全级别 II 或 III）（见 Figure 4）。

3.3.1. Stability Certification. 本节介绍了在基于学习的控制策略下保证闭环稳定性的认证方法。

Lipschitz-Based Safety Certification for DNN-Based Learning Controllers. 这些方法利用深度神经网络（DNN）在策略参数化方面的表达能力，并通过对DNN施加Lipschitz约束来保证闭环稳定性。设 ρ 为DNN策略的Lipschitz常数（参见M12），可通过使用小增益稳定性分析（109）、求解半正定规划（SDP）（110）或应用滑模控制框架（111）建立保证闭环稳定性（安全级别III）的Lipschitz常数 ρ 的上界。该 ρ 的上界可以应用于(i)一种被动的迭代执行方法，其中Lipschitz常数 ρ 首先被估计（例如，通过基于SDP的估计（112）），然后用于指导重新训练，直到满足Lipschitz约束，或者(ii)一种主动执行方法，其中Lipschitz约束由DNN的训练算法直接执行（例如，通过谱归一化（111））。虽然能够保证稳定性，但基于Lipschitz的认证方法

Learning Regions of Attraction for Safety Certification. 闭环系统（见 M13）的吸引域（ROA）在基于学习的控制文献中被用作保证安全性的一种手段。对于一个具有给定状态反馈控制器的非线性系统，ROA 是指保证收敛到平衡点（被视为安全状态）的状态集合。这种安全性概念提供了一种认证基于学习控制器的方法。它保证在状态空间中存在一个区域，从该区域出发，控制器可以将系统驱回安全状态（安全等级 III）（113）。例如，ROA 可以用于指导模型或控制器学习的数据采集（68, 114）。

我们考虑确定性的闭环系统 $\mathbf{x}_{k+1} = \mathbf{f}_\pi(\mathbf{x}_k) = \mathbf{f}(\mathbf{x}_k, \pi(\mathbf{x}_k))$ ，其中 $\mathbf{f}_\pi(\mathbf{x}_k)$ 是利普希茨连续的。Lyapunov 神经网络（LNN）可以用于从系统的输入输出数据（113）迭代学习受控非线性系统的 ROA。与控制中典型的 Lyapunov 函数（例如二次 Lyapunov 函数）相比，所提出的方法使用 LNN 作为更灵活的 Lyapunov 函数表示，以提供系统 ROA 的较不保守估计。Lyapunov 函数的必要特性通过网络的架构设计得以保留。在（114）中，提出了一种用于高维系统的 ROA 估计方法，将平方和（SOS）规划方法用于 ROA 计算（115）与动力学模型阶次降阶技术结合，以降低计算复杂性（116）。

3.3.2. Constraint Set Certification. This section summarizes approaches that provide constraint set certification to a learning-based controller based on the notion of *robust positive control invariant* (RPCI) safe sets $\Omega_{\text{safe}} \subseteq \mathbb{X}_c$ (see M23). Certified learning, which can be achieved through a safety filter (7) or shielding (117), finds the minimal modification of a learning-based control input $\mathbf{u}_{\text{learn}}$ (see Figure 2) such that the system's state stays inside

M23. Robust Positive Control Invariant Safe Set: A robust positive control invariant (RPCI) safe set Ω_{safe} is a set contained in $\mathbb{X}_c$ such that, if we start within the safe set, there exists a feedback policy $\pi(\mathbf{x})$ to keep the system in the safe set despite all possible model errors and process noise, captured by $\mathbb{D}$.

M24. Control Lyapunov Functions: *Control Lyapunov Functions (CLFs)* extend the notion of Lyapunov stability to continuous-time, control-affine systems (Equation 17). A positive definite function L_c is a CLF, if there exists a state-feedback control input $\mathbf{u}$, such that the time derivative $\dot{L}_c$ satisfies: $\dot{L}_c(\mathbf{x}) \leq -L_c(\mathbf{x})$. The existence of a CLF guarantees that there also exists a state-feedback controller $\boldsymbol{\pi}(\mathbf{x})$ that asymptotically stabilizes the system.

the set Ω_{safe} :

$$\mathbf{u}_{\text{safe},k} = \arg \min_{\mathbf{u}_k \in \mathcal{U}_c} \|\mathbf{u}_k - \mathbf{u}_{\text{learn},k}\|_2^2 \quad 16a.$$

$$\begin{aligned} \text{s.t. } \mathbf{x}_{k+1} &= \bar{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k) + \hat{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \in \Omega_{\text{safe}}, \\ \forall \hat{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) &\in \mathbb{D}(\mathbf{x}_k, \mathbf{u}_k), \end{aligned} \quad 16b.$$

where the range of possible disturbances $\mathbb{D}(\mathbf{x}_k, \mathbf{u}_k)$ is given. Since the safety filter and controller are usually decoupled, suboptimal behavior can emerge as the learning-based controller may try to violate the constraints (65).

Control Barrier Functions (CBFs). CBFs are used to define safe sets. More specifically, the safe set Ω_{safe} is defined as the superlevel set of a continuously differentiable CBF B_c , $B_c : \mathbb{R}^{n_x} \rightarrow \mathbb{R}$, as $\Omega_{\text{safe}} = \{\mathbf{x} \in \mathbb{R}^{n_x} : B_c(\mathbf{x}) \geq 0\}$. The function B_c is generally considered for continuous-time, control-affine systems (see also M4) of the form:

$$\dot{\mathbf{x}} = \mathbf{f}_x(\mathbf{x}) + \mathbf{f}_u(\mathbf{x})\mathbf{u}, \quad 17.$$

where $\mathbf{f}_x$ and $\mathbf{f}_u$ are locally Lipschitz and $\mathbf{x}, \mathbf{u}$ are functions of time (118). In the simplest form, the function B_c is a CBF if there exists a state-feedback control input $\mathbf{u}$, such that the time derivative $\dot{B}_c(\mathbf{x}) = \frac{\partial B_c}{\partial \mathbf{x}} \dot{\mathbf{x}}$ satisfies (118):

$$\sup_{\mathbf{u} \in \mathcal{U}} \frac{\partial B_c}{\partial \mathbf{x}} (\mathbf{f}_x(\mathbf{x}) + \mathbf{f}_u(\mathbf{x})\mathbf{u}) \geq -B_c(\mathbf{x}). \quad 18.$$

The CBF condition in Equation 18 is a continuous-time version of the robust positive control invariance constraint in Equation 16b. In addition to the RPCI constraint, a similar constraint for asymptotic stability can be added to Equation 16 in the form of a constraint on the time derivative of a control Lyapunov function (CLF, see M24) L_c . However, uncertain dynamics also yield uncertain time derivatives of B_c and L_c .

Learning-based approaches extend CBF and CLF analyses to control-affine systems in Equation 17 with known nominal $\bar{\mathbf{f}}_x, \bar{\mathbf{f}}_u$ and unknown $\hat{\mathbf{f}}_x, \hat{\mathbf{f}}_u$. The time derivative of the unknown dynamics for CLFs and/or CBFs (e.g., $\frac{\partial B_c}{\partial \mathbf{x}} (\hat{\mathbf{f}}_x(\mathbf{x}) + \hat{\mathbf{f}}_u(\mathbf{x})\mathbf{u})$) can be learned from iterative trials (119, 120) or data collected by an RL agent (121, 122). Given a CBF and/or CLF for the true dynamics $\mathbf{f}$, improving the estimate of the CBF's or CLF's time derivative for the unknown dynamics $\hat{\mathbf{f}}$ using data, collected either offline or online, yields a more precise estimate of the constraint in Equation 18. However, any learning error in the CBF's or CLF's time derivative of the unknown dynamics can still lead to applying falsely certified control inputs. To this end, ideas from robust control can be used to guarantee set invariance (Safety Level III) during the learning process by mapping a bounded uncertainty in the dynamics to a bounded uncertainty in the time derivatives of the CBF or CLF (123, 124), or by accounting for all model errors consistent with the collected data (125). In addition, adaptive control approaches have also been proposed to allow safe adaptation of parametric uncertainties in the time derivatives of the CBF or CLF (126, 127). Probabilistic learning techniques for CBFs and CLFs have been used to achieve set invariance probabilistically (Safety Level II) with varying assumptions on the system dynamics: the function $\mathbf{f}_u(\mathbf{x})$ is fully known (128, 129), a nominal model is known (130), and no nominal model is available (131).

Hamilton-Jacobi Reachability Analysis. Another approach for state constraint set certification of a learning-based controller is via the Hamilton-Jacobi (HJ) reachability analysis. The HJ reachability analysis provides a means to estimate a robust positive control invariant safe set Ω_{safe} under dynamics uncertainties (see M23 and Equation 16). Consider a nonlinear system subject to unknown but bounded disturbances $\hat{\mathbf{f}}(\mathbf{x}) \in \mathbb{D}(\mathbf{x})$, where $\mathbb{D}(\mathbf{x})$ is

M24. Control Lyapunov Functions: *Control Lyapunov Functions (CLFs)* 将李雅普诺夫稳定性的概念扩展到连续时间、控制仿射系统（方程17）。如果存在一个状态反馈控制输入 $\mathbf{u}$ ，使得其时间导数 $\dot{L}_c$ 满足： $\dot{L}_c(\mathbf{x}) \leq -L_c(\mathbf{x})$ ，则一个正定函数 L_c 是一个 CLF。CLF 的存在保证也存在一个状态反馈控制器 $\pi(\mathbf{x})$ ，能够渐近稳定该系统。

the set Ω_{safe} :

$$\mathbf{u}_{\text{safe},k} = \arg \min_{\mathbf{u}_k \in \mathcal{U}_c} \|\mathbf{u}_k - \mathbf{u}_{\text{learn},k}\|_2^2 \quad 16a.$$

$$\begin{aligned} \text{s.t. } \mathbf{x}_{k+1} &= \bar{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k) + \hat{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) \in \Omega_{\text{safe}}, \\ \forall \hat{\mathbf{f}}_k(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}_k) &\in \mathbb{D}(\mathbf{x}_k, \mathbf{u}_k), \end{aligned} \quad 16b.$$

给出了可能干扰 $\mathbb{D}(\mathbf{x}_k, \mathbf{u}_k)$ 的范围。由于安全过滤器和控制器通常是解耦的，学习型控制器可能会尝试违反约束（65），因此可能会出现次优行为。

Control Barrier Functions (CBFs). CBF用于定义安全集。更具体地说，安全集 Ω_{safe} 被定义为连续可微CBF B_c 的上超集合， $B_c: \mathbb{R}^{n_x} \rightarrow \mathbb{R}$ ，为 $\Omega_{\text{safe}} = \{\mathbf{x} \in \mathbb{R}^{n_x} : B_c(\mathbf{x}) \geq 0\}$ 。函数 B_c 通常用于连续时间、控制仿射系统（参见M4），其形式为：

$$\dot{\mathbf{x}} = \mathbf{f}_x(\mathbf{x}) + \mathbf{f}_u(\mathbf{x})\mathbf{u}, \quad 17.$$

其中 $\mathbf{f}_x$ 和 $\mathbf{f}_u$ 是局部利普希茨连续的，而 $\mathbf{x}, \mathbf{u}$ 是时间的函数 (118)。在最简单的形式中，如果存在一个状态反馈控制输入 $\mathbf{u}$ ，使得时间导数 $\dot{B}_c(\mathbf{x}) = \frac{\partial B_c}{\partial \mathbf{x}} \dot{\mathbf{x}}$ 满足 (118)，则函数 B_c 是一个 CBF：

$$\sup_{\mathbf{u} \in \mathcal{U}} \frac{\partial B_c}{\partial \mathbf{x}} (\mathbf{f}_x(\mathbf{x}) + \mathbf{f}_u(\mathbf{x})\mathbf{u}) \geq -B_c(\mathbf{x}). \quad 18.$$

方程18中的CBF条件是方程16b中鲁棒正控制不变性约束的连续时间版本。除了RPCI约束之外，一个类似的

渐近稳定性的约束可以以控制李雅普诺夫函数（CLF，见 M24） L_c 的时间导数的约束形式添加到方程16中。然而，不确定的动力学也会导致 B_c 和 L_c 的时间导数不确定。

基于学习的方法将 CBF 和 CLF 分析扩展到方程 17 中的控制仿射系统，系统具有已知的 $\bar{\mathbf{f}}_x$ 、 $\bar{\mathbf{f}}_u$ 和未知的 $\hat{\mathbf{f}}_x$ 、 $\hat{\mathbf{f}}_u$ 。可以通过迭代试验 (119, 120) 或由强化学习代理收集的数据 (121, 122) 学习 CLF 和/或 CBF 的未知动力学的时间导数（例如， $\frac{\partial B_c}{\partial \mathbf{x}} (\hat{\mathbf{f}}_x(\mathbf{x}) + \hat{\mathbf{f}}_u(\mathbf{x})\mathbf{u})$ ）。给定真实动力学 $\mathbf{f}$ 的 CBF 和/或 CLF，通过使用离线或在线收集的数据改进未知动力学 $\hat{\mathbf{f}}$ 的 CBF 或 CLF 时间导数的估计，可以得到方程 18 中约束的更精确估计。然而，CBF 或 CLF 对未知动力学的时间导数的任何学习误差仍可能导致应用错误认证的控制输入。为此，可以利用鲁棒控制的思想，通过将动力学中的有界不确定性映射到 CBF 或 CLF 时间导数的有界不确定性 (123, 124)，或通过考虑 ...，在学习过程中保证集合不变性（安全级别 III）。

Hamilton-Jacobi Reachability Analysis. 另一种基于学习的控制器状态约束集合认证的方法是通过Hamilton-Jacobi (HJ) 可达性分析。HJ可达性分析提供了一种在动力学不确定性下估计鲁棒正控制不变安全集合 Ω_{safe} 的方法（参见M23和方程16）。考虑一个受未知但有界扰动 $\hat{\mathbf{f}}(\mathbf{x}) \in \mathbb{D}(\mathbf{x})$ 影响的非线性系统，其中 $\mathbb{D}(\mathbf{x})$ 是

assumed to be known but possibly conservative. To compute Ω_{safe} , a two-player, zero-sum differential game is formulated:

$$V(\mathbf{x}) = \max_{\mathbf{u}_{\text{sig}} \in \mathbb{U}_{\text{sig}}} \min_{\hat{\mathbf{f}}_{\text{sig}} \in \mathbb{D}_{\text{sig}}} \left(\inf_{k \geq 0} l_c \left(\phi(\mathbf{x}, k; \mathbf{u}_{\text{sig}}, \hat{\mathbf{f}}_{\text{sig}}) \right) \right), \quad 19.$$

where V is the value function associated with a point $\mathbf{x} \in \mathbb{X}$, $l_c : \mathbb{X} \mapsto \mathbb{R}$ is a cost function that is non-negative for $\mathbf{x} \in \mathbb{X}_c$ and negative otherwise, $\phi(\mathbf{x}, k; \mathbf{u}_{\text{sig}}, \hat{\mathbf{f}}_{\text{sig}})$ denotes the state at k along a trajectory initialized at $\mathbf{x}$ following input signal $\mathbf{u}_{\text{sig}}$ and disturbance signal $\hat{\mathbf{f}}_{\text{sig}}$, and $\mathbb{U}_{\text{sig}}$ and $\mathbb{D}_{\text{sig}}$ are collections of input and disturbance signals such that each time instance is in $\mathbb{U}$ and $\mathbb{D}$, respectively. The value function V can be found as the unique viscosity solution of the Hamilton-Jacobi Isaacs (HJI) variational inequality (132). The safe set is then $\Omega_{\text{safe}} = \{\mathbf{x} \in \mathbb{X} \mid V(\mathbf{x}) \geq 0\}$. Based on this formulation, we can also obtain an optimally safe policy π_{safe}^* that maximally steers the system towards the safe set Ω_{safe} (i.e., in the greatest ascent direction of V). The HJ reachability analysis allows us to define a safety filter for learning-based control approaches to guarantee constraint set satisfaction (Safety Level II or III). In particular, given Ω_{safe} and π_{safe}^* , one can safely learn in the interior of Ω_{safe} and apply the optimally safe policy π_{safe}^* if the system reaches the boundary of Ω_{safe} . To reduce the conservativeness of the approach, a GP-based learning scheme is proposed in (133) to adapt (and shrink) the unknown dynamics set $\mathbb{D}(\mathbf{x})$ based on observed data.

The general HJ reachability analysis framework (132) has also been combined with on-line dynamics model learning for a target tracking task (134), with online planning for safe exploration (135), and with temporal difference algorithms for safe RL (136). In (137), HJ reachability analysis and CBFs have been integrated to compute smoother control policies while circumventing the need to hand-design appropriate CBFs. Another recent extension (138) proposed modifications that improve the scalability of the HJ safety analysis approach for higher-dimensional systems and was demonstrated on a ten-dimensional quadrotor tracking problem.

Predictive Safety Filters. Predictive safety filters can augment any learning-based controller to enforce state constraints, $\mathbf{x} \in \mathbb{X}_c$, and input constraints, $\mathbf{u} \in \mathbb{U}_c$. They do this by defining the safe invariant set Ω_{safe} in Equation 16b as the set of states (at the next time step) where a sequence of safe control inputs (e.g., from a backup controller) exists that allows the return to (i) a terminal safe set $\mathbb{X}_{\text{term}}$ (see M25), or (ii) to previously visited safe states.

Model Predictive Safety Certification (MPSC) uses the theory of robust MPC in Sec. 2.2 and learning-based robust MPC in Sec. 3.1.3 to filter the output of any learning-based controller, such as of an RL method, to ensure robust constraint satisfaction. The simplest implementation of MPSC (139) uses tube-based MPC and considers the constraints in Equations 12b–12d but replaces the cost in Equation 12a with the cost in Equation 16a to find the closest input $\mathbf{u}_k$ to the learned input $\mathbf{u}_{\text{learn},k}$ at the current time step that guarantees that we will continue to satisfy state and input constraints in the future. The main difference between MPSC and learning-based robust MPC described in Sec. 3.1.3 is that the terminal safe set $\mathbb{X}_{\text{term}}$ in Equation 12d is not coupled with the selection of the cost function in Equation 12a. Instead, the terminal safe set is conservatively initialized with $\mathbb{X}_{\text{term}} = \Omega_{\text{tube}}$ and can grow to include state trajectories from previous iterations. This approach has been extended to probabilistic constraints by considering a probabilistic tube Ω_{tube} in (140), and to nonlinear nominal models in (141) (Safety Level II).

Backup control for safe exploration ensures hard state constraint satisfaction (Safety Level III) by finding a safe backup controller for any given RL policy π (142). Under the assumptions of a known bound $\mathbb{D}(\mathbf{x}, \mathbf{u})$ on the dynamics $\mathbf{f}$ and a distance measure to the state constraints $\mathbb{X}_c$, the backup controller is used to obtain a future state in the

M25. Terminal Safe Set: A *terminal safe set*, denoted by $\mathbb{X}_{\text{term}}$, is a subset of the safe state constraint space wherein a known auxiliary controller is guaranteed to preserve the system's state. Entering the terminal safe set at some fixed horizon H is enforced as a constraint.

假定已知但可能是保守的。为了计算 Ω_{safe} ，构建了一个双人零和微分博弈：

$$V(\mathbf{x}) = \max_{\mathbf{u}_{\text{sig}} \in \mathbb{U}_{\text{sig}}} \min_{\hat{\mathbf{f}}_{\text{sig}} \in \mathbb{D}_{\text{sig}}} \left(\inf_{k \geq 0} l_c \left(\phi(\mathbf{x}, k; \mathbf{u}_{\text{sig}}, \hat{\mathbf{f}}_{\text{sig}}) \right) \right), \quad 19.$$

其中 V 是与点 $\mathbf{x} \in \mathbb{X}$ 相关联的值函数， $l_c : \mathbb{X} \mapsto \mathbb{R}$ 是一个成本函数，对于 $\mathbf{x} \in \mathbb{X}_c$ 非负，否则为负， $\phi(\mathbf{x}, k; \mathbf{u}_{\text{sig}}, \hat{\mathbf{f}}_{\text{sig}})$ 表示沿着从 $\mathbf{x}$ 初始化并遵循输入信号 $\mathbf{u}_{\text{sig}}$ 和干扰信号 $\hat{\mathbf{f}}_{\text{sig}}$ 的轨迹在 k 时的状态， $\mathbb{U}_{\text{sig}}$ 和 $\mathbb{D}_{\text{sig}}$ 是输入和干扰信号的集合，使得每个时间点分别在 $\mathbb{U}$ 和 $\mathbb{D}$ 。值函数 V 可以作为哈密顿-雅可比-伊萨克斯 (HJI) 变分不等式 (132) 的唯一粘性解求得。安全集合随后为 $\Omega_{\text{safe}} = \{\mathbf{x} \in \mathbb{X} \mid V(\mathbf{x}) \geq 0\}$ 。基于该公式，我们还可以获得最优安全策略 π_{safe}^* ，该策略最大限度地系统将引向安全集 Ω_{safe} （即沿 V ）的最大上升方向。HJ 可达性分析使我们能够为基于学习的控制方法定义安全滤波器，以保证约束集的满足（安全等级 II 或 III）。特别是，给定 Ω_{safe} 和 π_{safe}^* ，可以安全地

一般的 HJ 可达性分析框架 (132) 也已与目标跟踪任务的在线动态模型学习相结合 (134)，与安全探索的在线规划相结合 (135)，以及与安全强化学习的时序差分算法相结合 (136)。在 (137) 中，HJ 可达性分析和 CBF 被整合以计算更平滑的控制策略，同时避免手动设计适当的 CBF。另一项最近的扩展 (138) 提出了改进 HJ 安全分析方法在高维系统中可扩展性的修改，并在一个十维四旋翼跟踪问题中进行了演示。

Predictive Safety Filters. 预测安全滤波器可以增强任何基于学习的控制器，以执行状态约束 $\mathbf{x} \in \mathbb{X}_c$ 和输入约束 $\mathbf{u} \in \mathbb{U}_c$ 。它们通过在方程 16b 中将安全不变集合 Ω_{safe} 定义为下一个时间步的状态集合来实现这一点，其中存在一系列安全控制输入（例如来自备用控制器），允许返回 (i) 终端安全集合 $\mathbb{X}_{\text{term}}$ （见 M25），或 (ii) 返回先前访问过的安全状态。

Model Predictive Safety Certification (MPSC) 在第2.2节中使用稳健MPC理论，在第3.1.3节中使用基于学习的稳健MPC来过滤任何基于学习的控制器的输出，例如强化学习方法，以确保稳健的约束满足。MPSC (139) 的最简单实现方式是使用基于管道的MPC，并考虑方程12b–12d中的约束，但将方程12a中的代价替换为方程16a中的代价，以找到当前时间步最接近学习输入 $\mathbf{u}_{\text{learn},k}$ 的输入 $\mathbf{u}_k$ ，保证我们在未来继续满足状态和输入约束。MPSC 与第3.1.3节中描述的基于学习的稳健MPC的主要区别在于，方程12d中的终端安全集 $\mathbb{X}_{\text{term}}$ 不与方程12a中代价函数的选择耦合。相反，终端安全集被保守地初始化为 $\mathbb{X}_{\text{term}} = \Omega_{\text{tube}}$ ，并且可以增长以包含前几次迭代的状态轨迹。这种方法已经通过考虑概率约束得到了扩展。

M25. Terminal Safe Set: 一个 *terminal*，表示为 $\mathbb{X}_{\text{term}}$ ，是安全状态约束空间的一个子集，其中已知的辅助控制器能够保证系统状态的保持。在某个固定的时域 H 内进入终端安全集被作为约束强制执行。

Backup control for safe exploration 通过为任何给定的强化学习策略找到一个安全的备用控制器，确保硬状态约束的满足（安全等级 III） π (142)。在已知动力学边界 $\mathbb{D}(\mathbf{x}, \mathbf{u})$ 和状态约束的距离测量 $\mathbb{X}_c$ 的假设下，备用控制器用于获得未来状态

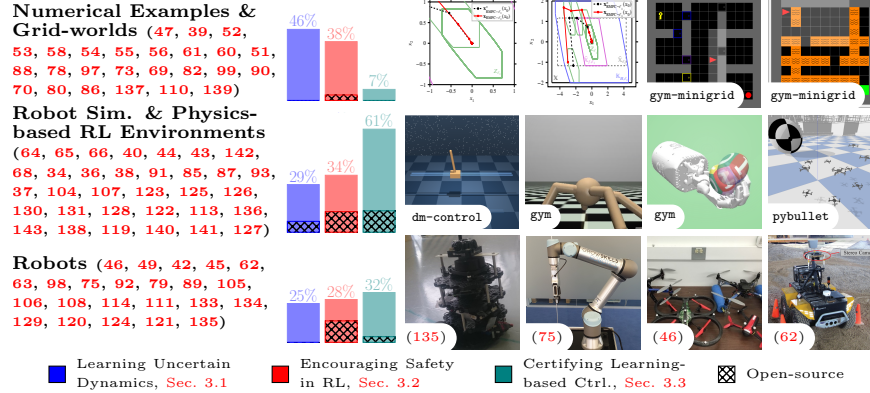


Figure 5: Summary of the environments used for evaluation. With increasing complexity, they can be classified as: abstract numerical examples, robot simulations, and real-world robot experiments. The histograms show the prevalence of each in Sec. 3.1, Sec. 3.2, and Sec. 3.3, as well as the fraction of those whose code is open-sourced.

neighborhood of a previously visited safe state in some prediction horizon. Before a control input $\mathbf{u}_k$ from π is applied to the system, all possible predicted states $\mathbf{x}_{k+1}$ must satisfy (i) $\mathbf{x}_{k+1} \in \mathbb{X}_c$ and (ii) the existence of a safe backup action $\mathbf{u}_{\text{certified}}(\mathbf{x}_{k+1})$. Otherwise, the previous backup control input $\mathbf{u}_{\text{certified}}(\mathbf{x}_k)$ is applied. This procedure guarantees that the system state stays inside a robust positive control invariant set $\Omega_{\text{safe}} \subseteq \mathbb{X}_c$.

4. BENCHMARKS

The approaches presented in Sec. 3 have been evaluated in vastly different ways, see Figure 5. The trends we observe are: The works learning uncertain dynamics (Sec. 3.1) include a preponderance of abstract numerical examples; encouraging safety in RL (Sec. 3.2) mostly leverages simulations—often based on physics engines (144) like MuJoCo—but grid worlds are also common; and works certifying learning-based control (Sec. 3.3) are still mostly simulated but also has the largest fraction of real-world experiments. While numerical examples make it difficult to gauge the practical applicability of a method, we also note that even many RL environments, including physics-based ones, are not representative of existing robotic platforms—for example, they often are deterministic and do not account for variations in the environment.

In an ideal world, all research would be demonstrated in simulations that closely resemble the target system—and brought to (ideally different) real robots, whenever possible. Furthermore, only a minority of the published research open-sourced their software implementations. Even in RL, where sharing code is more common (see, e.g., red bars in Figure 5)—and standard environments and interfaces such as gym (145) have been proposed—the reproducibility of results (that often rely on careful hyper-parameter tuning) remains challenging (71). With regard to safety, simple RL environments augmented with constraint evaluation (18) and disturbances (5) have been proposed, but lack a unified simulation interface for both safe RL and learning-based control approaches—notably, one that also exposes the available *a priori* knowledge of a dynamical system.

We believe that a necessary stepping stone for the advancement of safe learning control is to create physics-based environments that are (i) simple enough to promote adoption, (ii) realistic enough to represent meaningful robotic platforms, (iii) equipped with intuitive interfaces for both control and RL researchers, and (iv) provide useful comparison metrics (e.g., the amount of data required by different approaches).

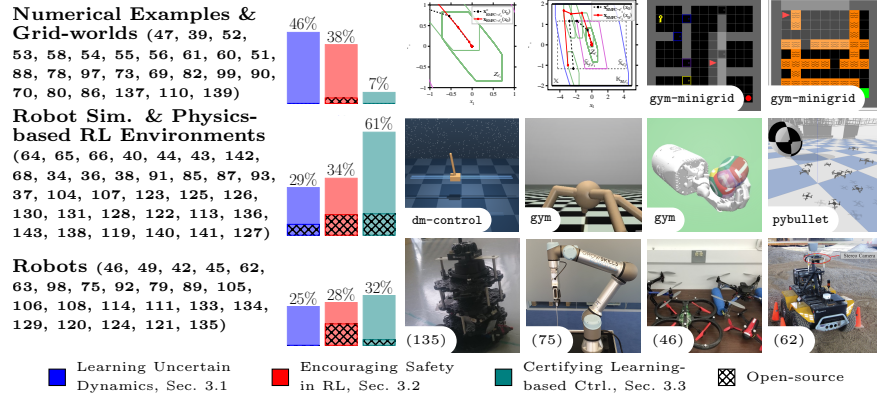


图5: 用于评估的环境总结。随着复杂性的增加, 它们可以被分类为: 抽象数值例子、机器人模拟和真实世界的机器人实验。直方图显示了它们在第3.1节、第3.2节和第3.3节中的普遍程度, 以及其中代码开源的比例。

在某个预测范围内, 先前访问过的安全状态的邻域。在将来自 π 的控制输入 $\mathbf{u}_k$ 应用于系统之前, 所有可能的预测状态 $\mathbf{x}_{k+1}$ 必须满足 (i) $\mathbf{x}_{k+1} \in \mathbb{X}_c$ 和 (ii) 的安全备份操作 $\mathbf{u}_{\text{certified}}(\mathbf{x}_{k+1})$ 的存在性。否则, 将应用先前的备份控制输入 $\mathbf{u}_{\text{certified}}(\mathbf{x}_k)$ 。该过程保证系统状态保持在稳健的正控制不变集 $\Omega_{\text{safe}} \subseteq \mathbb{X}_c$ 内。

4. BENCHMARKS

第3节中提出的方法已经通过非常不同的方式进行了评估, 见 **Figure 5**。我们观察到的趋势是: 学习不确定动力学的工作 (第3.1节) 大量采用抽象的数值示例; 在强化学习中鼓励安全性 (第3.2节) 主要依赖模拟——通常基于像 MuJoCo (144) 这样的物理引擎——但网格世界也很常见; 而认证基于学习的控制的工作 (第3.3节) 仍然主要是模拟, 但也有最大比例的真实世界实验。虽然数值示例很难评估一种方法的实际适用性, 我们还注意到, 即使是许多强化学习环境, 包括基于物理的环境, 也不能代表现有的机器人平台——例如, 它们通常是确定性的, 并且没有考虑环境的变化。

在一个理想的世界里, 所有的研究都会在与目标系统高度相似的模拟中进行展示——并且在可能的情况下, 应用到 (理想情况下不同的) 真实机器人上。此外, 只有少数公开发表的研究开源了他们的软件实现。即使在强化学习 (RL) 领域, 共享代码比较普遍 (例如, 参见 **Figure 5** 中的红色条)——并且已经提出了标准环境和接口如 gym (145)——结果的可重复性 (通常依赖于精细的超参数调优) 仍然具有挑战性 (71)。关于安全性, 已经提出了在简单的 RL 环境中增加约束评估 (18) 和干扰 (5) 的做法, 但缺乏针对安全强化学习和基于学习的控制方法的统一仿真接口——特别是一个能够同时暴露动力系统可用 *a priori* 知识的接口。

我们认为, 推动安全学习控制发展的必要前提是创建基于物理的环境, 这些环境 (i) 足够简单以促进采用, (ii) 足够真实以代表有意义的机器人平台, (iii) 配备直观的接口以供控制和强化学习研究人员使用, 并且 (iv) 提供有用的比较指标 (例如, 不同方法所需的数据量)。

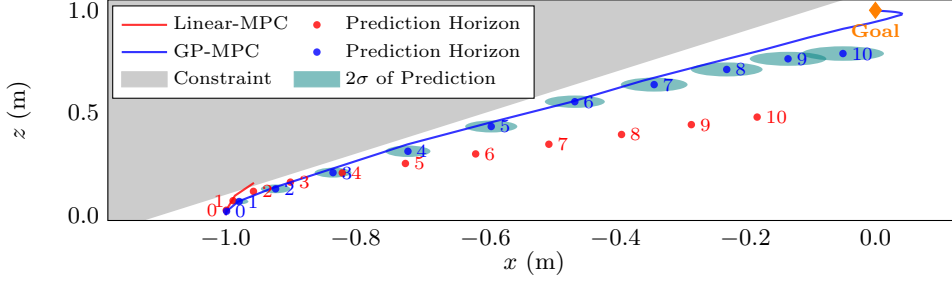


Figure 6: A position comparison of the 2D quadrotor stabilization using linear MPC (red) and GP-MPC (blue), along with the prediction horizons at the second time step, subject to a diagonal state constraint (gray) and input constraints.

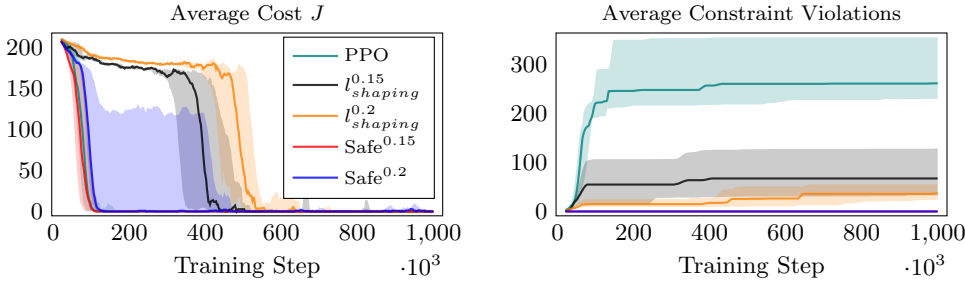


Figure 7: Total cost and constraint violations during learning for PPO, PPO with cost shaping, and PPO with safety layer (safe exploration). Plotted are medians with upper and lower quantiles over 10 seeds. The parameters in superscript represent values of the slack variable, which controls responsiveness to near constraint violations.

Cart-Pole and Quadrotor Benchmark Environments. For this reason, we have created an open-source benchmark suite² simulating two popular platforms for the evaluation of control and RL research: (i) the cart-pole (64, 104, 136) and (ii) a quadrotor (142, 66, 44, 130, 146). Our simulation environments are based on the open-source Bullet physics engine (147) and we adopt OpenAI’s gym interface (API) (145) for seamless integration with many of the current RL libraries. What sets apart our implementation from previous attempts to create safety-aware RL environments (18, 5) is the extension of the traditional API (145) with features to facilitate: (i) the evaluation of safe learning control approaches (such as the randomization of inertial properties and of initial positions and velocities, random and adversarial disturbances, and the specification and evaluation of state and input constraints) and (ii) the integration with approaches developed by the control theory community that leverage the knowledge of nominal models—notably, we use the symbolic framework CasADi (148) to enrich our gym environments with symbolic *a priori* dynamics, constraints, and cost functions.

Safe Learning Control Results. We focus on a constrained stabilization task, in each of our environments (`cartpole` and `quadrotor`). The results we present here are not meant to establish the superiority of one approach over another. Instead, we show how the approaches in (63, 70, 139), taken from each of Sec. 3.1 (learning uncertain dynamics), Sec. 3.2 (encouraging safety in RL), and Sec. 3.3 (certifying learning-based control), can improve

²Safe control benchmark suite on GitHub: <https://github.com/utiasDSL/safe-control-gym>

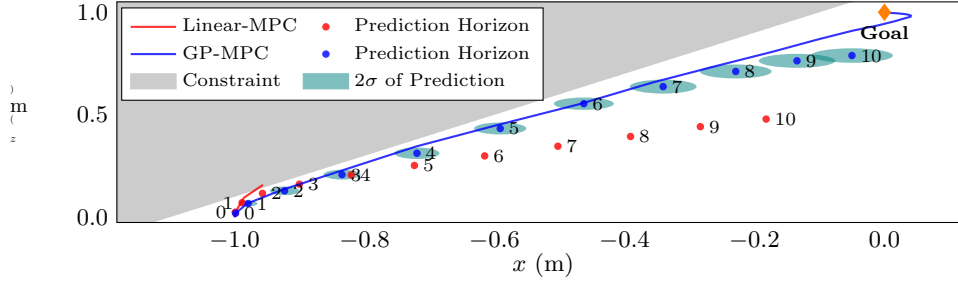


图6: 二维四旋翼使用线性MPC（红色）和GP-MPC（蓝色）进行稳定的位姿比较，以及在第二个时间步的预测视野，同时受到对角状态约束（灰色）和输入约束。

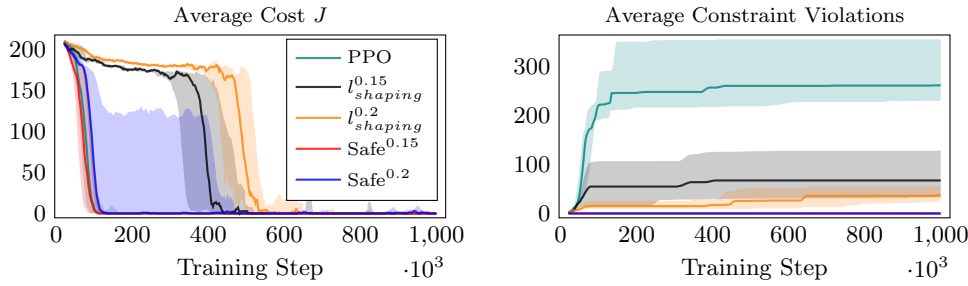


图7: 在学习过程中PPO、带成本塑形的PPO以及带安全层（安全探索）的PPO的总成本和约束违规情况。绘制了10个种子的中位数及上下四分位数。上标中的参数表示松弛变量的取值，该变量控制对接近约束违规的响应程度。

Cart-Pole and Quadrotor Benchmark Environments. 因此，我们创建了一个开源基准套件²，模拟两个用于控制和强化学习研究评估的流行平台：(i) 立式摆（64, 104, 136）和 (ii) 四旋翼（142, 66, 44, 130, 146）。我们的仿真环境基于开源的 Bullet 物理引擎（147），并采用 OpenAI 的 gym 接口（API）（145），以便与当前许多强化学习库无缝集成。将我们的实现与之前尝试创建安全感知强化学习环境（18, 5）区分开来的，是在传统 API（145）的基础上扩展了功能，以便于：(i) 评估安全学习控制方法（例如惯性属性以及初始位置和速度的随机化、随机和对抗扰动，以及状态和输入约束的指定和评估）以及 (ii) 与控制理论社区开发的方法集成，这些方法利用对称模型的知识——尤其是，我们

Safe Learning Control Results. 我们关注于一个受约束的稳定化任务，在我们每个环境中（cartpole 和 quadrotor）。我们在此呈现的结果并不是为了证明某种方法优于另一种方法。相反，我们展示了从第 3.1 节（学习不确定动力学）、第 3.2 节（鼓励强化学习中的安全性）和第 3.3 节（基于学习的控制认证）中选取的（63、70、139）中的方法如何可以改进

²安全控制本

chmark suite on GitHub: <https://github.com/utiasDSL/safe-control-gym>

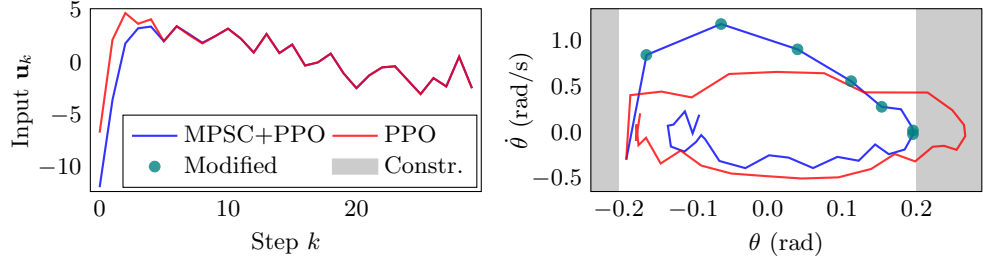


Figure 8: Left: The uncertified PPO input (red) is plotted against the certified MPSC+PPO input (blue). Right: The cart-pole state diagram (θ and $\dot{\theta}$) comparing the MPSC+PPO certified trajectory (blue) and the uncertified PPO trajectory (red). The MPSC is most active (green dots) when the system is about to leave the constraint boundary (gray) or the set of states from which the MPSC can correct the system.

control performance while pursuing constraint satisfaction. By doing so, we demonstrate that our benchmark can be easily integrated with learnable policies developed by either the control or RL research community. This also allows us to fairly compare the data hungriness of the different safe learning control approaches.

A learning-based robust MPC with a GP estimate of $\hat{\mathbf{f}}$ (GP-MPC), representative of [Sec. 3.1](#) (learning uncertain dynamics), was implemented to stabilize a quadrotor subject to a state and input constraints, following the approach in [\(63\)](#). For this experiment, a linearization about hover with a mass and moment of inertia 150% of the true values was used as the prior model. Eight-hundred randomly selected state-action pairs (or 80 seconds), sampled from a couple of minutes of training data, were used for hyper-parameter optimization, which was performed offline. [Figure 6](#) compares the performance of a linear MPC, using the inaccurate prior model, with the GP-MPC approach. We see that the Linear MPC, using the heavier prior model, predicts the trajectory of the quadrotor will be relatively shallow when maximum thrust is applied. This results in the quadrotor quickly violating the position constraint. In contrast, the GP-MPC is able to account for the inaccurate prior model and respects the constraint boundary by a margin proportional to the 95% confidence interval on its dynamics prediction, stabilizing the quadrotor.

We also applied the Safe Exploration approach [\(70\)](#) to the popular deep RL algorithm PPO [\(149\)](#), as a representative of [Sec. 3.2](#) (encouraging safety in RL), and tested it on the cart-pole stabilization task with constraints on the cart position. Notably, the task terminates upon any constraint violation. We compared it against two baselines, standard PPO and PPO with naive cost shaping (adding a penalty when close to constraint violation). Each RL approach used over 9 hours of simulation time collecting data for training. [Figure 7](#) shows that the two constraint-aware approaches (cost shaping and safe exploration [\(70\)](#)) achieve the same performance after learning, but with substantially lower constraint violations than the standard PPO. The safe exploration approach outperforms cost shaping in terms of constraint satisfaction, while not compromising convergence speed towards reaching the optimal cost. Safe Exploration, however, requires careful parameter tuning, especially on the slack variable that determines its responsiveness to near constraint violation.

Finally, a Model Predictive Safety Certification (MPSC) algorithm, based on [\(139\)](#), was chosen as a representative of [Sec. 3.3](#) (certifying learning-based control). This particular formulation uses an MPC framework to modify an unsafe learning controller’s actions. Here, a sub-optimal PPO controller provides the uncertified inputs trying to stabilize the cart-pole. The advantages of using MPSC are highlighted in [Figure 8](#). In [Figure 8](#) (right), the inputs are modified by the MPSC early in the stabilization to keep the cart-pole within the

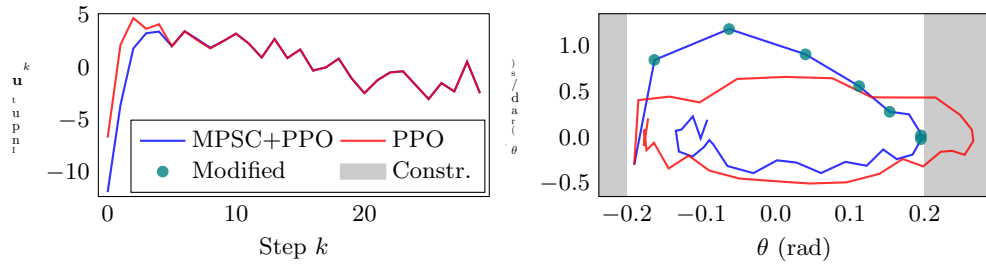


图 8: 左图: 未经认证的 PPO 输入 (红色) 与经过认证的 MPSC+PPO 输入 (蓝色) 进行绘图对比。右图: 小车-摆状态图 (θ 和 $\dot{\theta}$) , 比较 MPSC+PPO 认证轨迹 (蓝色) 与未经认证的 PPO 轨迹 (红色)。当系统即将离开约束边界 (灰色) 或 MPSC 可以纠正系统的状态集合时, MPSC 的活动最为活跃 (绿色点)。

在追求约束满足的同时控制性能。通过这样做, 我们证明了我们的基准可以轻松地与控制或强化学习研究社区开发的可学习策略集成。这也使我们能够公平地比较不同安全学习控制方法的数据需求量。

一种基于学习的鲁棒MPC, 使用 $\hat{f}$ (GP-MPC)的高斯过程估计, 代表第3.1节 (学习不确定动力学), 被实现用于在状态和输入约束下稳定四旋翼, 遵循(63)中的方法。在这个实验中, 使用关于悬停的线性化模型, 并将质量和惯性矩设为真实值的150%作为先验模型。使用从几分钟训练数据中随机选取的八百个状态-动作对 (或80秒) 进行超参数优化, 该优化是在离线完成的。**Figure 6**比较了使用不准确先验模型的线性MPC与GP-MPC方法的性能。我们可以看到, 使用更重先验模型的线性MPC预测四旋翼在施加最大推力时的轨迹相对平缓, 这导致四旋翼很快违反位置约束。相比之下, GP-MPC能够考虑不准确的先验模型并遵守约束。

我们还将安全探索方法 (70) 应用于流行的深度强化学习算法PPO (149), 作为第3.2节 (在强化学习中鼓励安全) 的代表, 并在考虑小车位置约束的倒立摆稳定任务上进行了测试。值得注意的是, 一旦违反任何约束, 任务即终止。我们将其与两个基线方法进行了比较, 即标准PPO和带有简单成本塑造的PPO (在接近约束违规时添加惩罚)。每种强化学习方法都使用了超过9小时的模拟时间来收集训练数据。**Figure 7**显示, 两种考虑约束的方法 (成本塑造和安全探索 (70)) 在学习后达到相同的性能, 但比标准PPO的约束违规明显更少。安全探索方法在约束满足方面优于成本塑造, 同时不会妨碍向达到最优成本的收敛速度。然而, 安全探索需要仔细调整参数, 特别是在松弛变量方面。

最后, 基于 (139) 的模型预测安全认证 (MPSC) 算法被选为第3.3节 (认证基于学习的控制) 的代表。该特定形式使用MPC框架来修改不安全的学习控制器的动作。在这里, 一个次优的PPO控制器提供未认证的输入, 试图稳定推车-摆系统。使用MPSC的优势在**Figure 8**中得到了强调。在**Figure 8 (right)**中, 这些输入在稳定初期被MPSC修改, 以保持推车-摆系统在

constraint boundaries. **Figure 8** (left) shows that, without the MPSC, PPO violates the constraints, but with MPSC, it manages to stay within the boundaries. The plot also shows that MPSC is most active when the system is close to the constraint boundaries (the green dots show when MPSC modified the learning controller’s input). This provides a proof of concept of how safety filters can be combined with RL control for safer performance.

In our future work, we intend to use our `cartpole` and `quadrotor` benchmark environments to evaluate the robustness, performance, safety and data efficiency of different control and learning approaches.

5. DISCUSSION AND PERSPECTIVES FOR FUTURE DIRECTIONS

The problem of safe learning control is emerging as a crucial topic for next-generation robotics. In this review, we summarized approaches from the control and the machine learning communities that allow data to be safely used to improve the closed-loop performance of robot control systems. We show that machine learning techniques, particularly RL, can help generalization towards larger classes of systems (i.e., fewer prior model assumptions), while control theory provides the insights and frameworks necessary to provide constraint satisfaction guarantees and closed-loop stability guarantees during the learning. Despite the many advances to date, there remain many opportunities for future research.

OPEN CHALLENGES

1. **Capturing a Broader Class of Systems.** Work to date has focused on nonlinear systems in the form of **Equation 1**. While they can model many robotic platforms, robots can also exhibit hybrid dynamics (e.g., legged robots or other contact dynamics with the environment (150)), time-varying dynamics (e.g., operation in changing environments (151, 152)), time delays (e.g., in actuation, sensing, or observing the reward (6)), partial differential dependencies in the dynamics (e.g., in continuum robotics (153)). Expanding safe learning control approaches to these scenarios is essential for their broader applicability in robotics (see **Figure 4**).
2. **Accounting for Imperfect State Measurements.** The majority of safe learning control approaches assume direct access to (possibly noisy) state measurements and neglect the problem of state estimation. In practice, obtaining accurate state information is challenging due to sensors that do not provide state measurements directly (e.g., images as measurements), inaccurate process and observation models used for state estimation, and/or improper state feature representations. One open challenge is to account for state estimation errors and learned process and observation models in safe learning control (143). Expanding existing approaches to work with (possibly high-dimensional) sensor data is essential for a broad applicability of these methods in robotics.
3. **Considering Scalability, and Sampling and Computational Efficiency.** Many of the approaches presented here have only been demonstrated on small toy problems and applying them to high-dimensional robotics problems is not trivial. Moreover, in practice, we often face issues such as data sparsity, distribution shifts, and the optimality-complexity trade-off for real-time implementations. Efficient robot learning relies on multiple factors including control architecture design (154), systematic training data collection (155), and appropriate function class selection (156). While current approaches focus on providing theoretical safety guarantees, formal analysis of sampling complexity and computational complexity is indispensable to facilitate the implementation of safe learning control algorithms in real-world robot applications.

约束边界。**Figure 8** (左)显示, 没有MPSC时, PPO会违反约束, 但有了MPSC, 它能够保持在边界内。该图还显示, 当系统接近约束边界时, MPSC的活跃度最高(绿色点表示MPSC修改了学习控制器的输入)。这为安全滤波器如何与RL控制结合以实现更安全的性能提供了概念验证。

在我们未来的工作中, 我们打算使用我们的 `cartpole` 和 `quadrotor` 基准环境来评估不同控制和学习方法的稳健性、性能、安全性和数据效率。

5. DISCUSSION AND PERSPECTIVES FOR FUTURE DIRECTIONS

安全学习控制问题正成为下一代机器人技术的一个关键话题。在本综述中, 我们总结了控制和机器学习领域的方法, 这些方法能够安全地使用数据来提高机器人控制系统的闭环性能。我们表明, 机器学习技术, 特别是强化学习, 可以帮助推广到更大类别的系统(即减少先验模型假设), 而控制理论则提供了在学习过程中实现约束满足保证和闭环稳定性保证所需的见解和框架。尽管迄今已有许多进展, 未来的研究仍有许多机会。

OPEN CHALLENGES

1. Capturing a Broader Class of Systems. 迄今为止的工作主要集中在以方程1形式表示的非线性系统上。虽然它们可以建模许多机器人平台, 但机器人也可能表现出混合动力学(例如, 有腿机器人或与环境有接触动力学(150))、时变动力学(例如, 在不断变化的环境中操作(151, 152))、时间延迟(例如, 在执行、感知或观察奖励时(6))、动力学中的偏微分依赖(例如, 在连续体机器人中(153))。将安全学习控制方法扩展到这些场景对于其在机器人学中的更广泛适用性至关重要(见 **Figure 4**)。

2. Accounting for Imperfect State Measurements. 大多数安全学习控制方法假设可以直接获取(可能带噪声的)状态测量, 并且忽略状态估计问题。在实际应用中, 由于传感器不能直接提供状态测量(例如, 将图像作为测量)、用于状态估计的过程和观测模型不准确, 和/或状态特征表示不当, 获取准确的状态信息是具有挑战性的。一个开放性难题是如何在安全学习控制中考虑状态估计误差以及学习到的过程和观测模型(143)。将现有方法扩展以处理(可能是高维的)传感器数据, 对于这些方法在机器人学中的广泛应用至关重要。

3. Considering Scalability, and Sampling and Computational Efficiency. 这里介绍的许多方法仅在小型示例问题上进行了验证, 将它们应用于高维机器人问题并不简单。此外, 在实践中, 我们经常面临数据稀疏、分布变化以及实时实现中的最优性-复杂性折衷等问题。高效的机器人学习依赖于多种因素, 包括控制架构设计(154)、系统的训练数据收集(155)以及适当的函数类选择(156)。虽然当前的方法侧重于提供理论上的安全保证, 但对采样复杂性和计算复杂性的形式分析对于促进在实际机器人应用中实现安全学习控制算法是不可或缺的。

4. **Verifying System and Modeling Assumptions.** The safety guarantees provided often rely on a set of assumptions (e.g., Lipschitz continuous *true* dynamics with a known Lipschitz constant or bounded disturbance sets). It is difficult to verify these assumptions prior to a robot’s operation. To facilitate algorithm implementation, we also see other approximations being made (e.g., linearization, data assumed to be independent and identically distributed (i.i.d.) Gaussian samples). Systematic approaches to verify or quantify the impact of the assumptions and the approximations with minimal (online) data are crucial to allow the safe learning approaches to be applied in real-world applications. This can also include investigations into the interpretability of trained models, especially black-box models such as DNNs, for safe closed-loop operation (6).

This list of challenges is by no means complete. Other important and open questions pertinent to safe learning control are:

- What are the appropriate **benchmarks, evaluation metrics, and practices** to provide practical insights and perform fair comparisons of algorithms that rely on different assumptions?
- What should be the **role of simulation** in the offline design and evaluation phases? Can simulation be used to find safe hyperparameters?
- How do we define safety in **human-robot interaction**?
- How can data be safely used in **multi-agent learning** settings?

Efforts that combine control theory and machine learning for safe learning control have been shown to result in improved control performance and system safety. Recent successes and growing interest should motivate the further development of a systematic body of theory, advanced methodologies, and computational methods for safe learning in robotics, bringing the large potential of learning into safety-critical control and robotics applications. The open challenges and questions are intended to push us further, towards a future of real-world safe autonomy where robots reliably and safely function in complex environments.

ACKNOWLEDGMENTS

The authors would like to acknowledge the early contributions to this work by Karime Pereida and Sepehr Samavi, the invaluable suggestions and feedback by Hallie Siegel, as well as the support from the Natural Sciences and Engineering Research Council of Canada (NSERC), the Canada Research Chairs Program, and the CIFAR AI Chair.

LITERATURE CITED

1. Burnett K, Qian J, Du X, Liu L, Yoon DJ, et al. 2021. Zeus: A system description of the two-time winner of the collegiate SAE autodrive competition. *Journal of Field Robotics* 38:139–166
2. Boutilier JJ, Brooks SC, Janmohamed A, Byers A, Buick JE, et al. 2017. Optimizing a drone network to deliver automated external defibrillators. *Circulation* 135:2454–2465
3. Dong K, Pereida K, Shkurti F, Schoellig AP. 2020. Catch the ball: Accurate high-speed motions for mobile manipulators via inverse dynamics learning. [arXiv:2003.07489](#) [cs.RO]
4. García J, Fern, o Fernández. 2015. A comprehensive survey on safe reinforcement learning. *Journal of Machine Learning Research* 16:1437–1480
5. Dulac-Arnold G, Levine N, Mankowitz DJ, Li J, Paduraru C, et al. 2021. An empirical investigation of the challenges of real-world reinforcement learning. [arXiv:2003.11881](#) [cs.LG]
6. Dulac-Arnold G, Mankowitz D, Hester T. 2019. Challenges of real-world reinforcement learning. [arXiv:1904.12901](#) [cs.LG]

4. Verifying System and Modeling Assumptions. 提供的安全保障通常依赖于一组假设（例如，具有已知Lipschitz常数的Lipschitz连续 $true$ 动力学或有界扰动集）。在机器人操作之前，很难验证这些假设。为了方便算法实现，我们还会看到其他近似的做法（例如，线性化，假设数据为独立同分布（i.i.d.）的高斯样本）。系统化的方法来验证或量化这些假设和近似的影响，同时使用最少的（在线）数据，对于使安全学习方法能够应用于现实世界的场景至关重要。这也可以包括对训练模型的可解释性进行研究，尤其是像DNN这样的黑箱模型，以实现安全的闭环操作(6)。

这个挑战列表绝不是完整的。其他与安全学习控制相关的重要且未解决的问题包括：

- 提供实际见解并对依赖不同假设的算法进行公平比较的适当 **benchmarks, evaluation metrics, and practices** 是什么？
- 在离线设计和评估阶段，**role of simulation** 应该是什么？可以使用仿真来寻找安全的超参数吗？
- 我们如何在 **human-robot interaction** 中定义安全？
- 数据如何能在 **multi-agent learning** 环境中被安全使用？

将控制理论与机器学习相结合进行安全学习控制的努力已被证明能够提高控制性能和系统安全性。近期的成功和日益增长的兴趣应该促使我们进一步开发系统化的理论体系、先进的方法论和计算方法，以实现机器人领域的安全学习，将学习的巨大潜力引入安全关键的控制和机器人应用中。提出的开放性挑战和问题旨在推动我们前进，迈向一个现实世界中机器人能够在复杂环境中可靠且安全运行的安全自主未来。

ACKNOWLEDGMENTS

作者们想要感谢 Karime Pereida 和 Sepehr Samavi 对本工作的早期贡献，Hallie Siegel 提出的宝贵建议和反馈，以及来自加拿大自然科学与工程研究理事会（NSERC）、加拿大研究主席项目和 CIFAR 人工智能主席的支持。

LITERATURE CITED

1. Burnett K, Qian J, Du X, Liu L, Yoon DJ 等人. 2021. Zeus：一项关于大学 SAE 自动驾驶竞赛两次获奖系统的描述。 *Journal of Field Robotics* 38:139–1662. Boutilier JJ, Brooks SC, Janmohamed A, Byers A, Buick JE 等人. 2017. 优化无人机网络以交付自动体外除颤器。 *Circulation* 135:2454–24653. Dong K, Pereida K, Shkurti F, Schoellig AP. 2020. 接球：通过逆向动力学学习实现移动机械臂的高速精确运动。 arXiv:2003.07489 [cs.RO]4. Garc ía J, Fern, o Fernández. 2015. 关于安全强化学习的综合调查。 *Journal of Machine Learning Research* 16:1437–14805. Dulac-Arnold G, Levine N, Mankowitz DJ, Li J, Paduraru C 等人. 2021. 关于现实世界强化学习挑战的实证研究。 arXiv:2003.11881 [cs.LG]6. Dulac-Arnold G, Mankowitz D, Hester T. 2019. 现实世界强化学习的挑战。 arXiv:1904.12901 [cs.LG]

7. Hewing L, Wabersich KP, Menner M, Zeilinger MN. 2020. Learning-based model predictive control: Toward safe learning in control. *Annual Review of Control, Robotics, and Autonomous Systems* 3:269–296
8. Bristow D, Tharayil M, Alleyne A. 2006. A survey of iterative learning control. *IEEE Control Systems Magazine* 26:96–114
9. Ahn HS, Chen Y, Moore KL. 2007. Iterative learning control: Brief survey and categorization. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* 37:1099–1121
10. Polydoros AS, Nalpantidis L. 2017. Survey of model-based reinforcement learning: Applications on robotics. *Journal of Intelligent & Robotic Systems* 86:153–173
11. Chatzilygeroudis K, Vassiliades V, Stulp F, Calinon S, Mouret JB. 2020. A survey on policy search algorithms for learning robot controllers in a handful of trials. *IEEE Transactions on Robotics* 36:328–347
12. Ravichandar H, Polydoros AS, Chernova S, Billard A. 2020. Recent advances in robot learning from demonstration. *Annual Review of Control, Robotics, and Autonomous Systems* 3:297–330
13. Kober J, Bagnell JA, Peters J. 2013. Reinforcement learning in robotics: A survey. *The International Journal of Robotics Research* 32:1238–1274
14. Recht B. 2019. A tour of reinforcement learning: The view from continuous control. *Annual Review of Control, Robotics, and Autonomous Systems* 2:253–279
15. Kiumarsi B, Vamvoudakis KG, Modares H, Lewis FL. 2018. Optimal and autonomous control using reinforcement learning: A survey. *IEEE Transactions on Neural Networks and Learning Systems* 29:2042–2062
16. Osborne M, Shin HS, Tsourdos A. 2021. *A Review of Safe Online Learning for Nonlinear Control Systems*. In *2021 International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 794–803. Piscataway, NJ: IEEE
17. Tambon F, Laberge G, An L, Nikanjam A, Mindom PSN, et al. 2021. How to certify machine learning based safety-critical systems? a systematic literature review. [arXiv:2107.12045](https://arxiv.org/abs/2107.12045) [cs.LG]
18. Ray A, Achiam J, Amodei D. 2019. Benchmarking Safe Exploration in Deep Reinforcement Learning. <https://cdn.openai.com/safexp-short.pdf>
19. Leike J, Martic M, Krakovna V, Ortega PA, Everitt T, et al. 2017. AI safety gridworlds. [arXiv:1711.09883](https://arxiv.org/abs/1711.09883) [cs.LG]
20. Sutton RS, Barto AG. 2018. *Reinforcement Learning: An Introduction*. Cambridge, MA: MIT Press, 2nd ed.
21. Khalil H. 2002. *Nonlinear Systems*. Pearson Education. Prentice Hall
22. Åström K, Wittenmark B. 2011. *Computer-Controlled Systems: Theory and Design, Third Edition*. Dover Books on Electrical Engineering. Dover Publications
23. Sastry S, Bodson M. 2011. *Adaptive Control: Stability, Convergence and Robustness*. Dover Books on Electrical Engineering Series. Dover Publications
24. Nguyen-Tuong D, Peters J. 2011. Model learning for robot control: a survey. *Cognitive processing* 12:319–340
25. Zhou K, Doyle J, Glover K. 1996. *Robust and Optimal Control*. Prentice Hall
26. Dullerud G, Paganini F. 2005. *A Course in Robust Control Theory: A Convex Approach*. Texts in Applied Mathematics. Springer New York
27. Rawlings J, Mayne D, Diehl M. 2017. *Model Predictive Control: Theory, Computation, and Design*. Nob Hill Publishing
28. Mayne D, Seron M, Raković S. 2005. Robust model predictive control of constrained linear systems with bounded disturbances. *Automatica* 41:219–224
29. Arulkumaran K, Deisenroth MP, Brundage M, Bharath AA. 2017. Deep reinforcement learning: A brief survey. *IEEE Signal Processing Magazine* 34:26–38
30. Dai B, Shaw A, Li L, Xiao L, He N, et al. 2018. *SBEED: Convergent Reinforcement Learning with Nonlinear Function Approximation*. In *Proceedings of the 35th International Conference on Machine Learning*, ed. J Dy, A Krause, pp. 1125–1134, vol. 80 of *Proceedings of Machine Learning Research*, pp. 1125–1134. N.p.: PMLR
31. Cheng R, Verma A, Orosz G, Chaudhuri S, Yue Y, Burdick J. 2019. Control regularization for reduced variance reinforcement learning. In *Proceedings of the 36th International Conference on Machine Learning*, ed. K Chaudhuri, R Salakhutdinov, pp. 1141–1150, vol. 97 of

7. Hewing L, Wabersich KP, Menner M, Zeilinger MN. 2020. 基于学习的模型预测控制: 迈向控制中的安全学习。 *Annual Review of Control, Robotics, and Autonomous Systems* 3:269–296

8. Bristow D, Tharayil M, Alleyne A. 2006. 迭代学习控制的综述。 *IEEE Control Systems Magazine* 26:96–114

9. Ahn HS, Chen Y, Moore KL. 2007. 迭代学习控制: 简要综述与分类。 *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* 37:1099–1121

10. Polydoros AS, Nalpantidis L. 2017. 基于模型的强化学习综述: 机器人应用。 *Journal of Intelligent & Robotic Systems* 86:153–173

11. Chatzilygeroudis K, Vassiliades V, Stulp F, Calinon S, Mouret JB. 2020. 少量实验中学习机器人控制器的策略搜索算法综述。 *IEEE Transactions on Robotics* 36:328–347

12. Ravichandar H, Polydoros AS, Chernova S, Billard A. 2020. 机器人示教学习的最新进展。 *Annual Review of Control, Robotics, and Autonomous Systems* 3:297–330

13. Kober J, Bagnell JA, Peters J. 2013. 机器人中的强化学习: 综述。 *The International Journal of Robotics Research* 32:1238–1274

14. Recht B. 2019. 强化学习导览: 从控制视角出发

- Proceedings of Machine Learning Research*. N.p.: PMLR
32. Ghavamzadeh M, Mannor S, Pineau J, Tamar A. 2015. Bayesian reinforcement learning: A survey. *Foundations and Trends® in Machine Learning* 8:359–483
 33. Altman E. 1999. *Constrained Markov decision processes*, vol. 7. Boca Raton, FL: Chapman & Hall/CRC Press
 34. Achiam J, Held D, Tamar A, Abbeel P. 2017. Constrained policy optimization. In *Proceedings of the 34th International Conference on Machine Learning*, ed. D Precup, YW Teh, pp. 22–31, vol. 70 of *Proceedings of Machine Learning Research*. N.p.: PMLR
 35. Nilim A, El Ghaoui L. 2005. Robust control of Markov decision processes with uncertain transition matrices. *Operations Research* 53:780–798
 36. Pinto L, Davidson J, Sukthankar R, Gupta A. 2017. Robust adversarial reinforcement learning. In *Proceedings of the 34th International Conference on Machine Learning*, ed. D Precup, YW Teh, pp. 2817–2826, vol. 70 of *Proceedings of Machine Learning Research*. N.p.: PMLR
 37. Pan X, Seita D, Gao Y, Canny J. 2019. *Risk Averse Robust Adversarial Reinforcement Learning*. In *2019 International Conference on Robotics and Automation (ICRA)*, pp. 8522–8528. Piscataway, NJ: IEEE
 38. Vinitzky E, Du Y, Parvate K, Jang K, Abbeel P, Bayen A. 2020. Robust reinforcement learning using adversarial populations. [arXiv:2008.01825](https://arxiv.org/abs/2008.01825) [cs.LG]
 39. Cooper J, Che J, Cao C. 2014. The use of learning in fast adaptation algorithms. *International Journal of Adaptive Control and Signal Processing* 28:325–340
 40. Gahlawat A, Zhao P, Patterson A, Hovakimyan N, Theodorou E. 2020. L1-GP: L1 adaptive control with Bayesian learning. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, ed. AM Bayen, A Jadbabaie, G Pappas, PA Parrilo, B Recht, C Tomlin, M Zeilinger, pp. 826–837, vol. 120 of *Proceedings of Machine Learning Research*. N.p.: PMLR
 41. Hovakimyan N, Cao C. 2010. *L1 Adaptive Control Theory: Guaranteed Robustness with Fast Adaptation*. USA: Society for Industrial and Applied Mathematics
 42. Grande RC, Chowdhary G, How JP. 2014. Experimental validation of Bayesian nonparametric adaptive control using Gaussian processes. *Journal of Aerospace Information Systems* 11:565–578
 43. Chowdhary G, Kingravi HA, How JP, Vela PA. 2015. Bayesian nonparametric adaptive control using Gaussian processes. *IEEE Transactions on Neural Networks and Learning Systems* 26:537–550
 44. Joshi G, Chowdhary G. 2019. *Deep model reference adaptive control*. In *2019 IEEE 58th Conference on Decision and Control (CDC)*, pp. 4601–4608. Piscataway, NJ: IEEE
 45. Joshi G, Virdi J, Chowdhary G. 2020. Asynchronous deep model reference adaptive control. [arXiv:2011.02920](https://arxiv.org/abs/2011.02920) [cs.RO]
 46. Berkenkamp F, Schoellig AP. 2015. *Safe and robust learning control with Gaussian processes*. In *2015 European Control Conference (ECC)*, pp. 2496–2501. Piscataway, NJ: IEEE
 47. Holicki T, Scherer CW, Trimpe S. 2021. Controller design via experimental exploration with robustness guarantees. *IEEE Control Systems Letters* 5:641–646
 48. von Rohr A, Neumann-Brosig M, Trimpe S. 2021. Probabilistic robust linear quadratic regulators with Gaussian processes. In *Proceedings of the 3rd Conference on Learning for Dynamics and Control*, ed. A Jadbabaie, J Lygeros, GJ Pappas, PA Parrilo, B Recht, CJ Tomlin, MN Zeilinger, pp. 324–335, vol. 144 of *Proceedings of Machine Learning Research*. N.p.: PMLR
 49. Helwa MK, Heins A, Schoellig AP. 2019. Provably robust learning-based approach for high-accuracy tracking control of lagrangian systems. *IEEE Robotics and Automation Letters* 4:1587–1594
 50. Greeff M, Schoellig AP. 2021. Exploiting differential flatness for robust learning-based tracking control using Gaussian processes. *IEEE Control Systems Letters* 5:1121–1126
 51. Tanaskovic M, Fagiano L, Smith R, Morari M. 2014. Adaptive receding horizon control for constrained MIMO systems. *Automatica* 50:3019–3029
 52. Lorenzen M, Cannon M, Allgöwer F. 2019. Robust MPC with recursive model update. *Automatica* 103:461–471
 53. Bujarbaruah M, Zhang X, Borrelli F. 2018. *Adaptive MPC with Chance Constraints for FIR Systems*. In *2018 Annual American Control Conference (ACC)*, pp. 2312–2317. Piscataway, NJ: IEEE
 54. Bujarbaruah M, Zhang X, Tanaskovic M, Borrelli F. 2019. Adaptive MPC under Time Varying

Proceedings of Machine Learning Research. 无出版地: PMLR 32. Ghavamzadeh M, Mannor S, Pineau J, Tamar A. 2015. 贝叶斯强化学习: 综述。 *Foundations and Trends® in Machine Learning* 8:359–483 33. Altman E. 1999. *Constrained Markov decision processes*, 第7卷。佛罗里达州博卡拉顿: Chapman & Hall/CRC Press 34. Achiam J, Held D, Tamar A, Abbeel P. 2017. 受约束的策略优化。收录于 *Proceedings of the 34th International Conference on Machine Learning*, 编. D Precup, YW Teh, 页 22–31, 第70卷 *Proceedings of Machine Learning Research*. 无出版地: PMLR 35. Nilim A, El Ghaoui L. 2005. 具有不确定转移矩阵的马尔可夫决策过程的稳健控制。 *Operations Research* 53:780–798 36. Pinto L, Davidson J, Sukthankar R, Gupta A. 2017. 鲁棒对抗强化学习。收录于 *Proceedings of the 34th International Conference on Machine Learning*, 编. D Precup, YW Teh, 页 2817–2826, 第70卷 *Proceedings of Machine Learning Research*. 无出版地: PMLR 37. Pan X, Seita D, Gao Y, Canny J. 2019. *Risk Averse Robust Adversarial Reinforcement Learning*. 收录于 *2019 International Conference on Robotics and Automation (ICRA)*, 页 8522–8528. 新泽西州皮斯卡特维: IEEE 38. Vinitisky E, Du Y, Parvate K, Jang K, Abbeel P, Bayen A. 2020. 使用对抗群体的鲁棒强化学习。arXiv:2008.01825 [cs.LG] 39. Cooper J, Che J, Cao C. 2014. 在快速适应算法中使用学习。{v

- Uncertainty: Robust and Stochastic. [arXiv:1909.13473](https://arxiv.org/abs/1909.13473) [eess.SY]
55. Gonçalves GA, Guay M. 2016. Robust discrete-time set-based adaptive predictive control for nonlinear systems. *Journal of Process Control* 39:111–122
 56. Köhler J, Kötting P, Soloperto R, Allgöwer F, Müller MA. 2020. A robust adaptive model predictive control framework for nonlinear uncertain systems. *International Journal of Robust and Nonlinear Control* <https://doi.org/10.1002/rnc.5147>
 57. Rosolia U, Borrelli F. 2018. Learning model predictive control for iterative tasks. a data-driven control framework. *IEEE Transactions on Automatic Control* 63:1883–1896
 58. Bujarbaruah M, Zhang X, Rosolia U, Borrelli F. 2018. *Adaptive MPC for Iterative Tasks*. In *2018 IEEE Conference on Decision and Control (CDC)*, pp. 6322–6327. Piscataway, NJ: IEEE
 59. Pereida K, Brunke L, Schoellig AP. 2021. Robust adaptive model predictive control for guaranteed fast and accurate stabilization in the presence of model errors. *International Journal of Robust and Nonlinear Control* in-press
 60. Aswani A, Gonzalez H, Sastry SS, Tomlin C. 2013. Provably safe and robust learning-based model predictive control. *Automatica* 49:1216–1226
 61. Soloperto R, Müller MA, Trimpe S, Allgöwer F. 2018. Learning-based robust model predictive control with state-dependent uncertainty. *IFAC-PapersOnLine* 51(20):442–447
 62. Ostafew CJ, Schoellig AP, Barfoot TD. 2016. Robust constrained learning-based NMPC enabling reliable mobile robot path tracking. *The International Journal of Robotics Research* 35:1547–1563
 63. Hewing L, Kabzan J, Zeilinger MN. 2020. Cautious model predictive control using Gaussian process regression. *IEEE Transactions on Control Systems Technology* 28:2736–2743
 64. Kamthe S, Deisenroth M. 2018. Data-efficient reinforcement learning with probabilistic model predictive control. In *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, ed. A Storkey, F Perez-Cruz, pp. 1701–1710, vol. 84 of *Proceedings of Machine Learning Research*. N.p.: PMLR
 65. Koller T, Berkenkamp F, Turchetta M, Boedecker J, Krause A. 2019. Learning-based model predictive control for safe exploration and reinforcement learning. [arXiv:1906.12189](https://arxiv.org/abs/1906.12189) [eess.SY]
 66. Fan D, Agha A, Theodorou E. 2020. *Deep Learning Tubes for Tube MPC*. In *Proceedings of Robotics: Science and Systems XVI*, pp. pap. 87. N.p.: Robot. Sci. Syst. Found.
 67. McKinnon CD, Schoellig AP. 2020. *Context-aware Cost Shaping to Reduce the Impact of Model Error in Receding Horizon Control*. In *2020 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2386–2392. Piscataway, NJ: IEEE
 68. Berkenkamp F, Turchetta M, Schoellig A, Krause A. 2017. Safe model-based reinforcement learning with stability guarantees. In *Advances in Neural Information Processing Systems*, ed. I Guyon, UV Luxburg, S Bengio, H Wallach, R Fergus, S Vishwanathan, R Garnett, pp. 908–919, vol. 30. Red Hook, NY, USA: Curran Associates, Inc.
 69. Turchetta M, Berkenkamp F, Krause A. 2016. Safe exploration in finite Markov decision processes with Gaussian processes. In *Advances in Neural Information Processing Systems*, ed. DD Lee, M Sugiyama, UV Luxburg, I Guyon, R Garnett, pp. 4312–4320, vol. 29. Red Hook, NY, USA: Curran Associates, Inc.
 70. Dalal G, Dvijotham K, Vecerik M, Hester T, Paduraru C, Tassa Y. 2018. Safe exploration in continuous action spaces. [arXiv:1801.08757](https://arxiv.org/abs/1801.08757) [cs.AI]
 71. Henderson P, Islam R, Bachman P, Pineau J, Precup D, Meger D. 2018. *Deep Reinforcement Learning That Matters*. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32(1). Palo Alto, CA: AAAI Press. <https://ojs.aaai.org/index.php/AAAI/article/view/11694>
 72. Mezić I. 2003. Controllability, integrability and ergodicity. In *Multidisciplinary Research in Control: The Mohammed Dahleh Symposium 2002*, ed. L Giarre, B Bamieh, pp. 213–229. Berlin, Heidelberg: Springer Berlin Heidelberg
 73. Moldovan TM, Abbeel P. 2012. *Safe Exploration in Markov Decision Processes*. In *Proceedings of the 29th International Conference on Machine Learning (ICML)*, pp. 1451–1458. Madison, WI: Omnipress
 74. Brafman RI, Tennenholtz M. 2002. R-max-a general polynomial time algorithm for near-optimal reinforcement learning. *Journal of Machine Learning Research* 3:213–231
 75. Pham TH, De Magistris G, Tachibana R. 2018. *OptLayer - Practical Constrained Optimization for Deep Reinforcement Learning in the Real World*. In *2018 IEEE International Conference*

不确定性：鲁棒与随机。arXiv:1909.13473 [eess.SY] 55. Gonçalves GA, Guay M. 2016. 针对非线性系统的鲁棒离散时间基于集合的自适应预测控制。 *Journal of Process Control* 39:111–122 56. Köhler J, Kötting P, Soloperto R, Allgöwer F, Müller MA. 2020. 针对非线性不确定系统的鲁棒自适应模型预测控制框架。 *International Journal of Robust and Nonlinear Control* <https://doi.org/10.1002/rnc.5147> 57. Rosolia U, Borrelli F. 2018. 用于迭代任务的学习型模型预测控制。一个数据驱动的控制框架。 *IEEE Transactions on Automatic Control* 63:1883–1896 58. Bujarbaruah M, Zhang X, Rosolia U, Borrelli F. 2018. *Adaptive MPC for Iterative Tasks*. 收录于 *2018 IEEE Conference on Decision and Control (CDC)*, 第 6322–6327 页。新泽西州皮斯卡特韦：IEEE 59. Pereida K, Brunke L, Schoellig AP. 2021. 鲁棒自适应模型预测控制，用于在存在模型误差时保证快速且精确的稳定。 *International Journal of Robust and Nonlinear Control* 待出版 60. Aswani A, Gonzalez H, Sastry SS, Tomlin C. 2013. 可证明安全与鲁棒的基于学习的模型预测控制。 *Automatica* 49:1216–1226 61. Soloperto R,

- on *Robotics and Automation (ICRA)*, pp. 6236–6243. Piscataway, NJ: IEEE
76. Kim Y, Allmendinger R, López-Ibáñez M. 2021. Safe learning and optimization techniques: Towards a survey of the state of the art. [arXiv:2101.09505](#) [cs.LG]
 77. Duivenvoorden RR, Berkenkamp F, Carion N, Krause A, Schoellig AP. 2017. Constrained Bayesian optimization with particle swarms for safe adaptive controller tuning. *IFAC-PapersOnLine* 50(1):11800–11807
 78. Sui Y, Gotovos A, Burdick J, Krause A. 2015. Safe exploration for optimization with Gaussian processes. In *Proceedings of the 32nd International Conference on Machine Learning*, ed. F Bach, D Blei, pp. 997–1005, vol. 37 of *Proceedings of Machine Learning Research*. N.p.: PMLR
 79. Berkenkamp F, Krause A, Schoellig AP. 2020. Bayesian optimization with safety constraints: Safe and automatic parameter tuning in robotics. [arXiv:1602.04450](#) [cs.RO]
 80. Sui Y, Vincent Zhuang, Burdick J, Yue Y. 2018. Stagewise safe Bayesian optimization with Gaussian processes. In *Proceedings of the 35th International Conference on Machine Learning*, ed. J Dy, A Krause, pp. 4781–4789, vol. 80 of *Proceedings of Machine Learning Research*. N.p.: PMLR
 81. Baumann D, Marco A, Turchetta M, Trimpe S. 2021. GoSafe: Globally optimal safe robot learning. [arXiv:2105.13281](#) [cs.RO]
 82. Wachi A, Sui Y, Yue Y, Ono M. 2018. *Safe Exploration and Optimization of Constrained MDPs Using Gaussian Processes*. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32(1). Palo Alto, CA: AAAI Press. <https://ojs.aaai.org/index.php/AAAI/article/view/12103>
 83. Kumar A, Zhou A, Tucker G, Levine S. 2020. Conservative Q-learning for offline reinforcement learning. [arXiv:2006.04779](#) [cs.LG]
 84. Chua K, Calandra R, McAllister R, Levine S. 2018. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems*, ed. S Bengio, H Wallach, H Larochelle, K Grauman, N Cesa-Bianchi, R Garnett, pp. 4759–4770, vol. 31. Red Hook, NY, USA: Curran Associates, Inc.
 85. Srinivasan K, Eysenbach B, Ha S, Tan J, Finn C. 2020. Learning to be safe: Deep RL with a safety critic. [arXiv:2010.14603](#) [cs.LG]
 86. Thananjeyan B, Balakrishna A, Nair S, Luo M, Srinivasan K, et al. 2021. Recovery RL: Safe reinforcement learning with learned recovery zones. *IEEE Robotics and Automation Letters* 6:4915–4922
 87. Bharadhwaj H, Kumar A, Rhinehart N, Levine S, Shkurti F, Garg A. 2021. Conservative safety critics for exploration. [arXiv:2010.14497](#) [cs.LG]
 88. Chow Y, Ghavamzadeh M, Janson L, Pavone M. 2017. Risk-constrained reinforcement learning with percentile risk criteria. *Journal of Machine Learning Research* 18:1–51
 89. Kahn G, Villaflor A, Pong V, Abbeel P, Levine S. 2017. Uncertainty-aware reinforcement learning for collision avoidance. [arXiv:1702.01182](#) [cs.LG]
 90. Lütjens B, Everett M, How JP. 2019. *Safe Reinforcement Learning With Model Uncertainty Estimates*. In *2019 International Conference on Robotics and Automation (ICRA)*, pp. 8662–8668. Piscataway, NJ: IEEE
 91. Zhang J, Cheung B, Finn C, Levine S, Jayaraman D. 2020. Cautious adaptation for reinforcement learning in safety-critical settings. In *Proceedings of the 37th International Conference on Machine Learning*, ed. HD III, A Singh, pp. 11055–11065, vol. 119 of *Proceedings of Machine Learning Research*. N.p.: PMLR
 92. Thananjeyan B, Balakrishna A, Rosolia U, Li F, McAllister R, et al. 2020. Safety augmented value estimation from demonstrations (SAVED): Safe deep model-based rl for sparse cost robotic tasks. *IEEE Robotics and Automation Letters* 5:3612–3619
 93. Urpí NA, Curi S, Krause A. 2021. Risk-averse offline reinforcement learning. [arXiv:2102.05371](#) [cs.LG]
 94. Bellemare MG, Dabney W, Munos R. 2017. *A Distributional Perspective on Reinforcement Learning*. In *Proceedings of the 34th International Conference on Machine Learning*, ed. D Precup, YW Teh, pp. 449–458, vol. 70 of *Proceedings of Machine Learning Research*, pp. 449–458. N.p.: PMLR
 95. Liang Q, Que F, Modiano E. 2018. Accelerated primal-dual policy optimization for safe reinforcement learning. [arXiv:1802.06480](#) [cs.AI]

on Robotics and Automation (ICRA), 第6236–6243页。新泽西州皮斯卡特维: IEEE 76. Kim Y, Allmendinger R, López-Ibáñez M. 2021. 安全学习与优化技术: 面向现有研究的综述。arXiv:2101.09505 [cs.LG] 77. Duivenvoorden RR, Berkenkamp F, Carion N, Krause A, Schoellig AP. 2017. 使用粒子群进行约束贝叶斯优化以实现安全自适应控制器调优。 *IFAC- PapersOnLine* 50(1):11800–11807 78. Sui Y, Gotovos A, Burdick J, Krause A. 2015. 高斯过程优化的安全探索。收录于 *Proceedings of the 32nd International Conference on Machine Learning*, 编辑 F Bach, D Blei, 第997–1005页, 第37卷 *Proceedings of Machine Learning Research*。无出版地: PMLR 79. Berkenkamp F, Krause A, Schoellig AP. 2020. 带安全约束的贝叶斯优化: 机器人中的安全与自动参数调优。arXiv:1602.04450 [cs.RO] 80. Sui Y, Vincent Zhuang, Burdick J, Yue Y. 2018. 高斯过程的分阶段安全贝叶斯优化。收录于 *Proceedings of the 35th International Conference on Machine Learning*, 编辑 J Dy, A Krause, 第4781–4789页, 第80卷 *Proceedings of Machine Learning Research*。无出版地: PMLR 81. Baumann D, Marco A, Turchetta M, Trimpe S. 2021. GoSafe: G

96. Schulman J, Levine S, Abbeel P, Jordan M, Moritz P. 2015. Trust region policy optimization. In *Proceedings of the 32nd International Conference on Machine Learning*, ed. F Bach, D Blei, pp. 1889–1897, vol. 37 of *Proceedings of Machine Learning Research*. N.p.: PMLR
97. Chow Y, Nachum O, Duenez-Guzman E, Ghavamzadeh M. 2018. A Lyapunov-based approach to safe reinforcement learning. In *Advances in Neural Information Processing Systems*, ed. S Bengio, H Wallach, H Larochelle, K Grauman, N Cesa-Bianchi, R Garnett, pp. 8103–8112, vol. 31. Red Hook, NY, USA: Curran Associates, Inc.
98. Chow Y, Nachum O, Faust A, Duenez-Guzman E, Ghavamzadeh M. 2019. Lyapunov-based safe policy optimization for continuous control. [arXiv:1901.10031](#) [cs.LG]
99. Satija H, Amortila P, Pineau J. 2020. Constrained Markov decision processes via backward value functions. In *Proceedings of the 37th International Conference on Machine Learning*, ed. HD III, A Singh, pp. 8502–8511, vol. 119 of *Proceedings of Machine Learning Research*. N.p.: PMLR
100. Morimoto J, Doya K. 2005. Robust reinforcement learning. *Neural Computation* 17:335–359
101. Turchetta M, Krause A, Trimpe S. 2020. *Robust Model-free Reinforcement Learning with Multi-objective Bayesian Optimization*. In *2020 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10702–10708. Piscataway, NJ: IEEE
102. Goodfellow IJ, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, et al. 2014. Generative adversarial networks. [arXiv:1406.2661](#) [stat.ML]
103. Mnih V, Kavukcuoglu K, Silver D, Rusu AA, Veness J, et al. 2015. Human-level control through deep reinforcement learning. *Nature* 518:529–533
104. Lütjens B, Everett M, How JP. 2020. Certified adversarial robustness for deep reinforcement learning. In *Proceedings of the Conference on Robot Learning*, ed. LP Kaelbling, D Kragic, K Sugiura, pp. 1328–1337, vol. 100 of *Proceedings of Machine Learning Research*. N.p.: PMLR
105. Sadeghi F, Levine S. 2017. *CAD2RL: Real Single-Image Flight Without a Single Real Image*. In *Proceedings of Robotics: Science and Systems XIII*, pp. pap. 34. N.p.: Robot. Sci. Syst. Found.
106. Loquercio A, Kaufmann E, Ranftl R, Dosovitskiy A, Koltun V, Scaramuzza D. 2020. Deep drone racing: From simulation to reality with domain randomization. *IEEE Transactions on Robotics* 36:1–14
107. Rajeswaran A, Ghotra S, Ravindran B, Levine S. 2017. EPOpt: Learning robust neural network policies using model ensembles. [arXiv:1610.01283](#) [cs.LG]
108. Mehta B, Diaz M, Golemo F, Pal CJ, Paull L. 2020. Active domain randomization. In *Proceedings of the Conference on Robot Learning*, ed. LP Kaelbling, D Kragic, K Sugiura, pp. 1162–1176, vol. 100 of *Proceedings of Machine Learning Research*. N.p.: PMLR
109. Zhou S, Helwa MK, Schoellig AP. 2020. Deep neural networks as add-on modules for enhancing robot performance in impromptu trajectory tracking. *The International Journal of Robotics Research* 39:1397–1418
110. Jin M, Lavaei J. 2020. Stability-certified reinforcement learning: A control-theoretic perspective. *IEEE Access* 8:229086–229100
111. Shi G, Shi X, O’Connell M, Yu R, Azizzadenesheli K, et al. 2019. *Neural Lander: Stable Drone Landing Control Using Learned Dynamics*. In *2019 International Conference on Robotics and Automation (ICRA)*, pp. 9784–9790. Piscataway, NJ: IEEE
112. Fazlyab M, Robey A, Hassani H, Morari M, Pappas GJ. 2019. Efficient and accurate estimation of Lipschitz constants for deep neural networks. [arXiv:1906.04893](#) [cs.LG]
113. Richards SM, Berkenkamp F, Krause A. 2018. The Lyapunov neural network: Adaptive stability certification for safe learning of dynamical systems. In *Proceedings of The 2nd Conference on Robot Learning*, ed. A Billard, A Dragan, J Peters, J Morimoto, pp. 466–476, vol. 87 of *Proceedings of Machine Learning Research*. N.p.: PMLR
114. Zhou Z, Oguz OS, Leibold M, Buss M. 2020. A general framework to increase safety of learning algorithms for dynamical systems based on region of attraction estimation. *IEEE Transactions on Robotics* 36:1472–1490
115. Jarvis-Wloszek Z, Feeley R, Tan W, Sun K, Packard A. 2003. *Some controls applications of sum of squares programming*. In *42nd IEEE International Conference on Decision and Control (CDC)*, vol. 5, pp. 4676–4681 Vol.5. Piscataway, NJ: IEEE
116. Schilders WH, Van der Vorst HA, Rommes J. 2008. *Model order reduction: theory, research aspects and applications*, vol. 13. Springer-Verlag Berlin Heidelberg

96. Schulman J, Levine S, Abbeel P, Jordan M, Moritz P. 2015. 信赖域策略优化. 收录于 *Proceedings of the 32nd International Conference on Machine Learning*, 编辑 F Bach, D Blei, 页码 1889–1897, 第 37 卷 *Proceedings of Machine Learning Research*. 未注明地点: PMLR 97. Chow Y, Nachum O, Duenez-Guzman E, Ghavamzadeh M. 2018. 基于李雅普诺夫的安全强化学习方法. 收录于 *Advances in Neural Information Processing Systems*, 编辑 S Bengio, H Wallach, H Larochelle, K Grauman, N Cesa-Bianchi, R Garnett, 页码 8103–8112, 第 31 卷. 美国纽约州 Red Hook: Curran Associates, Inc. 98. Chow Y, Nachum O, Faust A, Duenez-Guzman E, Ghavamzadeh M. 2019. 连续控制的基于李雅普诺夫的安全策略优化. arXiv:1901.10031 [cs.LG] 99. Satija H, Amortila P, Pineau J. 2020. 通过后向价值函数的约束马尔可夫决策过程. 收录于 *Proceedings of the 37th International Conference on Machine Learning*, 编辑 HD III, A Singh, 页码 8502–8511, 第 119 卷 *Proceedings of Machine Learning Research*. 未注明地点: PMLR 100. Morimoto J, Doya K. 2005. 鲁棒强化学习. *Neural Computation* 17:335–359 101. Turchetta M, Krause A, Trimpe S. 2020. *Robust Model-free Reinforcement Learning with Multi-objective Bayesian Optimization*. 收录于 *2020 IEEE International Conference on Robotics and Automation (ICRA)*, 页码 10702–10708. 美国新泽西州皮斯卡塔维: IEEE 102. Goodfellow IJ, Pouget-Abadie

117. Alshiekh M, Bloem R, Udiger Ehlers R, Könighofer B, Niekum S, Topcu U. 2018. *Safe Reinforcement Learning via Shielding*. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32(1). Palo Alto, CA: AAAI Press. <https://ojs.aaai.org/index.php/AAAI/article/view/11797>
118. Ames AD, Coogan S, Egerstedt M, Notomista G, Sreenath K, Tabuada P. 2019. *Control Barrier Functions: Theory and Applications*. In *2019 18th European Control Conference (ECC)*, pp. 3420–3431. Piscataway, NJ: IEEE
119. Taylor AJ, Dorobantu VD, Le HM, Yue Y, Ames AD. 2019. *Episodic Learning with Control Lyapunov Functions for Uncertain Robotic Systems**. In *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 6878–6884. Piscataway, NJ: IEEE
120. Taylor A, Singletary A, Yue Y, Ames A. 2020. Learning for safety-critical control with control barrier functions. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, ed. AM Bayen, A Jadbabaie, G Pappas, PA Parrilo, B Recht, C Tomlin, M Zeilinger, pp. 708–717, vol. 120 of *Proceedings of Machine Learning Research*. N.p.: PMLR
121. Ohnishi M, Wang L, Notomista G, Egerstedt M. 2019. Barrier-certified adaptive reinforcement learning with applications to brushbot navigation. *IEEE Transactions on Robotics* 35:1186–1205
122. Choi J, Castañeda F, Tomlin C, Sreenath K. 2020. *Reinforcement Learning for Safety-Critical Control under Model Uncertainty, using Control Lyapunov Functions and Control Barrier Functions*. In *Proceedings of Robotics: Science and Systems XVI*, pp. pap. 88. N.p.: Robot. Sci. Syst. Found.
123. Taylor AJ, Dorobantu VD, Krishnamoorthy M, Le HM, Yue Y, Ames AD. 2019. *A Control Lyapunov Perspective on Episodic Learning via Projection to State Stability*. In *2019 IEEE 58th Conference on Decision and Control (CDC)*, pp. 1448–1455. Piscataway, NJ: IEEE
124. Taylor AJ, Singletary A, Yue Y, Ames AD. 2020. A control barrier perspective on episodic learning via projection-to-state safety. [arXiv:2003.08028](https://arxiv.org/abs/2003.08028) [eess.SY]
125. Taylor AJ, Dorobantu VD, Dean S, Recht B, Yue Y, Ames AD. 2020. Towards robust data-driven control synthesis for nonlinear systems with actuation uncertainty. [arXiv:2011.10730](https://arxiv.org/abs/2011.10730) [eess.SY]
126. Taylor AJ, Ames AD. 2020. *Adaptive Safety with Control Barrier Functions*. In *2020 American Control Conference (ACC)*, pp. 1399–1405. Piscataway, NJ: IEEE
127. Lopez BT, Slotine JJE, How JP. 2021. Robust adaptive control barrier functions: An adaptive and data-driven approach to safety. *IEEE Control Systems Letters* 5:1031–1036
128. Cheng R, Orosz G, Murray RM, Burdick JW. 2019. *End-to-end safe reinforcement learning through barrier functions for safety-critical continuous control tasks*. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33(1), pp. 3387–3395. Palo Alto, CA: AAAI Press
129. Fan DD, Nguyen J, Thakker R, Alatur N, Agha-mohammadi Aa, Theodorou EA. 2019. Bayesian learning-based adaptive control for safety critical systems. [arXiv:1910.02325](https://arxiv.org/abs/1910.02325) [eess.SY]
130. Wang L, Theodorou EA, Egerstedt M. 2018. *Safe learning of quadrotor dynamics using barrier certificates*. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2460–2465. Piscataway, NJ: IEEE
131. Khojasteh MJ, Dhiman V, Franceschetti M, Atanasov N. 2020. Probabilistic safety constraints for learned high relative degree system dynamics. In *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, ed. AM Bayen, A Jadbabaie, G Pappas, PA Parrilo, B Recht, C Tomlin, M Zeilinger, pp. 781–792, vol. 120 of *Proceedings of Machine Learning Research*. N.p.: PMLR
132. Mitchell I, Bayen A, Tomlin C. 2005. A time-dependent Hamilton-Jacobi formulation of reachable sets for continuous dynamic games. *IEEE Transactions on Automatic Control* 50:947–957
133. Fisac JF, Akametalu AK, Zeilinger MN, Kaynama S, Gillula J, Tomlin CJ. 2019. A general safety framework for learning-based control in uncertain robotic systems. *IEEE Transactions on Automatic Control* 64:2737–2752
134. Gillula JH, Tomlin CJ. 2012. *Guaranteed safe online learning via reachability: tracking a ground target using a quadrotor*. In *2012 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2723–2730. Piscataway, NJ: IEEE
135. Bajcsy A, Bansal S, Bronstein E, Tolani V, Tomlin CJ. 2019. *An Efficient Reachability-Based Framework for Provably Safe Autonomous Navigation in Unknown Environments*. In *2019*

117. Alshiekh M, Bloem R, Udiger Ehlers R, Könighofer B, Niekum S, Topcu U. 2018. *Safe Reinforcement Learning via Shielding*. 载于 *Proceedings of the AAAI Conference on Artificial Intelligence*, 卷 32(1). 美国加利福尼亚州帕洛阿尔托: AAAI Press. <https://ojs.aaai.org/index.php/AAAI/article/view/11797>
118. Ames AD, Coogan S, Egerstedt M, Notomista G, Sreenath K, Tabuada P. 2019. *Control Barrier Functions: Theory and Applications*. 载于 *2019 18th European Control Conference (ECC)*, 页 3420–3431. 美国新泽西州皮斯卡特韦: IEEE
119. Taylor AJ, Dorobantu VD, Le HM, Yue Y, Ames AD. 2019. *Episodic Learning with Control Lyapunov Functions for Uncertain Robotic Systems**. 载于 *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 页 6878–6884. 美国新泽西州皮斯卡特韦: IEEE
120. Taylor A, Singletary A, Yue Y, Ames A. 2020. 基于控制屏障函数的安全关键控制学习. 载于 *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, 编. AM Bayen, A Jadbabaie, G Pappas, PA Parrilo, B Recht, C Tomlin, M Zeilinger, 页 708–717, 卷 120 of *Proceedings of Machine Learning Research*. 无出版地: PMLR
121. Ohnishi M, Wang L, Notomista G, Egerstedt M. 2019. 可屏障认证的自适应强化学习及其在并行机器人导航中的应用. *IEEE Transactions on Robotics* 35:1186–1205
122. Choi J, Castañeda F, Tomlin C, Sreenath K. 2020. *Reinforcement Learning for Safety-Critical Control under Model Uncertainty, using Control Lyapunov Functions and Control Barrier Functions*. 载于 *Proceedings of Robotics: Science and Systems XVI*, 页 pap. 88. 无出版地: Robot. Sci

- IEEE 58th Conference on Decision and Control (CDC)*, pp. 1758–1765. Piscataway, NJ: IEEE
136. Fisac JF, Lugovoy NF, Rubies-Royo V, Ghosh S, Tomlin CJ. 2019. *Bridging Hamilton-Jacobi safety analysis and reinforcement learning*. In *2019 International Conference on Robotics and Automation (ICRA)*, pp. 8550–8556. Piscataway, NJ: IEEE
137. Choi JJ, Lee D, Sreenath K, Tomlin CJ, Herbert SL. 2021. Robust control barrier-value functions for safety-critical control. [arXiv:2104.02808](#) [eess.SY]
138. Herbert S, Choi JJ, Sanjeev S, Gibson M, Sreenath K, Tomlin CJ. 2021. Scalable learning of safety guarantees for autonomous systems using Hamilton-Jacobi reachability. [arXiv:2101.05916](#) [cs.RO]
139. Wabersich KP, Zeilinger MN. 2018. *Linear Model Predictive Safety Certification for Learning-Based Control*. In *2018 IEEE Conference on Decision and Control (CDC)*, pp. 7130–7135. Piscataway, NJ: IEEE
140. Wabersich KP, Hewing L, Carron A, Zeilinger MN. 2019. Probabilistic model predictive safety certification for learning-based control. [arXiv:1906.10417](#) [eess.SY]
141. Wabersich KP, Zeilinger MN. 2021. A predictive safety filter for learning-based control of constrained nonlinear dynamical systems. *Automatica* 129:109597
142. Mannucci T, van Kampen E, de Visser C, Chu Q. 2018. Safe exploration algorithms for reinforcement learning controllers. *IEEE Transactions on Neural Networks and Learning Systems* 29:1069–1081
143. Dean S, Taylor AJ, Cosner RK, Recht B, Ames AD. 2020. Guaranteeing safety of learned perception modules via measurement-robust control barrier functions. [arXiv:2010.16001](#) [eess.SY]
144. Liu CK, Negrut D. 2021. The role of physics-based simulators in robotics. *Annual Review of Control, Robotics, and Autonomous Systems* 4:35–58
145. Brockman G, Cheung V, Pettersson L, Schneider J, Schulman J, et al. 2016. OpenAI Gym. [arXiv:1606.01540](#) [cs.LG]
146. Panerati J, Zheng H, Zhou S, Xu J, Prorok A, Schoellig AP. 2021. *Learning to Fly—a Gym Environment with PyBullet Physics for Reinforcement Learning of Multi-agent Quadcopter Control*. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Piscataway, NJ: IEEE
147. Coumans E, Bai Y. 2016–2021. PyBullet, a Python module for physics simulation for games, robotics and machine learning. <http://pybullet.org>
148. Andersson JAE, Gillis J, Horn G, Rawlings JB, Diehl M. 2019. CasADi – A software framework for nonlinear optimization and optimal control. *Mathematical Programming Computation* 11:1–36
149. Schulman J, Wolski F, Dhariwal P, Radford A, Klimov O. 2017. Proximal policy optimization algorithms. [arXiv:1707.06347](#) [cs.LG]
150. Wieber PB, Tedrake R, Kuindersma S. 2016. Modeling and control of legged robots. In *Springer Handbook of Robotics*, ed. B Siciliano, O Khatib, pp. 1203–1234. Cham: Springer International Publishing
151. McKinnon CD, Schoellig AP. 2018. *Experience-Based Model Selection to Enable Long-Term, Safe Control for Repetitive Tasks Under Changing Conditions*. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2977–2984. Piscataway, NJ: IEEE
152. Chandak Y, Jordan S, Theodorou G, White M, Thomas PS. 2020. Towards safe policy improvement for non-stationary MDPs. In *Advances in Neural Information Processing Systems*, ed. H Larochelle, M Ranzato, R Hadsell, MF Balcan, H Lin, pp. 9156–9168, vol. 33. Red Hook, NY, USA: Curran Associates, Inc.
153. Burgner-Kahrs J, Rucker DC, Choset H. 2015. Continuum robots for medical applications: A survey. *IEEE Transactions on Robotics* 31:1261–1280
154. Mueller FL, Schoellig AP, D’Andrea R. 2012. *Iterative learning of feed-forward corrections for high-performance tracking*. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 3276–3281. Piscataway, NJ: IEEE
155. Dean S, Tu S, Matni N, Recht B. 2018. Safely learning to control the constrained linear quadratic regulator. [arXiv :5582–5588](#)
156. McKinnon CD, Schoellig AP. 2019. *Learning Probabilistic Models for Safe Predictive Control in Unknown Environments*. In *2019 18th European Control Conference (ECC)*, pp. 2472–2479. Piscataway, NJ: IEEE

IEEE 58th Conference on Decision and Control (CDC), 第1758–1765页。新泽西州皮斯卡特维：IEEE 136. Fisac JF, Lugovoy NF, Rubies-Royo V, Ghosh S, Tomlin CJ. 2019. *Bridging Hamilton-Jacobi safety analysis and reinforcement learning*. 收录于 *2019 International Conference on Robotics and Automation (ICRA)*, 第8550–8556页。新泽西州皮斯卡特维：IEEE 137. Choi JJ, Lee D, Sreenath K, Tomlin CJ, Herbert SL. 2021. 面向安全关键控制的稳健控制屏障值函数。arXiv:2104.02808 [eess.SY] 138. Herbert S, Choi JJ, Sanjeev S, Gibson M, Sreenath K, Tomlin CJ. 2021. 使用Hamilton-Jacobi可达性对自主系统进行可扩展的安全保证学习。arXiv:2101.05916 [cs.RO] 139. Wabersich KP, Zeilinger MN. 2018. *Linear Model Predictive Safety Certification for Learning- Based Control*. 收录于 *2018 IEEE Conference on Decision and Control (CDC)*, 第7130–7135页。新泽西州皮斯卡特维：IEEE 140. Wabersich KP, Hewing L, Carron A, Zeilinger MN. 2019. 基于学习控制的概率模型预测安全认证。arXiv:1906.10417 [eess.SY] 141. Wabersich KP, Zeilinger MN. 2021. 用于约束非线性动力系统学习控制的预测安全滤波器。 *Automatica* 129:109597 142. Mannucci T, van Kampen E, de Visser C, Chu Q. 2018. 安全 ex

A. SUMMARY OF REVIEWED LITERATURE

Table 1 summarizes the safe learning control approaches reviewed in this paper. Learning approaches are classified as *Model-Based* if a dynamics model is used to produce control inputs or *Model-Free* if there is a direct mapping from states or measurements to control inputs. For model-based approaches, we categorize the structure of the *A Priori* and the *Learned* component of the model as Gaussian process (*GP*), neural network (*NN*), or *Other* for non-standard methods (e.g., set computation from data).

The *Safety Properties* category highlights which algorithms can handle state and/or input constraints (*Constraint Satisfaction*) and which ones provide stability guarantees (*Stability*). For both of these properties, we report whether they are enforced while the policy is being trained (*During Learning*) or by the resultant policy (*After Learning*). The *Robustness* category indicates whether the final policy is designed to be robust to *Input* disturbances; model *Parameter* uncertainty, and/or other *Dynamics* uncertainties broader than the previous two types of disturbances.

Finally, the last category reports the *Task(s)* to which each method was applied: *Stabilization* for stabilizing an equilibrium of the system; *Tracking* for the tracking of a given trajectory; *Navigation* for navigating or planning a sequence of actions to reach a given goal; *Locomotion* for the movement of legged robots like humanoids or quadrupeds; and *Manipulation* for pushing, reaching, or grasping operations using a robotic arm.

A. SUMMARY OF REVIEWED LITERATURE

Table 1 总结了本文回顾的安全学习控制方法。如果使用动力学模型来生成控制输入，则将学习方法分类为 *Model-Based*；如果存在从状态或测量直接映射到控制输入的情况，则分类为 *Model-Free*。对于基于模型的方法，我们将模型的 *A Priori* 结构和 *Learned* 组成部分分类为高斯过程（GP）、神经网络（NN）或 *Other*，用于非标准方法（例如，从数据进行集合计算）。

Safety Properties 类别突出了哪些算法可以处理状态和/或输入约束（*Constraint Satisfaction*），以及哪些算法提供稳定性保证（*Stability*）。对于这两种特性，我们都会报告它们是在策略训练过程中执行（*During Learning*），还是由生成的策略强制执行（*After Learning*）。*Robustness* 类别表明最终策略是否被设计为对 *Input* 干扰、模型 *Parameter* 不确定性和/或比前两种干扰更广泛的其他 *Dynamics* 不确定性保持鲁棒。

最后，最后一类报告了每种方法所应用的 *Task(s)*： *Stabilization* 用于稳定系统的平衡； *Localization* 用于追踪给定的轨迹； *Tracking* 用于导航或规划一系列动作以达到给定目标； *Navigation* 用于类人或四足机器人的运动； *Locomotion* 用于使用机器人手臂进行推、达到或抓取操作。

Table 1: Summary of the key properties of the works referenced in [Sec. 3](#)

Section		Learning			Safety Properties					Task
		Model-based		Model-free	During Learning		After Learning			
		<i>A Priori</i>	Learned		Constraint Satisfaction	Stability	Constraint Satisfaction	Stability	Robustness w.r.t.	
Learning Uncertain Dynamics Sec. 3.1	Sec. 3.1.1	Linear (39, 40, 44, 45), Nonlinear (42, 43)	NN (39, 44, 45), GP (40, 42, 43)			(39, 40, 42, 43, 44, 45)		(39, 40, 42, 43, 44, 45)	Input (39, 40), Dynamics (42, 43, 44, 45)	Tracking (39, 40, 42, 43, 44, 45)
	Sec. 3.1.2	Linear (46, 47, 48), Nonlinear (49, 50)	GP (46, 48, 49, 50) Other (47)			(46, 48, 49, 50)		(46, 47, 48, 49, 50)	Dynamics (46, 47, 48, 49, 50)	Stabilization (46, 47, 48), Tracking (49, 50)
	Sec. 3.1.3	Linear (51, 52, 53, 54, 58, 59, 60, 61), Nonlinear (55, 56, 62, 63, 64, 65, 66, 67)	NN (60, 66), GP (61, 62, 63, 64, 65), Other (51, 52, 53, 54, 55, 56, 58, 59, 67)		Input & state (51, 52, 53, 54, 55, 56, 58, 59, 58, 59, 60, 61, 62, 63, 64, 65, 66)	(51, 52, 53, 54, 55, 56, 58, 59, 60, 61, 62, 63, 64, 65, 66)	Input & state (51, 52, 53, 54, 55, 56, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67)	(51, 52, 53, 54, 55, 56, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67)	Input (59), Dynamics (60, 61, 62, 63, 64, 65, 66, 67), Parameters (51, 52, 53, 54, 55, 56, 58, 59)	Stabilization (52, 53, 54, 55, 56, 58, 59, 60, 61, 64, 65), Tracking (51, 62, 63, 66, 67)
	Sec. 3.1.4	Nonlinear (68)	GP (68)			(68)		(68)		Stabilization (68)
Encouraging Safety in RL Sec. 3.2	Sec. 3.2.1	Nonlinear (69, 79, 82),	GP (78, 79, 80, 81)	NN (70, 75, 85, 86, 87), Tabular (73)	Input & state (69, 70, 73, 75, 78, 79, 80, 81, 82) Trajectory (87)		Input & state (69, 70, 73, 75, 78, 79, 80, 81, 82) Trajectory (85, 86, 87)			Stabilization (81), Tracking (79), Navigation (69, 70, 73, 82, 85, 86), Locomotion (85, 87), Manipulation (75, 85, 86, 87)
	Sec. 3.2.2	Nonlinear (89, 90)	NN (89, 90, 91, 92)	NN (93)			Input & state (89, 90, 91, 92)		Dynamics (93)	Stabilization (91), Navigation (89, 90, 91, 92), Locomotion (91, 93), Manipulation (92)
	Sec. 3.2.3	Tabular (97, 99)		NN (34, 95, 97, 98, 99)	Trajectory (34, 97, 98, 99)		Trajectory (34, 95, 97, 98, 99)			Navigation (34, 95, 97, 98, 99), Locomotion (34, 98, 99)
	Sec. 3.2.4		NN (106)	NN (36, 37, 38, 104, 105, 107, 108)	Input & state (106)	(106)	Input & state (106)	(106)	Input (37, 38), Dynamics (36, 104, 105, 106), Parameters (107, 108)	Stabilization (36, 104, 108), Tracking (106), Navigation (104, 105), Locomotion (36, 37, 38, 107), Manipulation (108)
Certifying Learning- based Control Sec. 3.3	Sec. 3.3.1	Linear (110), Nonlinear (109, 111, 113, 114)	NN (109, 110, 111, 113), Other (114)			(110)		(109, 110, 111, 113, 114)	Dynamics (109, 110, 111, 113, 114)	Stabilization (113, 114), Tracking (109, 110, 111, 114), Locomotion (114)
	Sec. 3.3.2	Linear (139, 140), Nonlinear (119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 133, 134, 135, 141, 142)	GP (121, 128, 130, 131, 133, 140), NN (119, 120, 122, 123, 124, 128, 129, 135) Other (125, 126, 127, 134, 139, 140, 141)	NN (121, 122, 136)	Input & state (123, 124, 125, 126, 127, 128, 129, 130, 139, 140, 141, 142), State (133, 134, 135)	(123, 124, 125, 126, 127, 128, 129, 130, 133, 134, 135, 139, 140, 141)	Input & state (119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 139, 140, 141, 142), State (133, 134, 135, 136)	(119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 133, 134, 135, 136, 139, 140, 141)	Dynamics (119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 133, 134, 135, 142), Parameters (126, 127, 141)	Stabilization (128, 136, 139, 141, 142), Tracking (119, 120, 123, 124, 125, 126, 127, 129, 130, 131, 133, 134, 140), Navigation (121, 135), Locomotion (122)

表 1: 第 3 节中引用的工作的关键属性总结

Section	Learning			Safety Properties			
	Model-based		Model-free	During Learning		After Learning	
	<i>A Priori</i>	Learned		Constraint Satisfaction	Stability	Constraint Satisfaction	Stability
Learning Uncertain Dynamics Sec. 3.1	Sec. 3.1.1	Linear (39, 40, 44, 45), Nonlinear (42, 43)	NN (39, 44, 45), GP (40, 42, 43)		(39, 40, 42, 43, 44, 45)		Input (39, 40, 42, 43, 44, 45)
	Sec. 3.1.2	Linear (46, 47, 48), Nonlinear (49, 50)	GP (46, 48, 49, 50) Other (47)		(46, 48, 49, 50)		(46, 47, 48, 49, 50)
	Sec. 3.1.3	Linear (51, 52, 53, 54, 58, 59, 60, 61), Nonlinear (55, 56, 62, 63, 64, 65, 66, 67)	NN (60, 66), GP (61, 62, 63, 64, 65), Other (51, 52, 53, 54, 55, 56, 58, 59, 67)	Input & state (51, 52, 53, 54, 55, 56, 58, 59, 60, 61, 62, 63, 64, 65, 66)	(51, 52, 53, 54, 55, 56, 58, 59, 60, 61, 62, 63, 64, 65, 66)	Input & state (51, 52, 53, 54, 55, 56, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67)	(51, 52, 53, 54, 55, 56, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67)
	Sec. 3.1.4	Nonlinear (68)	GP (68)		(68)		(68)
Encouraging Safety in RL Sec. 3.2	Sec. 3.2.1	Nonlinear (69, 79, 82),	GP (78, 79, 80, 81)	NN (70, 75, 85, 86, 87), Tabular (73)	Input & state (69, 70, 73, 75, 78, 79, 80, 81, 82) Trajectory (87)	Input & state (69, 70, 73, 75, 78, 79, 80, 81, 82) Trajectory (85, 86, 87)	
	Sec. 3.2.2	Nonlinear (89, 90)	NN (89, 90, 91, 92)	NN (93)		Input & state (89, 90, 91, 92)	Dynami
	Sec. 3.2.3	Tabular (97, 99)		NN (34, 95, 97, 98, 99)	Trajectory (34, 97, 98, 99)	Trajectory (34, 95, 97, 98, 99)	
	Sec. 3.2.4		NN (106)	NN (36, 37, 38, 104, 105, 107, 108)	Input & state (106)	Input & state (106)	Input (37, 38, 104, 105, 107, 108)
Certifying Learning- based Control Sec. 3.3	Sec. 3.3.1	Linear (110), Nonlinear (109, 111, 113, 114)	NN (109, 110, 111, 113), Other (114)		(110)		(109, 110, 111, 113, 114)
	Sec. 3.3.2	Linear (139, 140), Nonlinear (119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 133, 134, 135, 141, 142)	GP (121, 128, 130, 131, 133, 140), NN (119, 120, 122, 123, 124, 128, 129, 135), Other (125, 126, 127, 134, 139, 140, 141)	Input & state (123, 125, 126, 127, 128, 129, 130, 139, 140, 141, 142), State (133, 134, 135)	(123, 124, 125, 126, 127, 128, 129, 130, 134, 135, 139, 140, 141)	Input & state (119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 139, 140, 141, 142), State (133, 134, 135, 136, 139, 140, 141)	Dynami (119, 120, 121, 122, 123, 124, 125, 126, 128, 129, 130, 136, 138, 142), P (131, 133, 134, 135, 136, 139, 141)

5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123
124
125
126
127
128
129
130
131
132
133
134
135
136
137
138
139
140
141
142
143
144
145
146
147
148
149
150
151
152
153
154
155
156
157
158
159
160
161
162
163
164
165
166
167
168
169
170
171
172
173
174
175
176
177
178
179
180
181
182
183
184
185
186
187
188
189
190
191
192
193
194
195
196
197
198
199
200

B. REVISION HISTORY

Send errata to angela.schoellig@robotics.utias.utoronto.ca.

Date	Change
December 6, 2021	Figure 6 update
August 13, 2021	Initial submission

B. REVISION HISTORY

将勘误表发送给 `angela.schoellig@robotics.utias.utoronto.ca`。

Date	Change
December 6, 2021	Figure 6 update
August 13, 2021	Initial submission